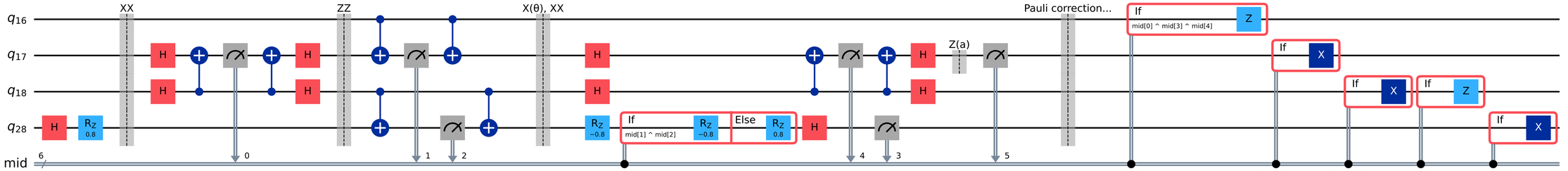
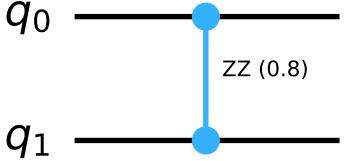
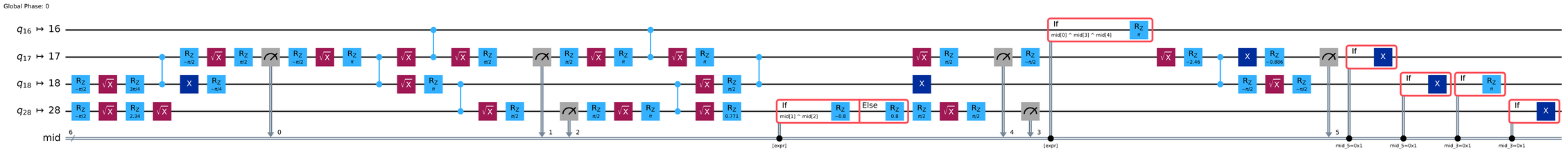
[Z2] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 6, 6 mid-circuit measurements, logical→physical [16, 18]



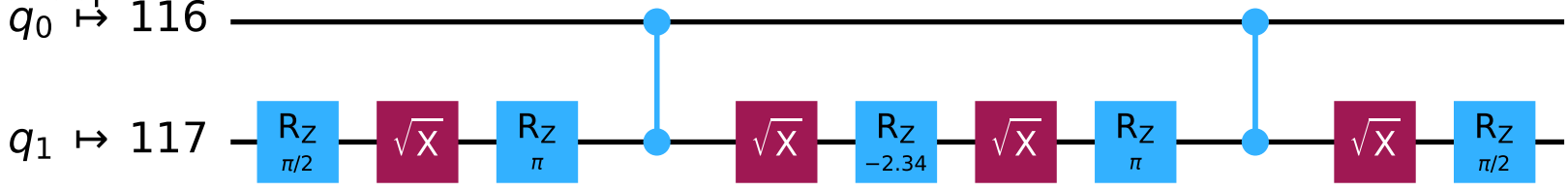
[Z2] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 1



[Z2] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:27 sx:18 cz:8 measure:6 if_else:6 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 6, total depth 39

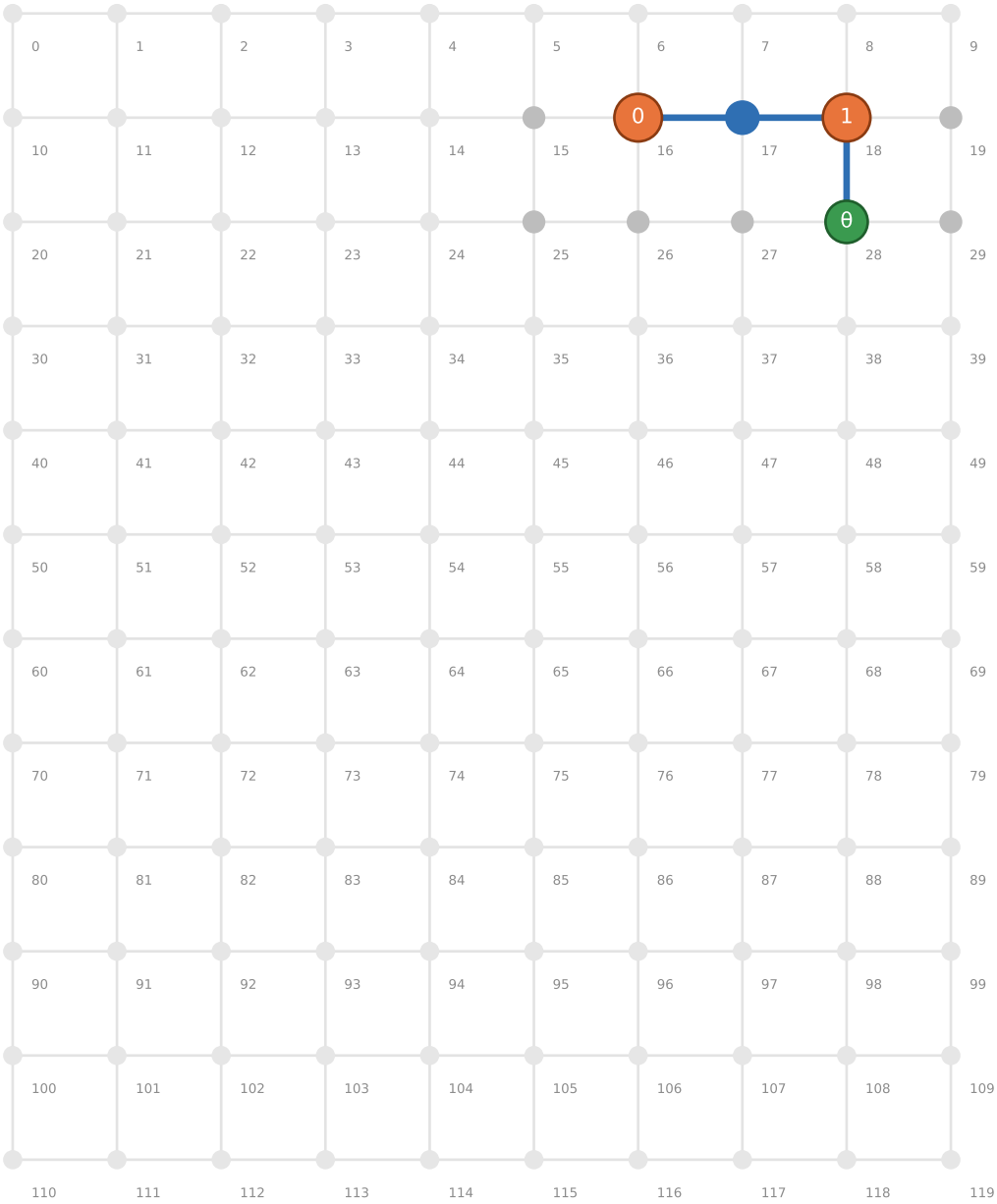


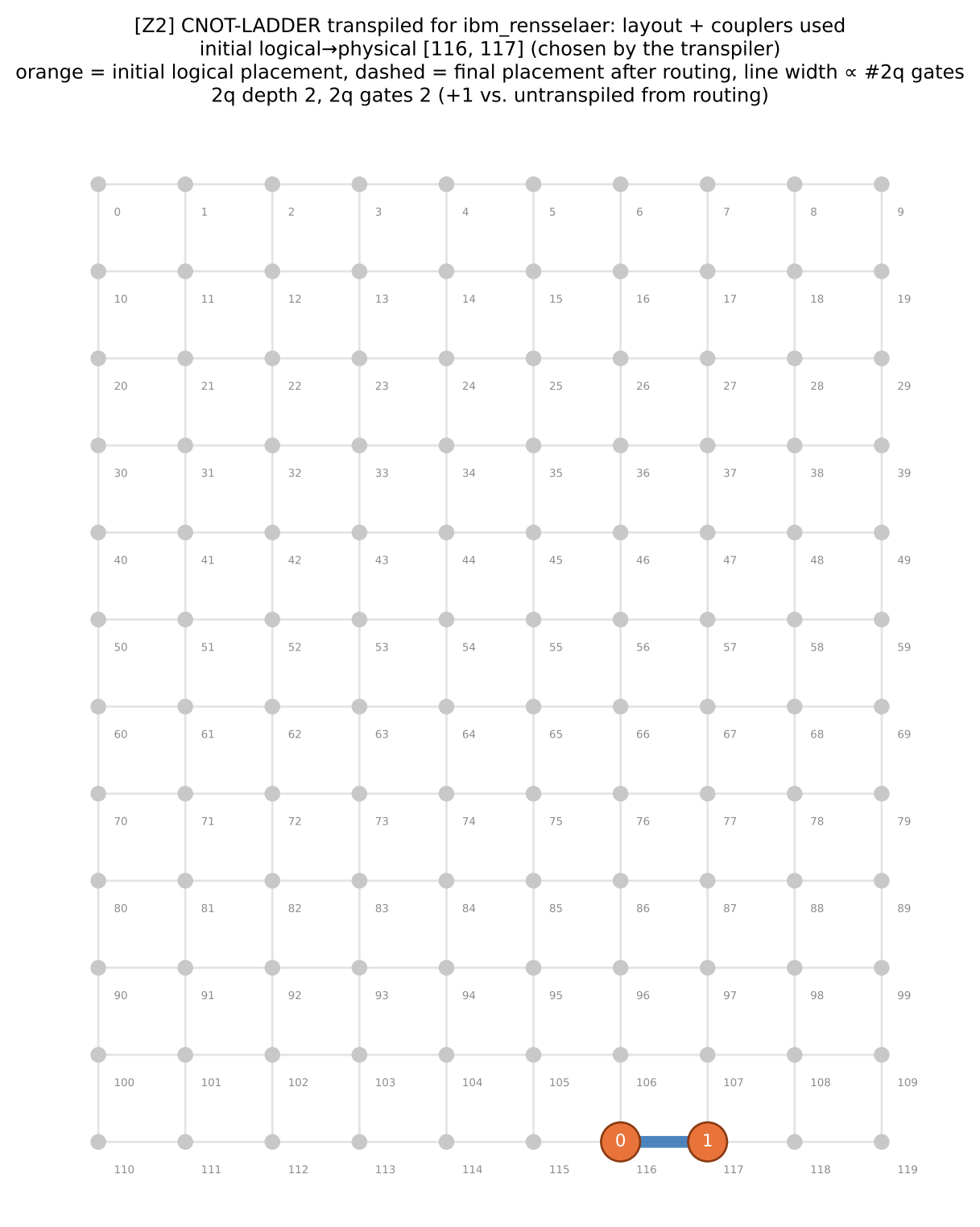
[Z2] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:5 sx:4 cz:2)
Global Phase: 0
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2, total depth 11



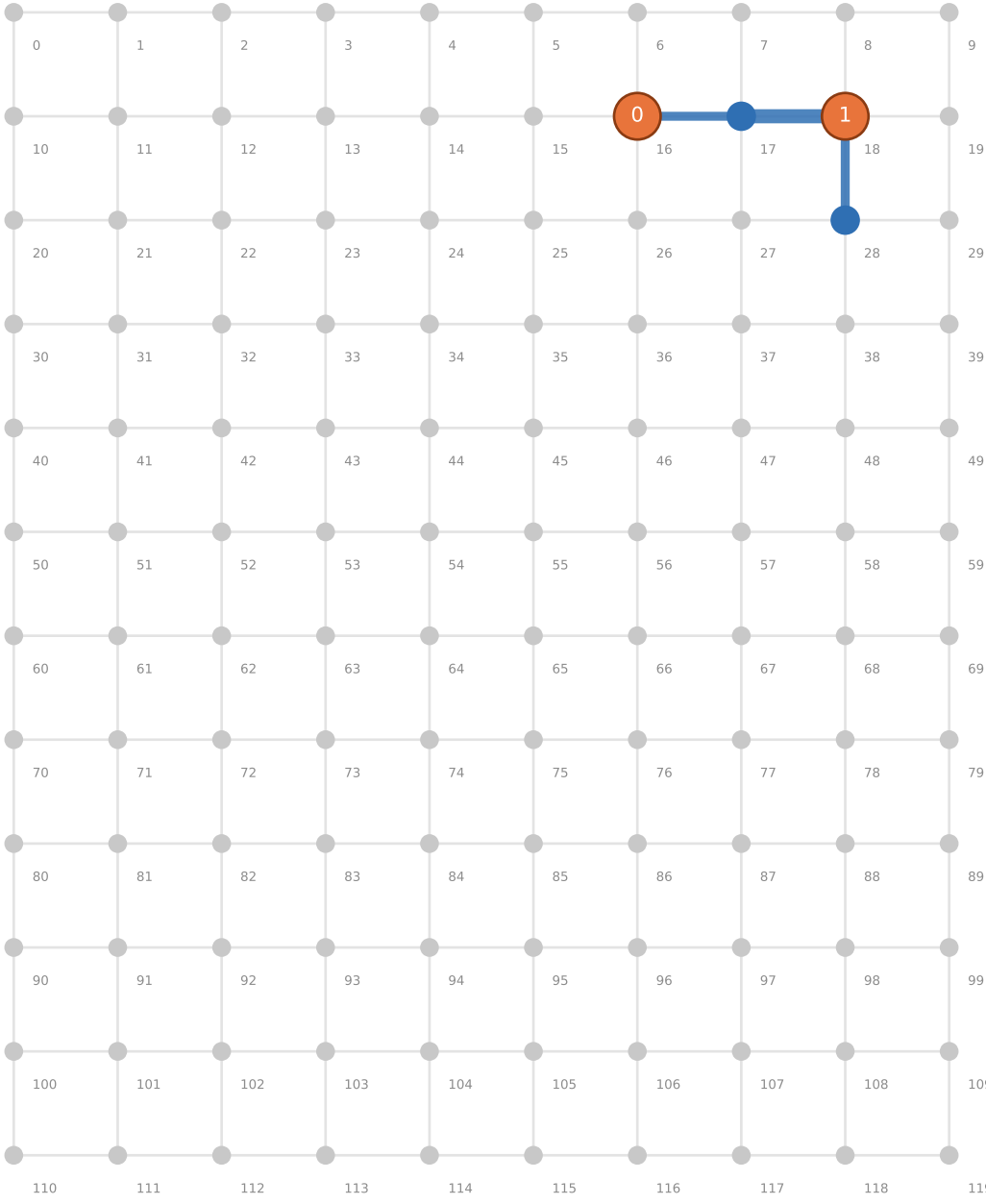
[Z2] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [16, 18]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1
chain length 3, 1 ancillas, 0 |0> fillers

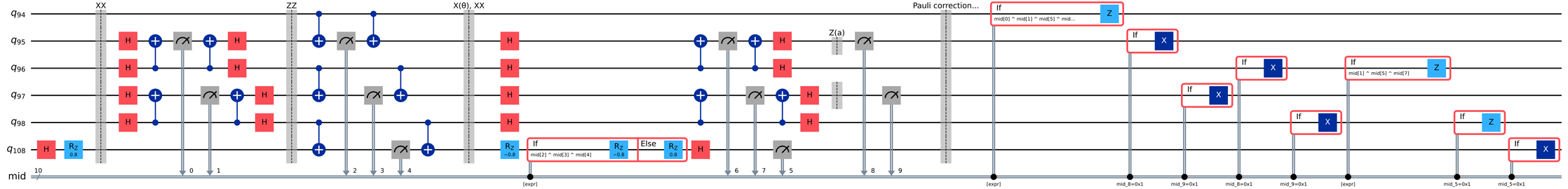



[Z2] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 2, 2q gates 2 (+1 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
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90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1

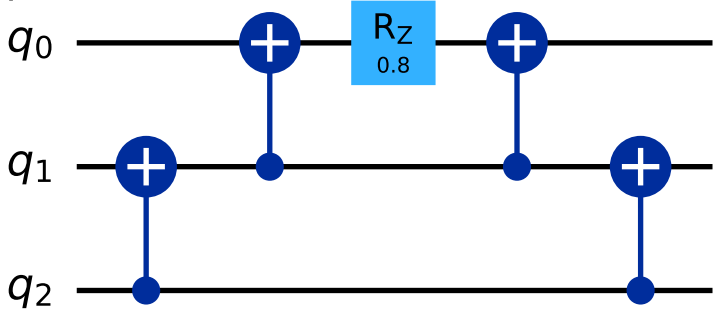
[Z2] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [16, 18]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 8



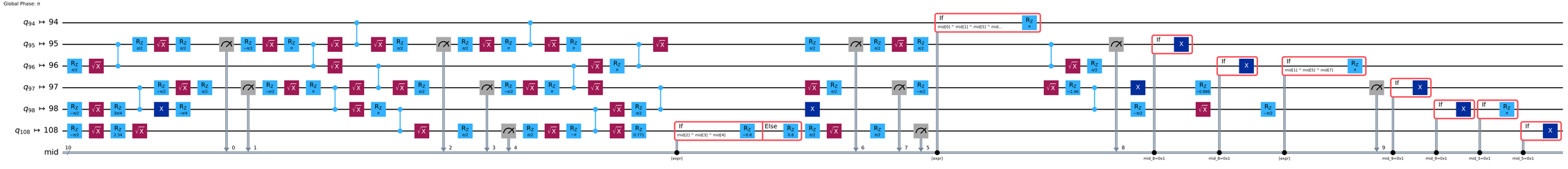
[Z3] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, 10 mid-circuit measurements, logical→physical [94, 96, 98]



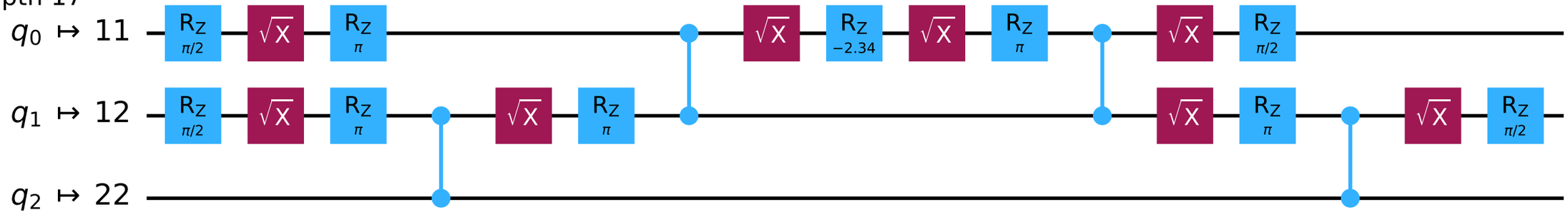
[Z3] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4



[Z3] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:40 sx:30 cz:14 measure:10 if_else:9 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, total depth 39

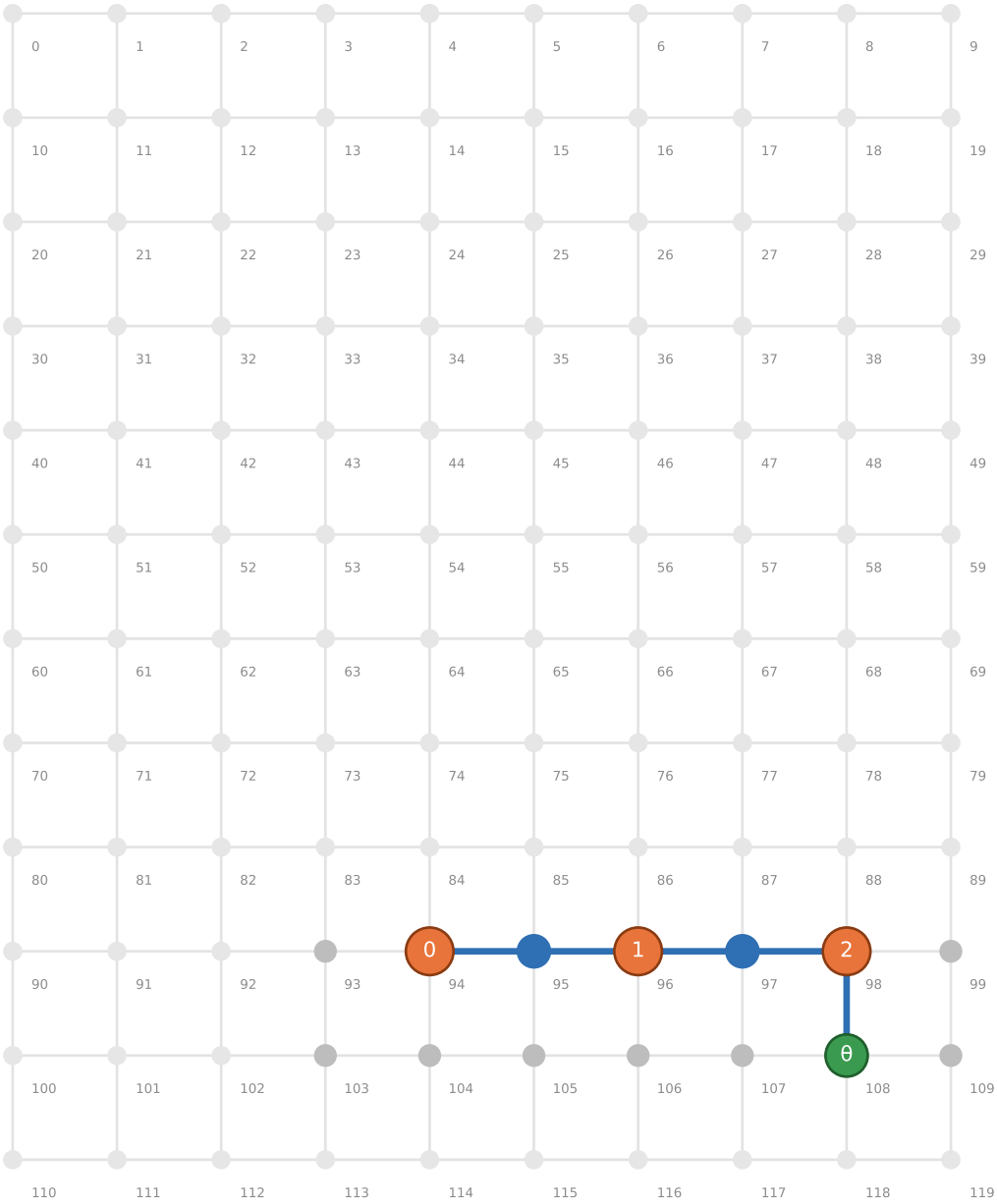


[Z3] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:10 sx:8 cz:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 (3 logical qubits)
2q depth 4, total depth 17

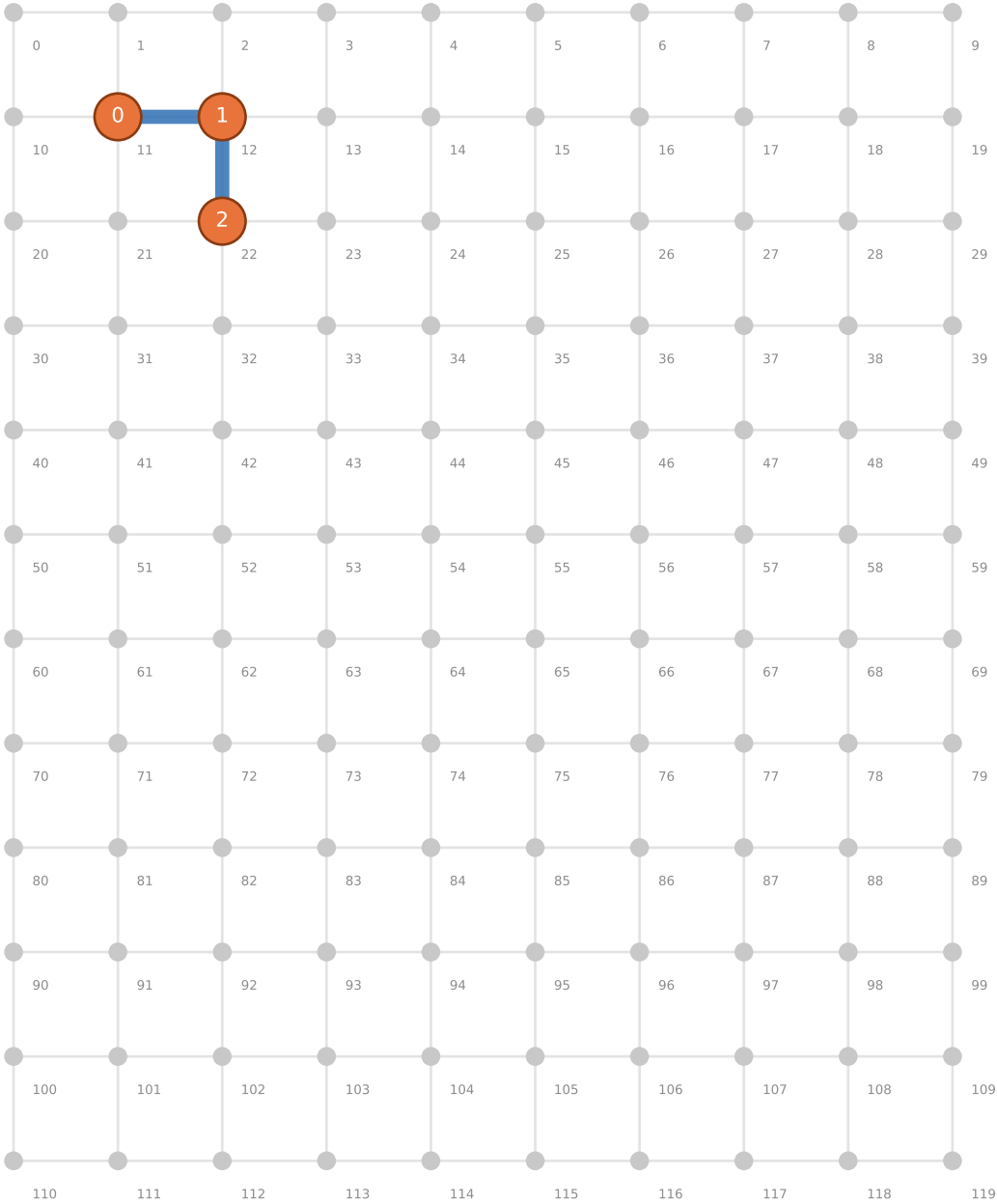


[Z3] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [94, 96, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

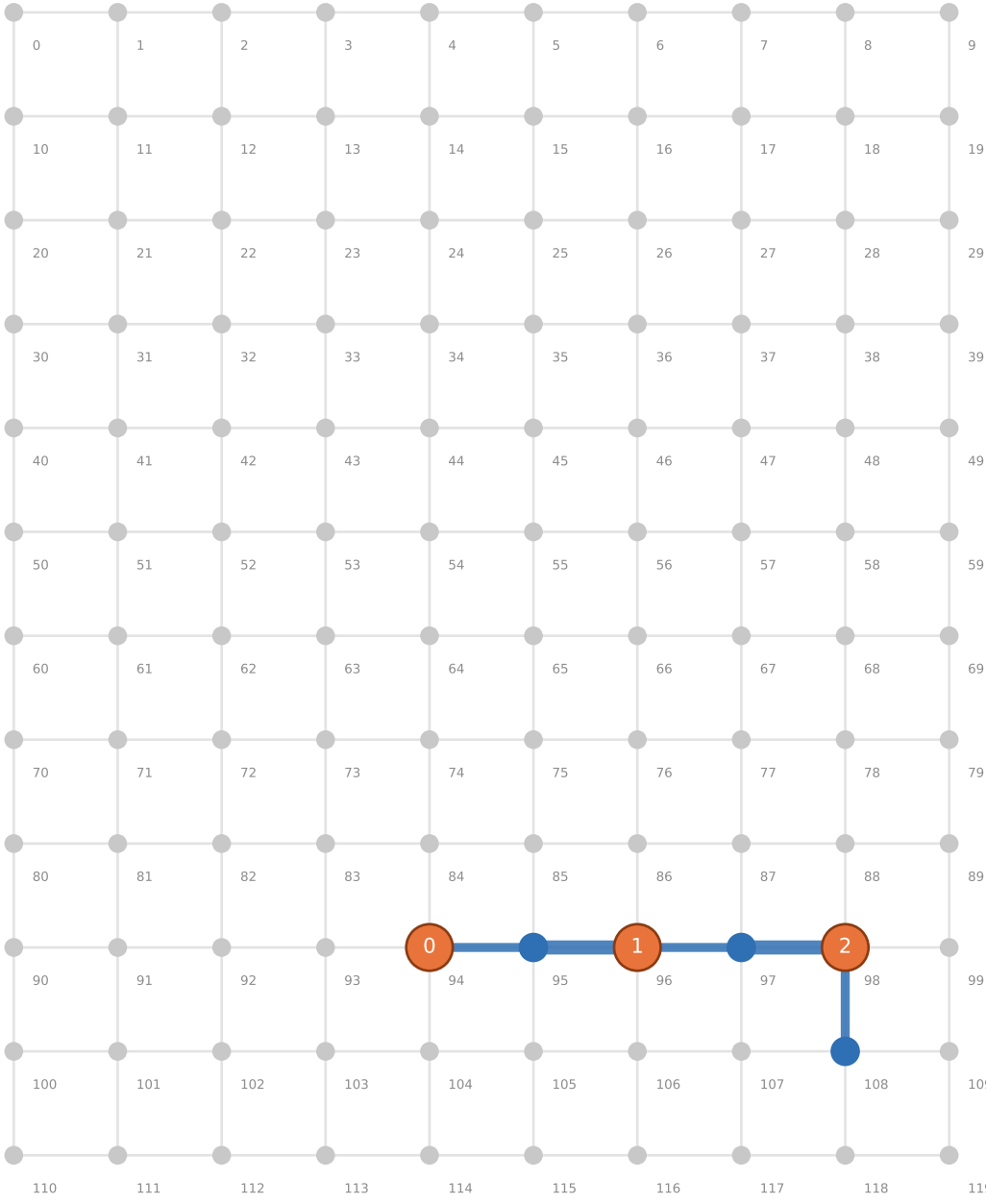
term 0: Z0 Z1 Z2
chain length 5, 2 ancillas, 0 |0> fillers



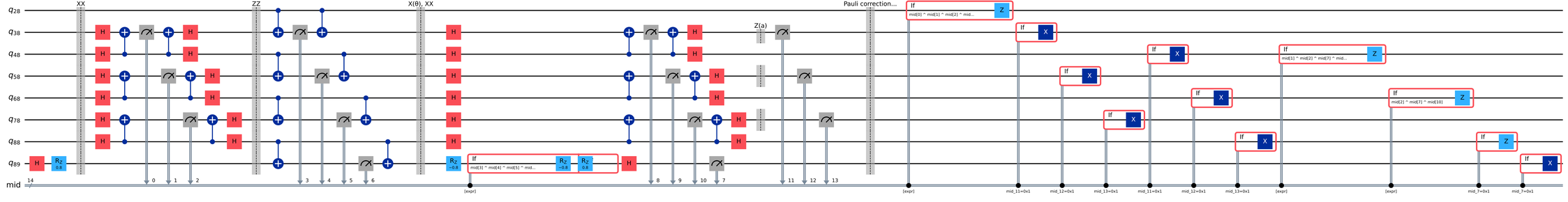
[Z3] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 4, 2q gates 4 (+0 vs. untranspiled from routing)



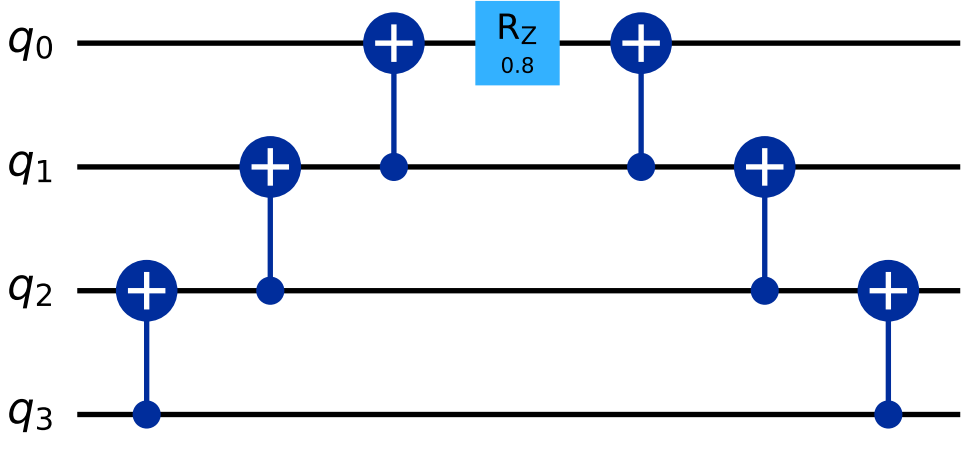
[Z3] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [94, 96, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 14



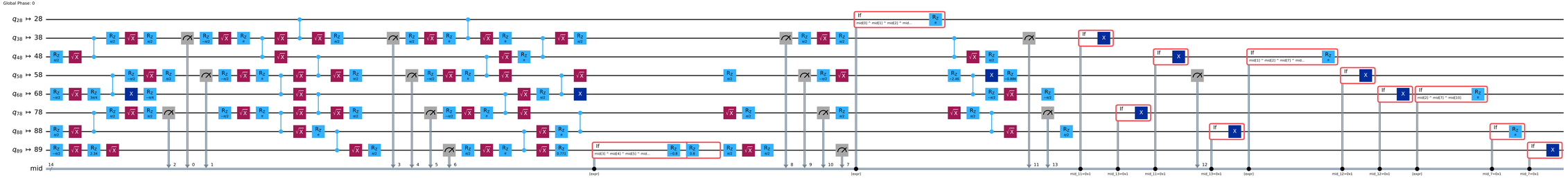
[Z4] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, 14 mid-circuit measurements, logical→physical [28, 48, 68, 88]



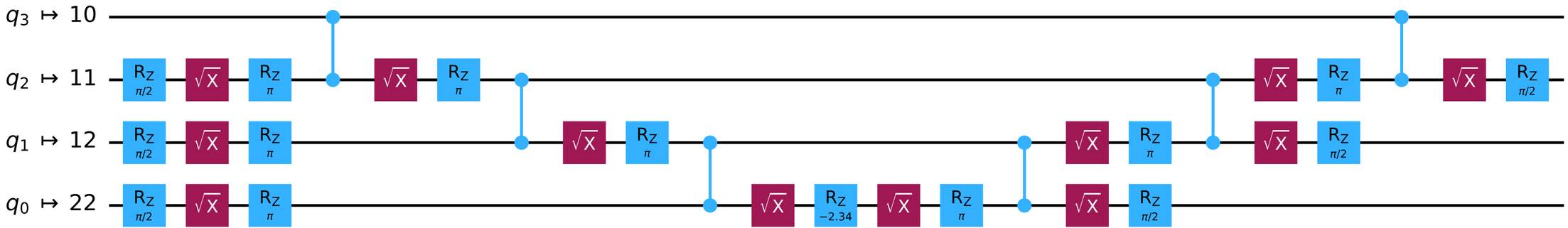
[Z4] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6



[Z4] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:53 sx:42 cz:20 measure:14 if_else:12 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, total depth 39

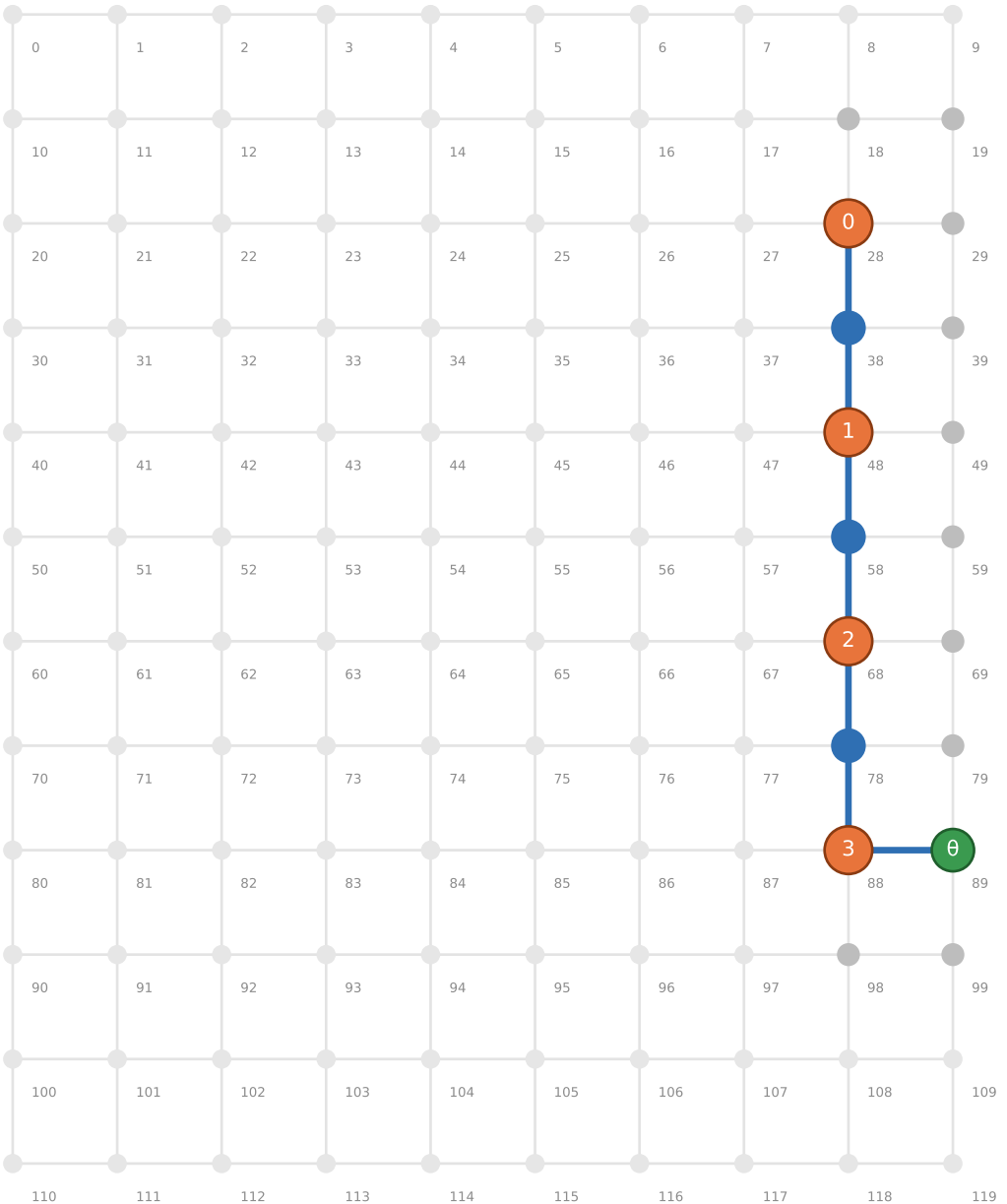


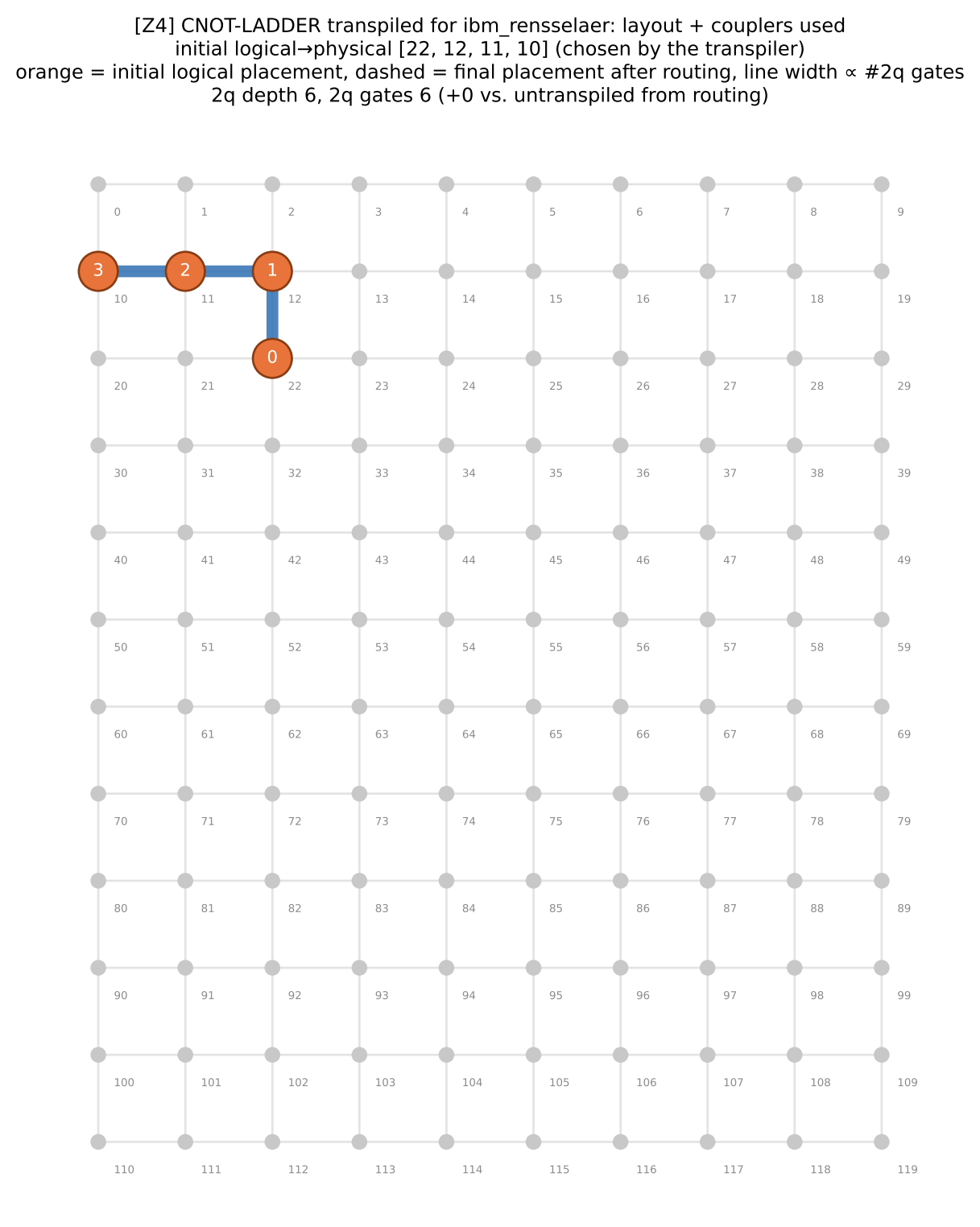
[Z4] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:15 sx:12 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
Global Phase: 0
2q depth 6, total depth 23



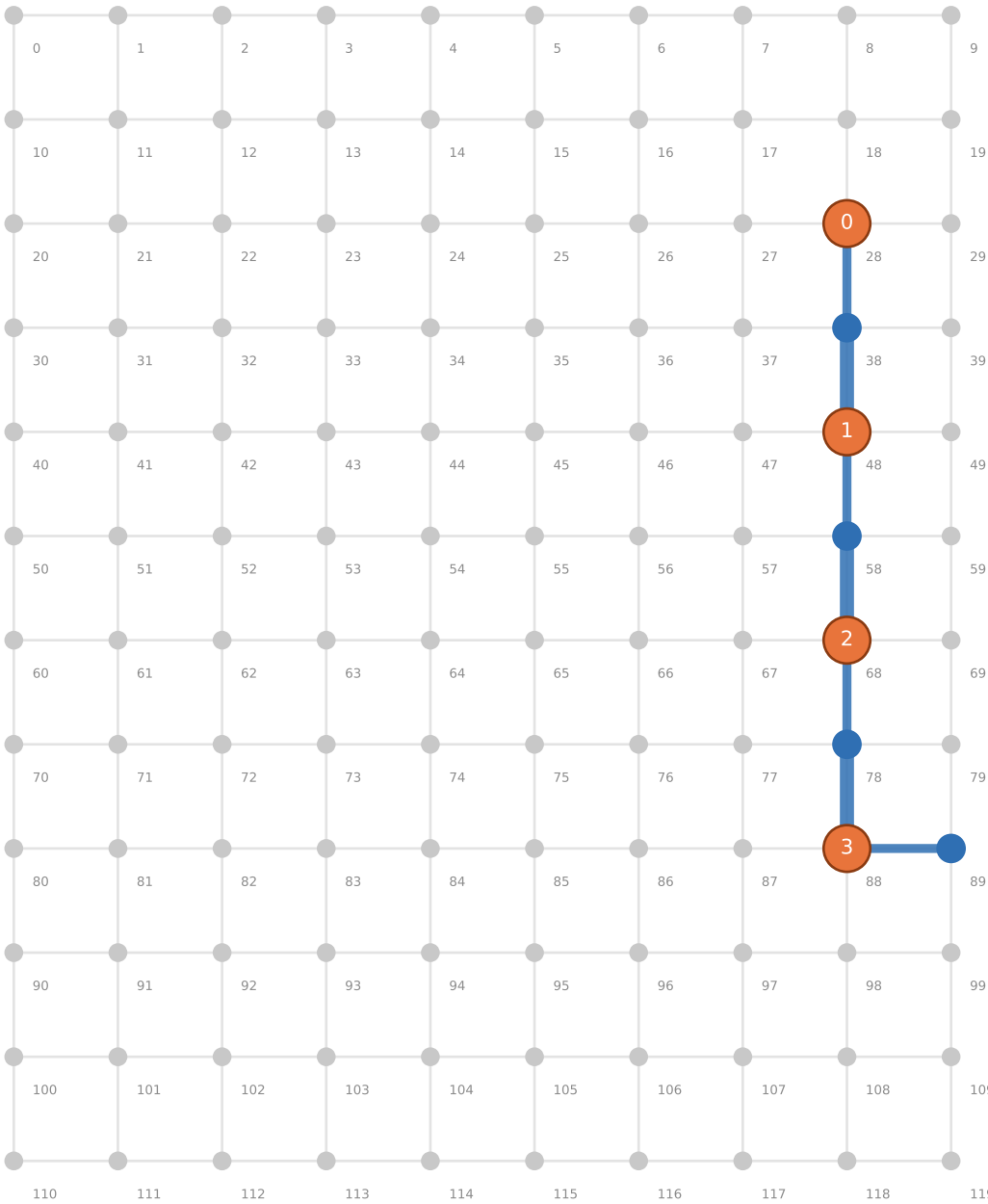
[Z4] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [28, 48, 68, 88]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3
chain length 7, 3 ancillas, 0 |0> fillers

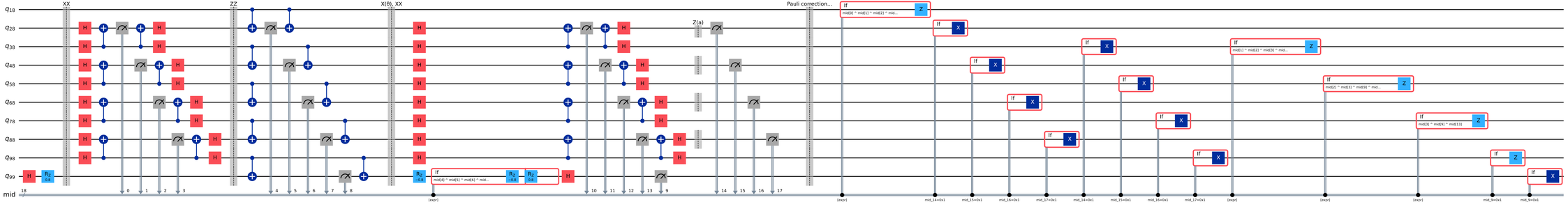



[Z4] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
3
2
1
0
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

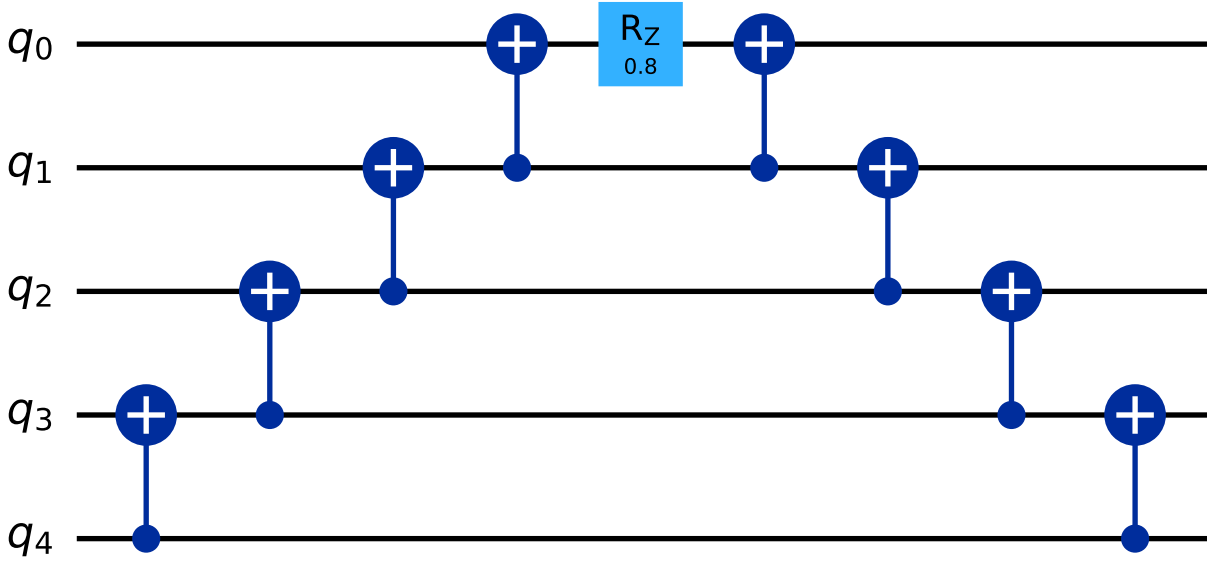
[Z4] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [28, 48, 68, 88]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 20



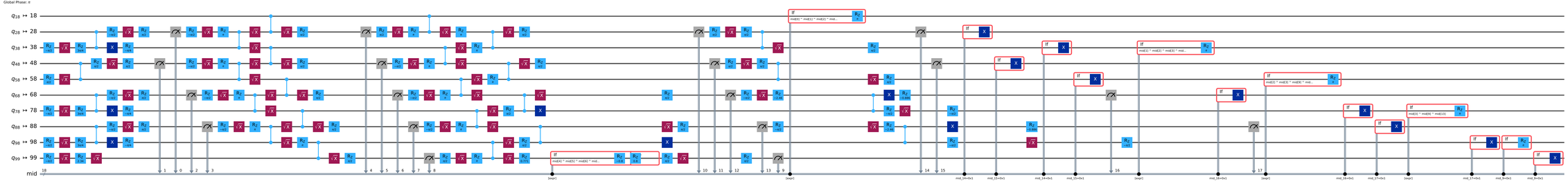
[Z5] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, 18 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



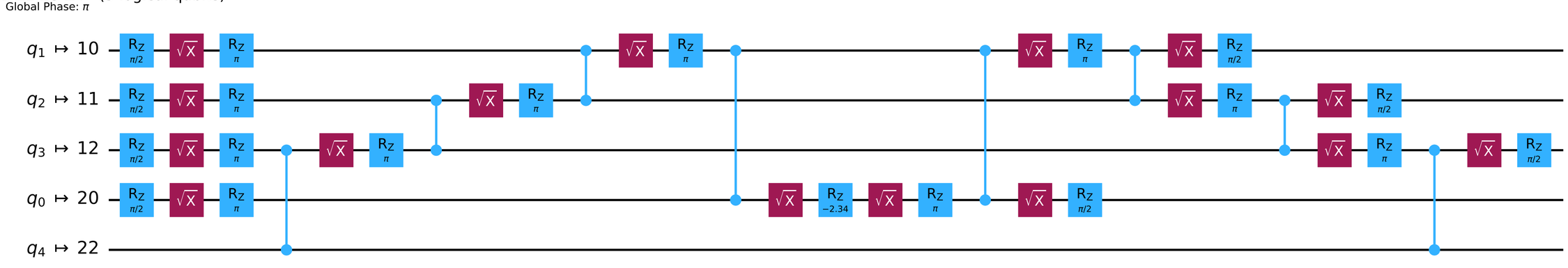
[Z5] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8



[Z5] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:72 sx:54 cz:26 measure:18 if_else:15 x:7)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, total depth 39

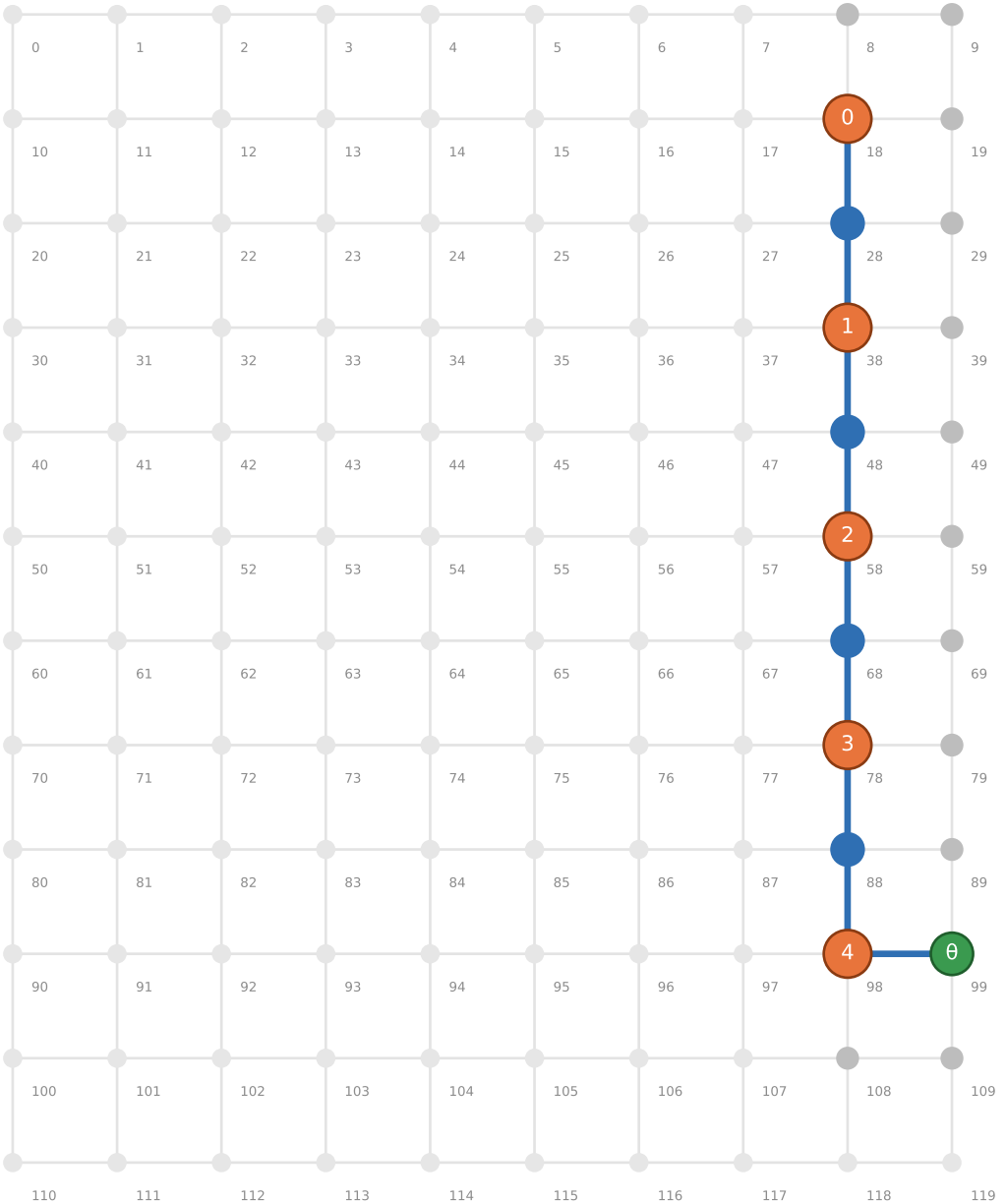


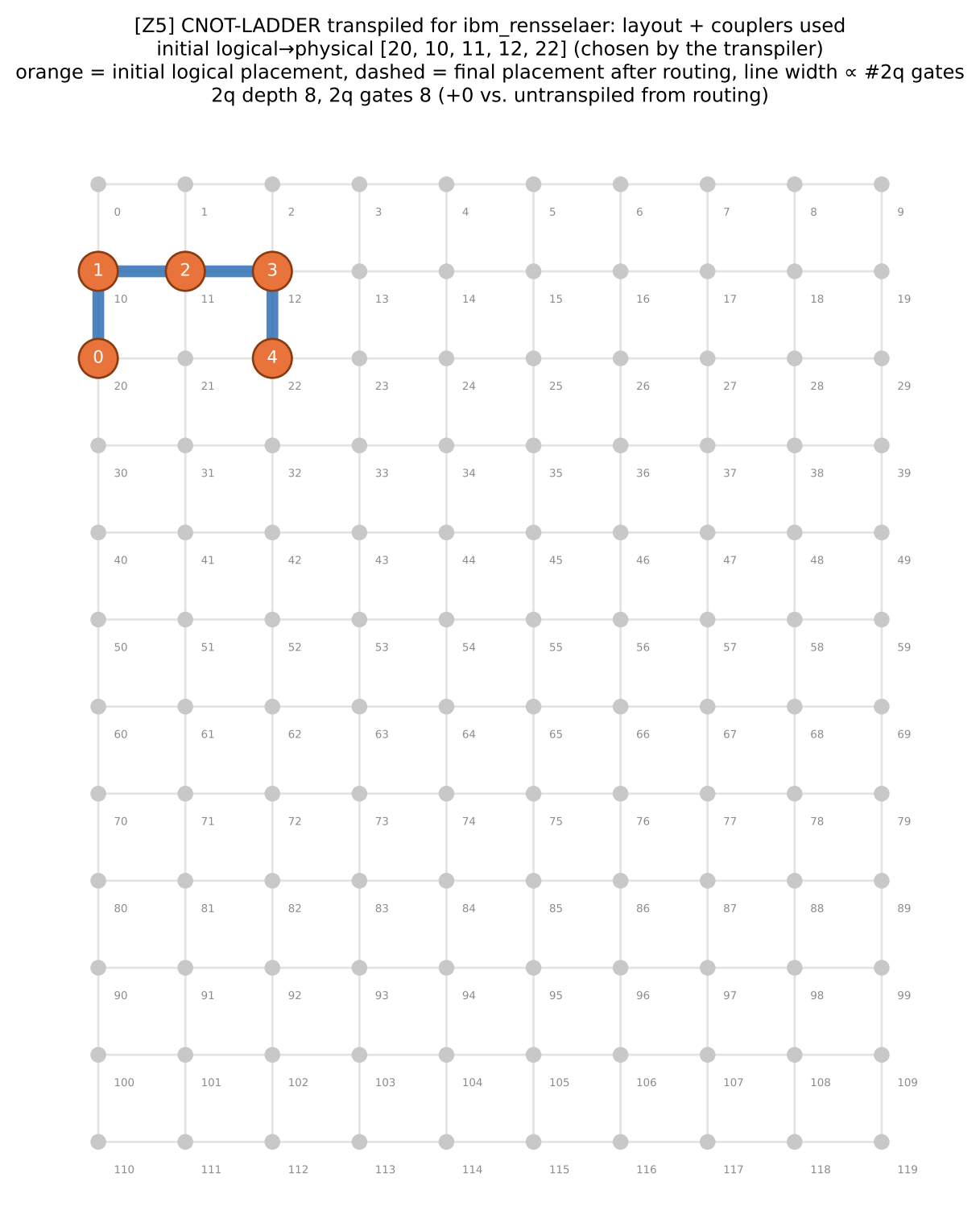
[Z5] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:20 sx:16 cz:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8, total depth 29



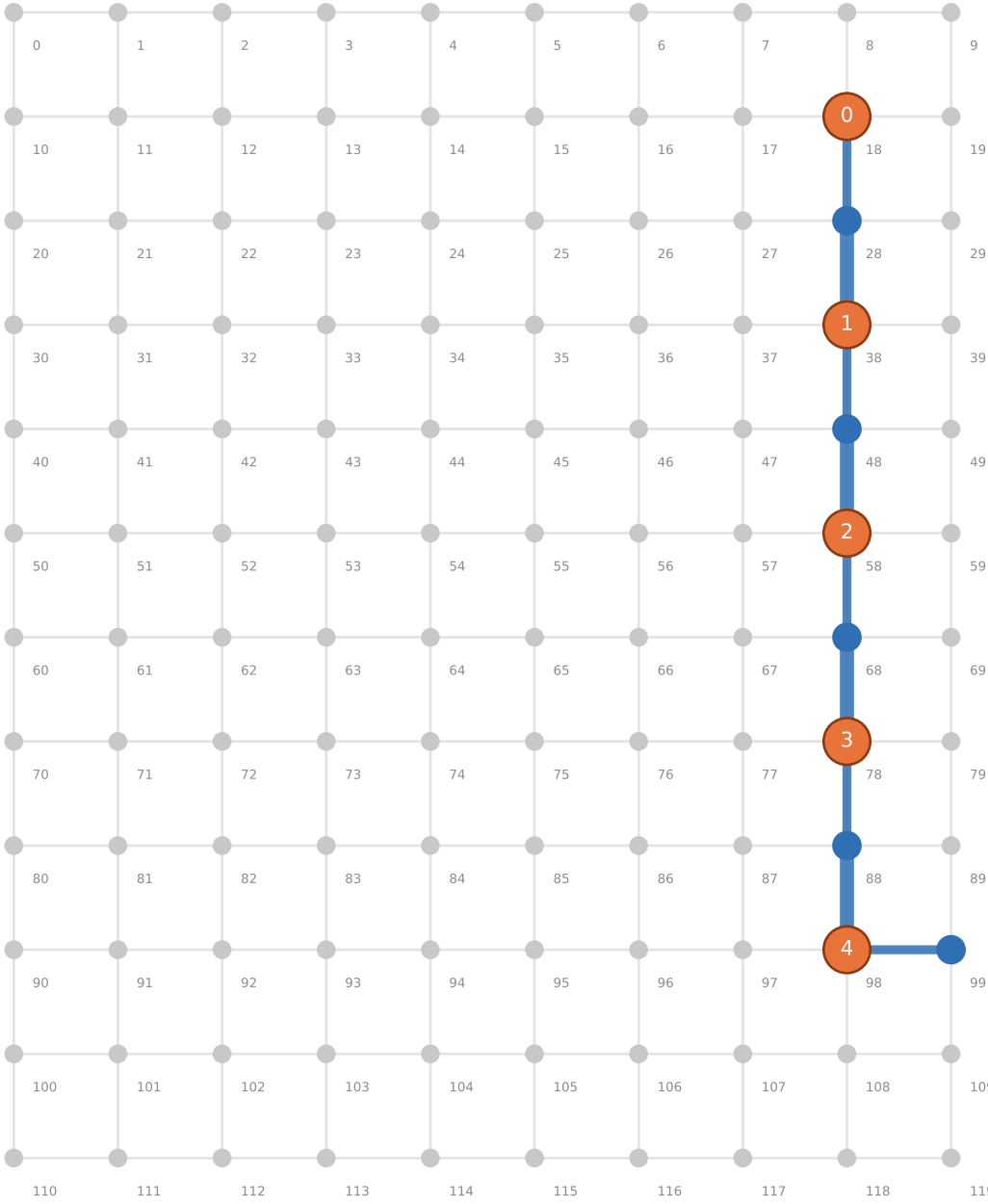
[Z5] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4
chain length 9, 4 ancillas, 0 |0> fillers




[Z5] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [20, 10, 11, 12, 22] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 8, 2q gates 8 (+0 vs. untranspiled from routing)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
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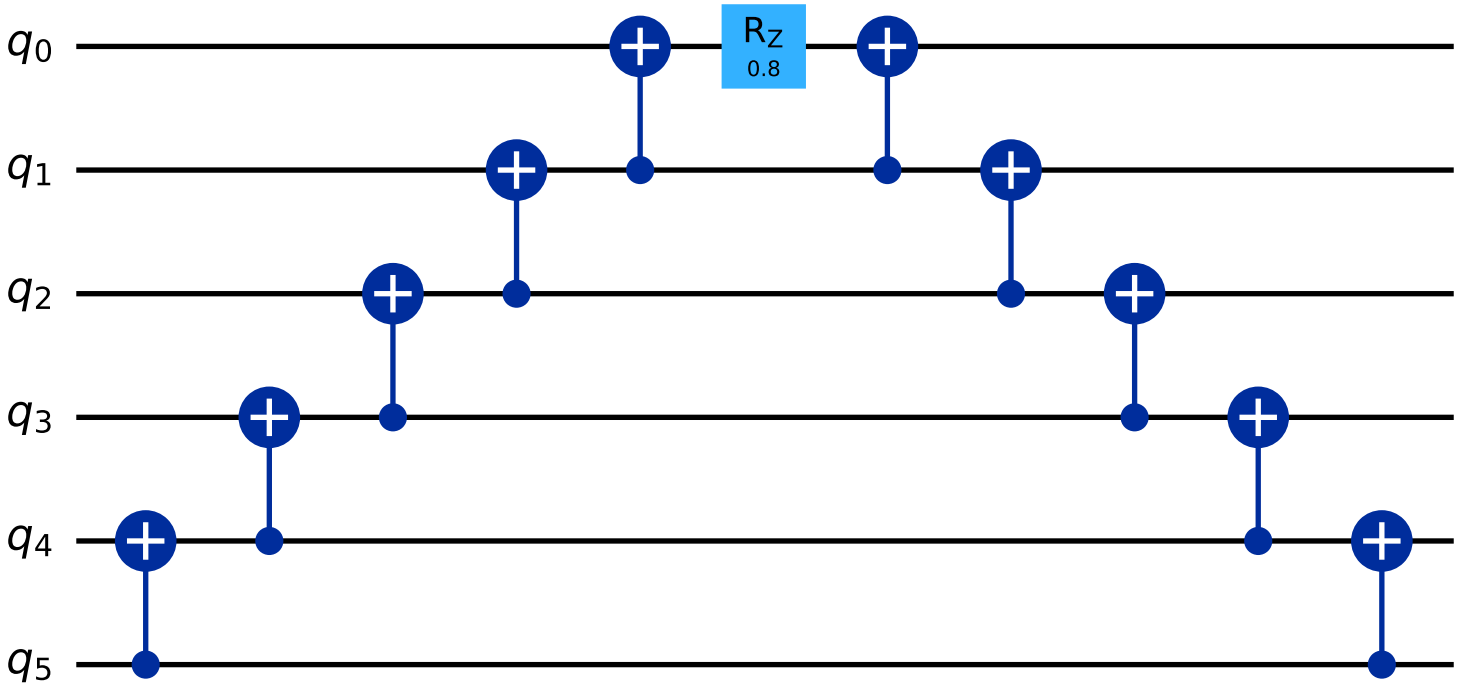
[Z5] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 26



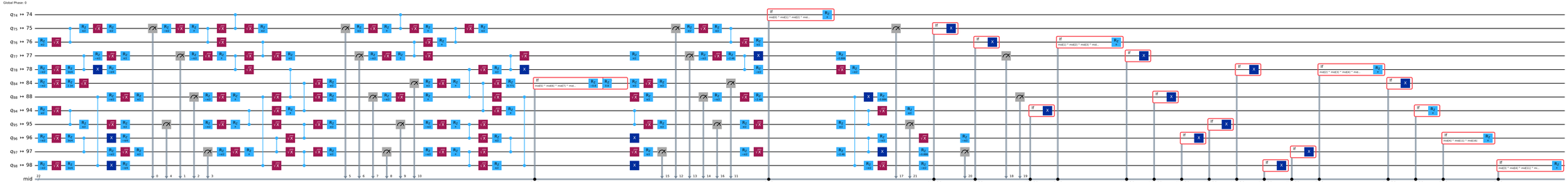
[Z6] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 6, 22 mid-circuit measurements, logical→physical [74, 76, 78, 98, 96, 94]



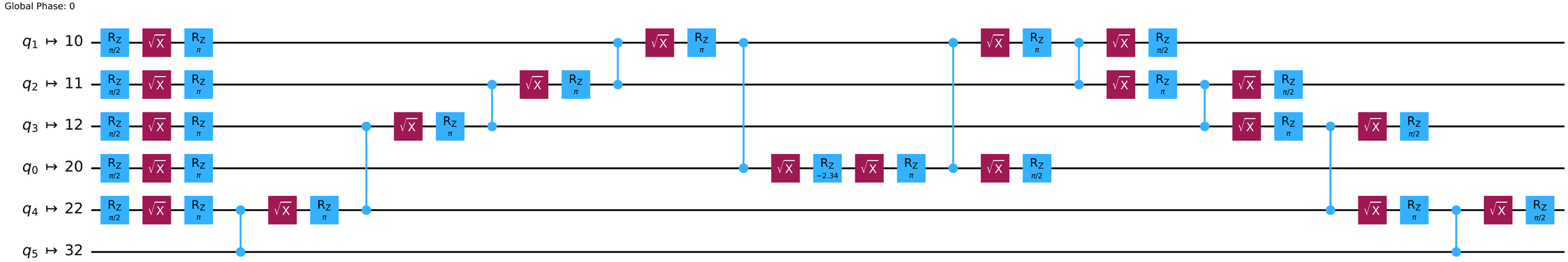
[Z6] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10



[Z6] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:87 sx:66 cz:32 measure:22 if_else:18 x:9)
exp(-i θ P), θ=0.4: P = ZQ Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 6, total depth 41

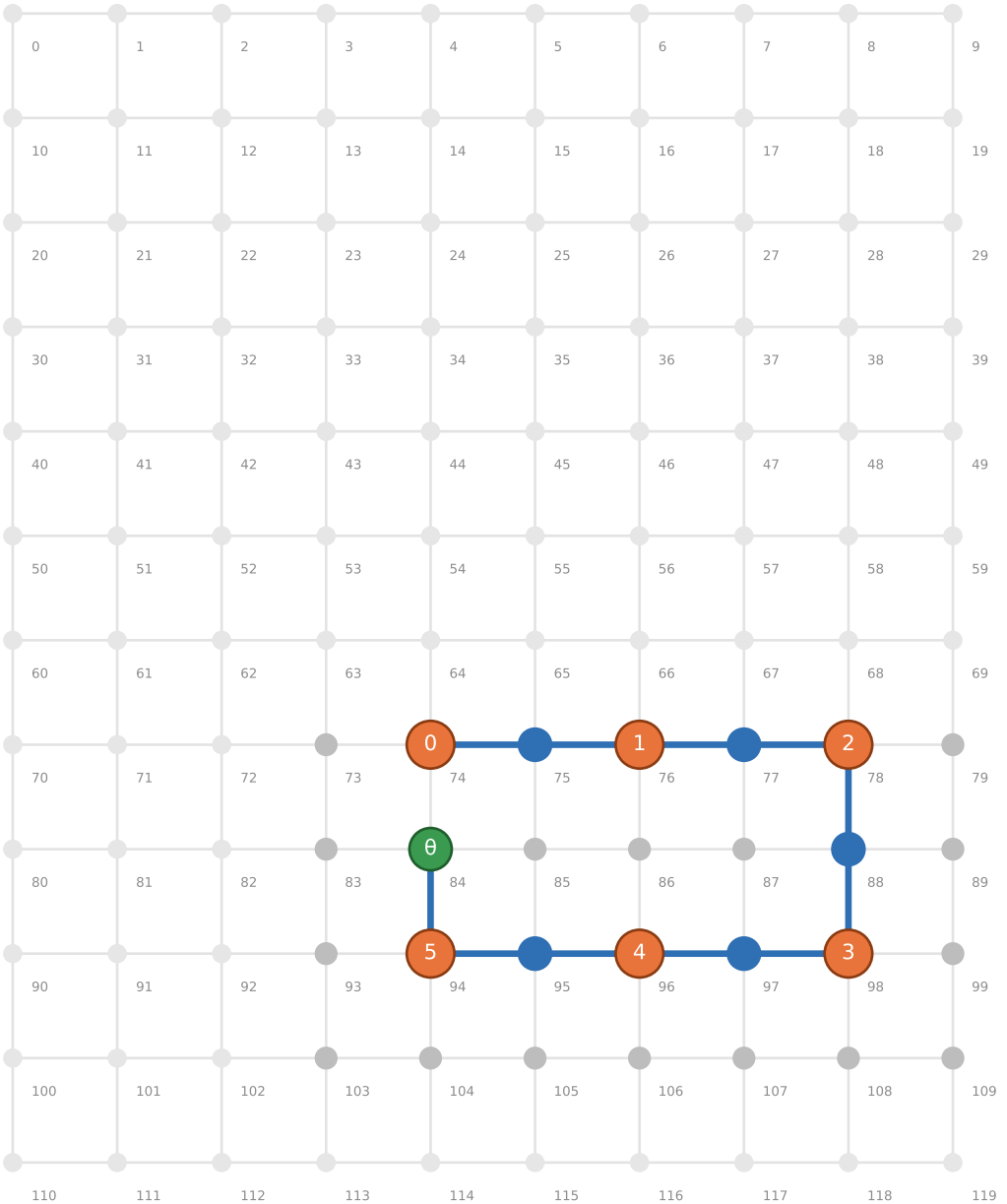


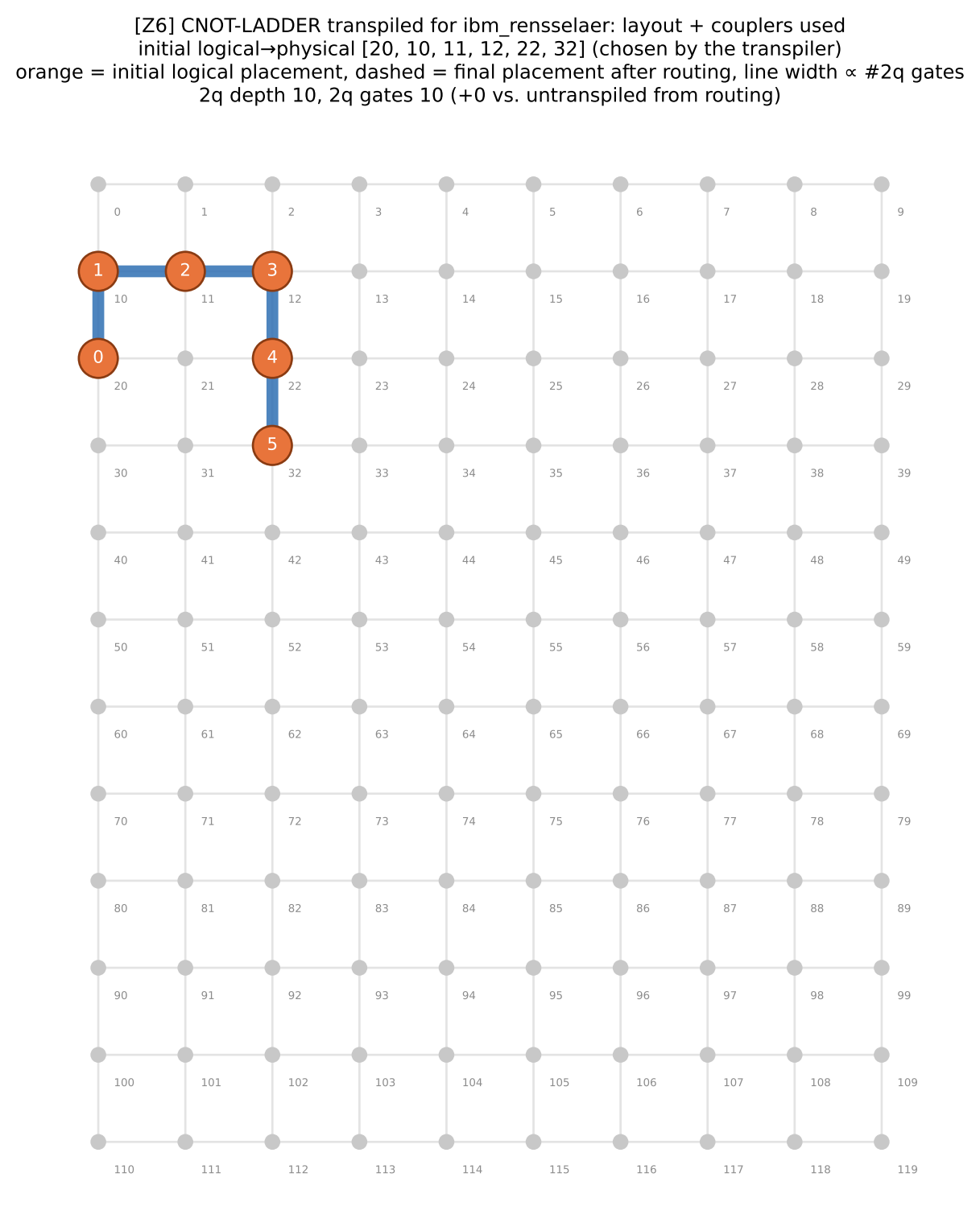
[Z6] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:25 sx:20 cz:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 10, total depth 35



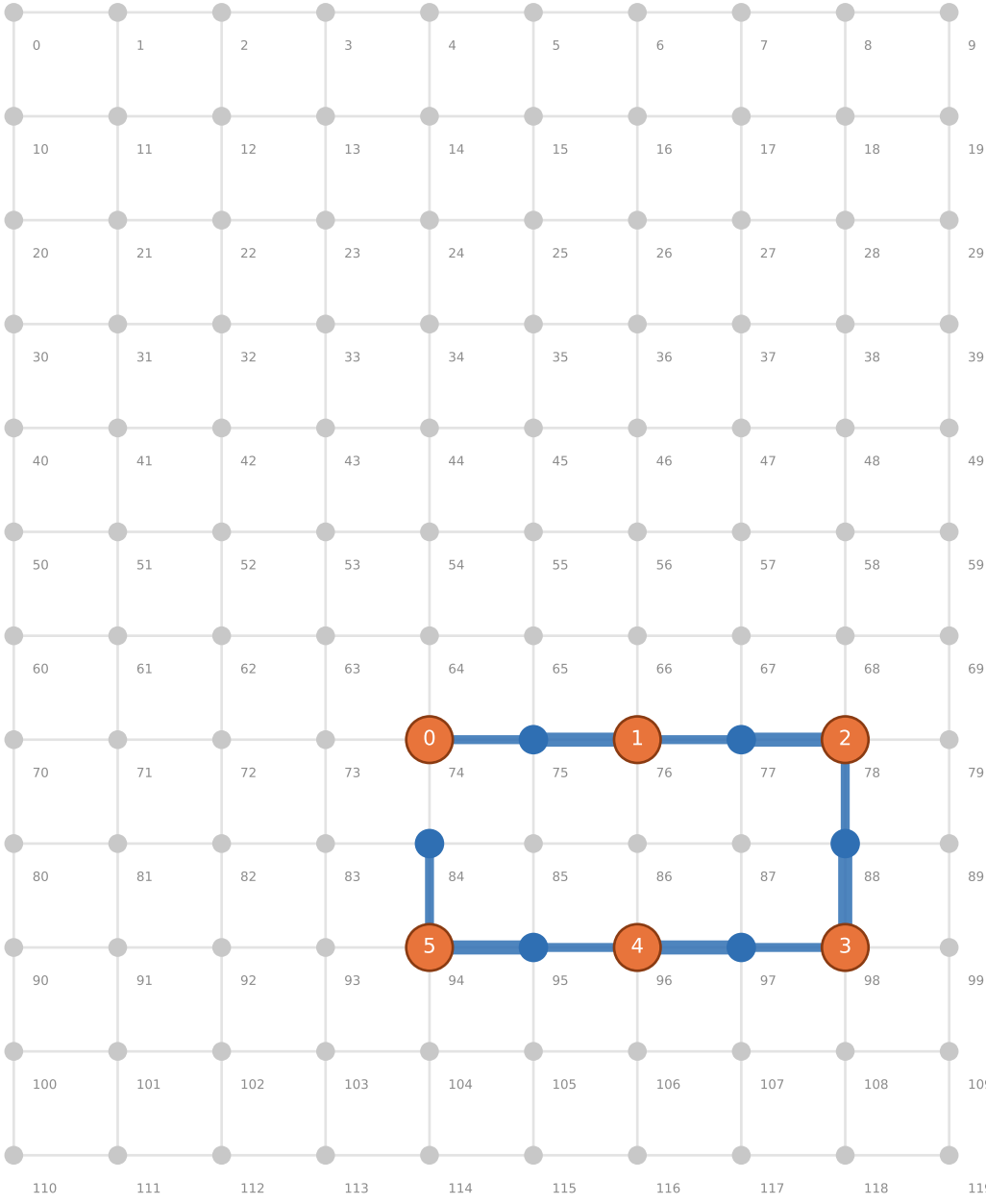
[Z6] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5
chain length 11, 5 ancillas, 0 |0> fillers

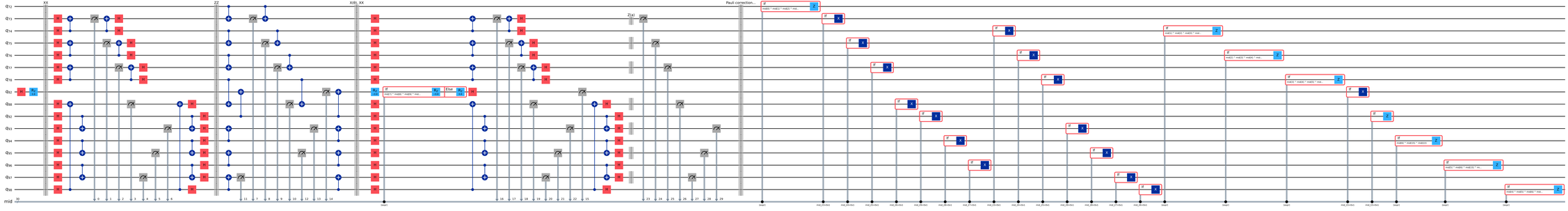



[Z6] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [20, 10, 11, 12, 22, 32] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 10, 2q gates 10 (+0 vs. untranspiled from routing)
0
1
2
3
4
5
6
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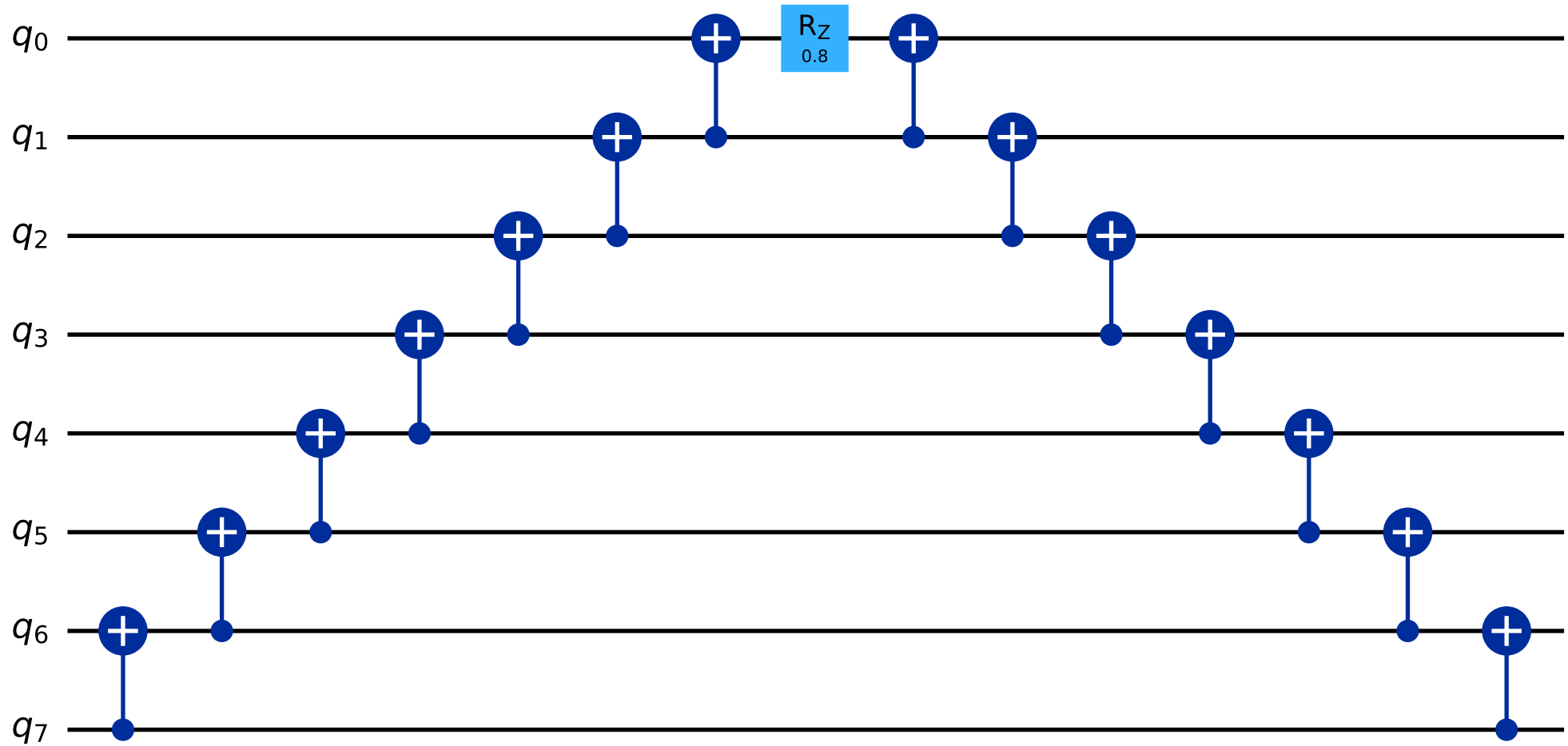
[Z6] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [74, 76, 78, 98, 96, 94]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 32



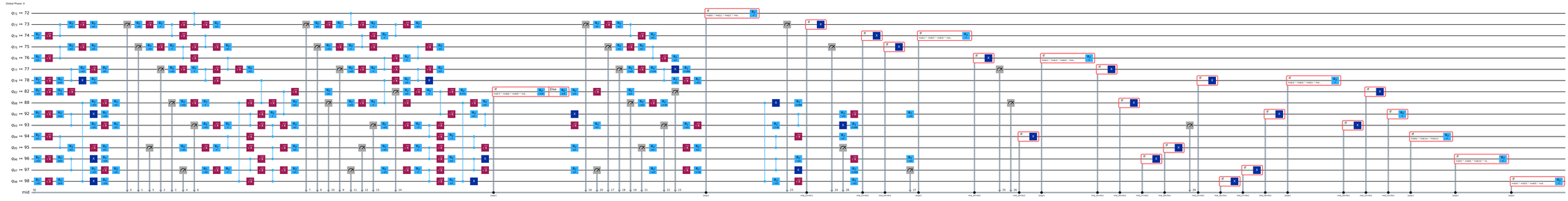
[ZB] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = Z0.Z1.Z2.Z3.Z4.Z5.Z6.Z7 · (8 logical qubits)
2q depth 6, 30 mid-circuit measurements, logical–physical [72, 74, 76, 78, 98, 96, 94, 92]



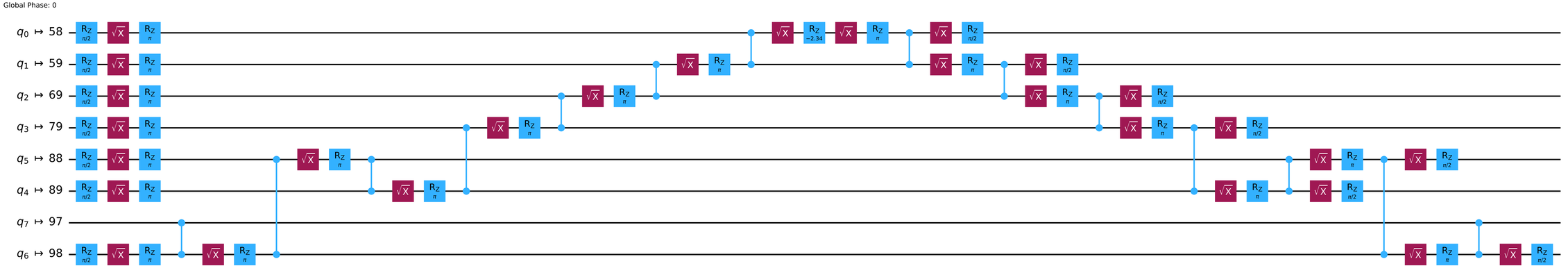
[Z8] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14



[ZB] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:117 sx:90 cz:44 measure:30 if_else:24 x:12)
expl i 9 ft, 8=0.4: P >= 25 21 22 23 24 25 26 27 (8 logical qubits)
2q depth 6, total depth 42

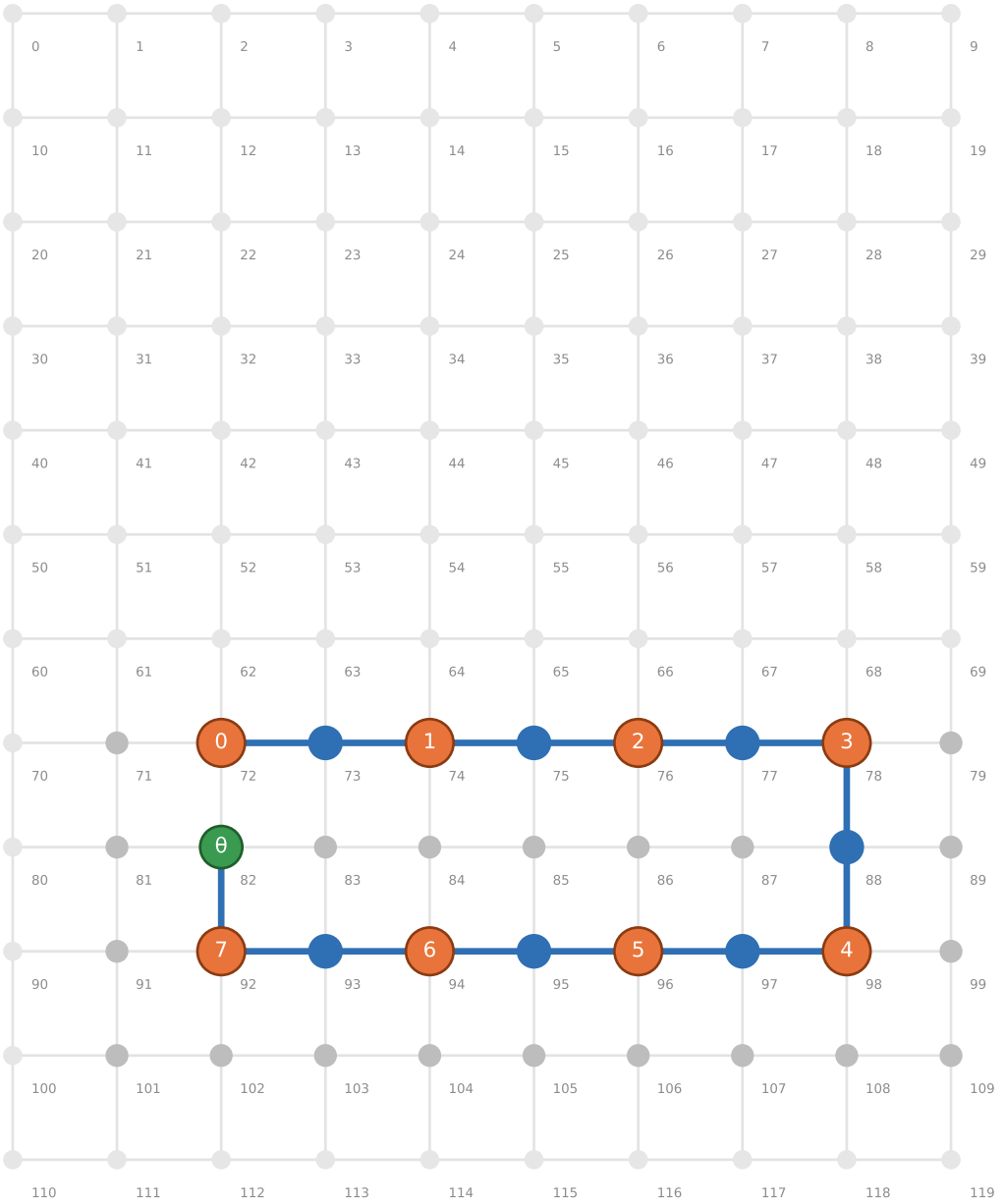


[Z8] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:35 sx:28 cz:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14, total depth 47

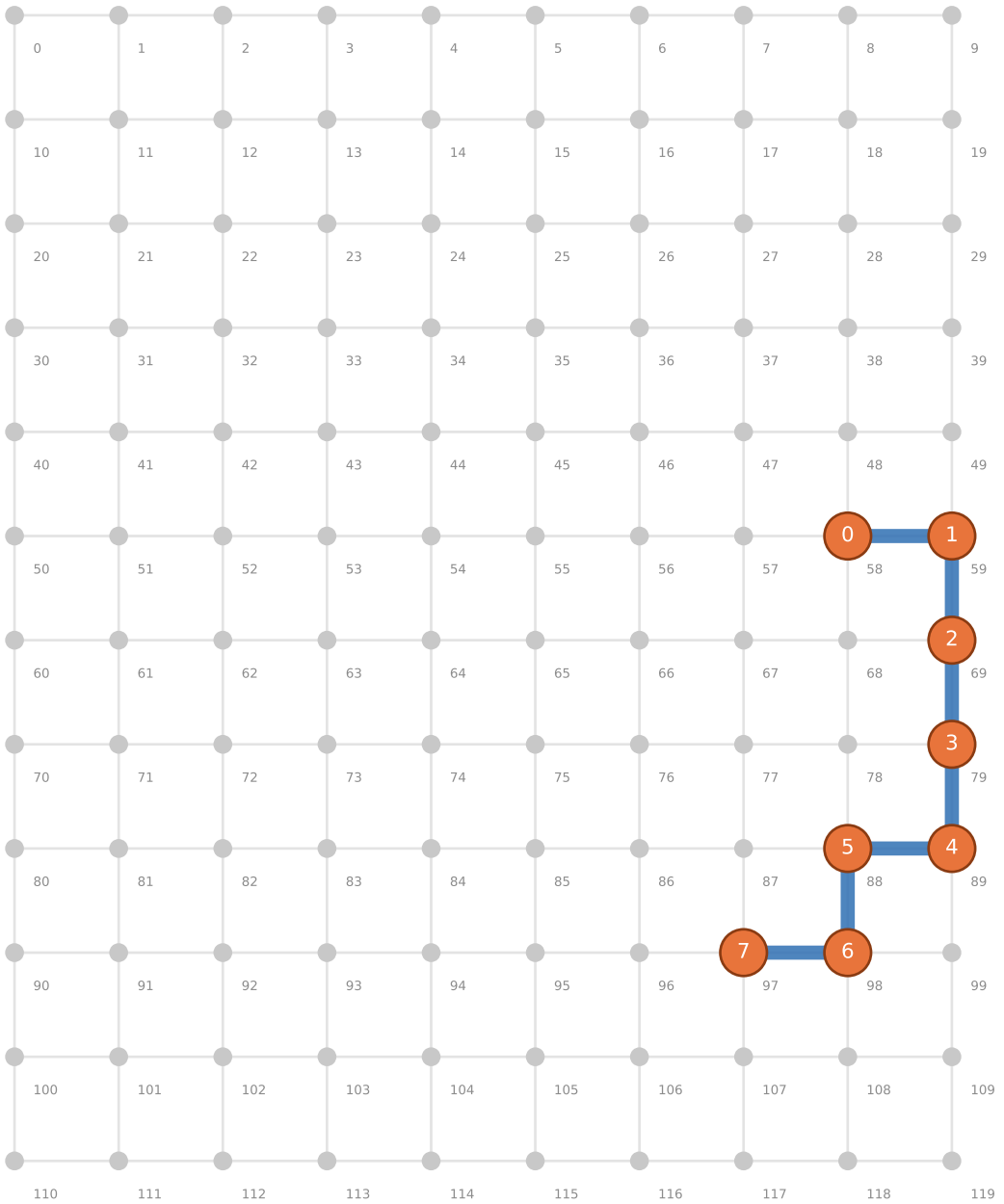


[Z8] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

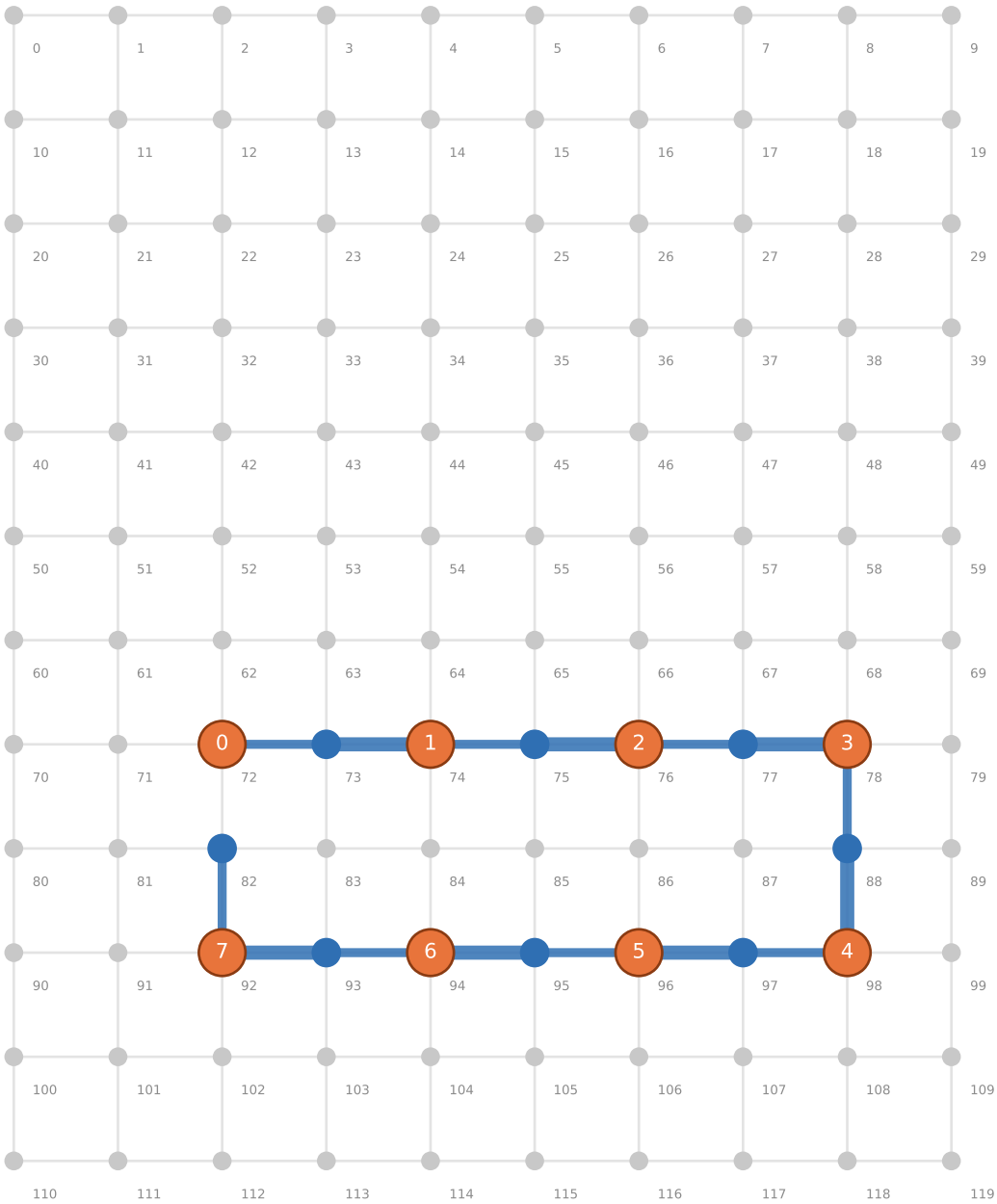
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7
chain length 15, 7 ancillas, 0 |0> fillers



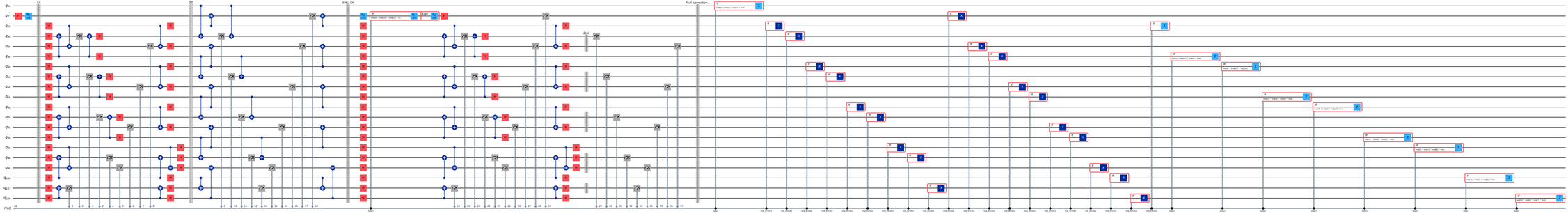
[Z8] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [58, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



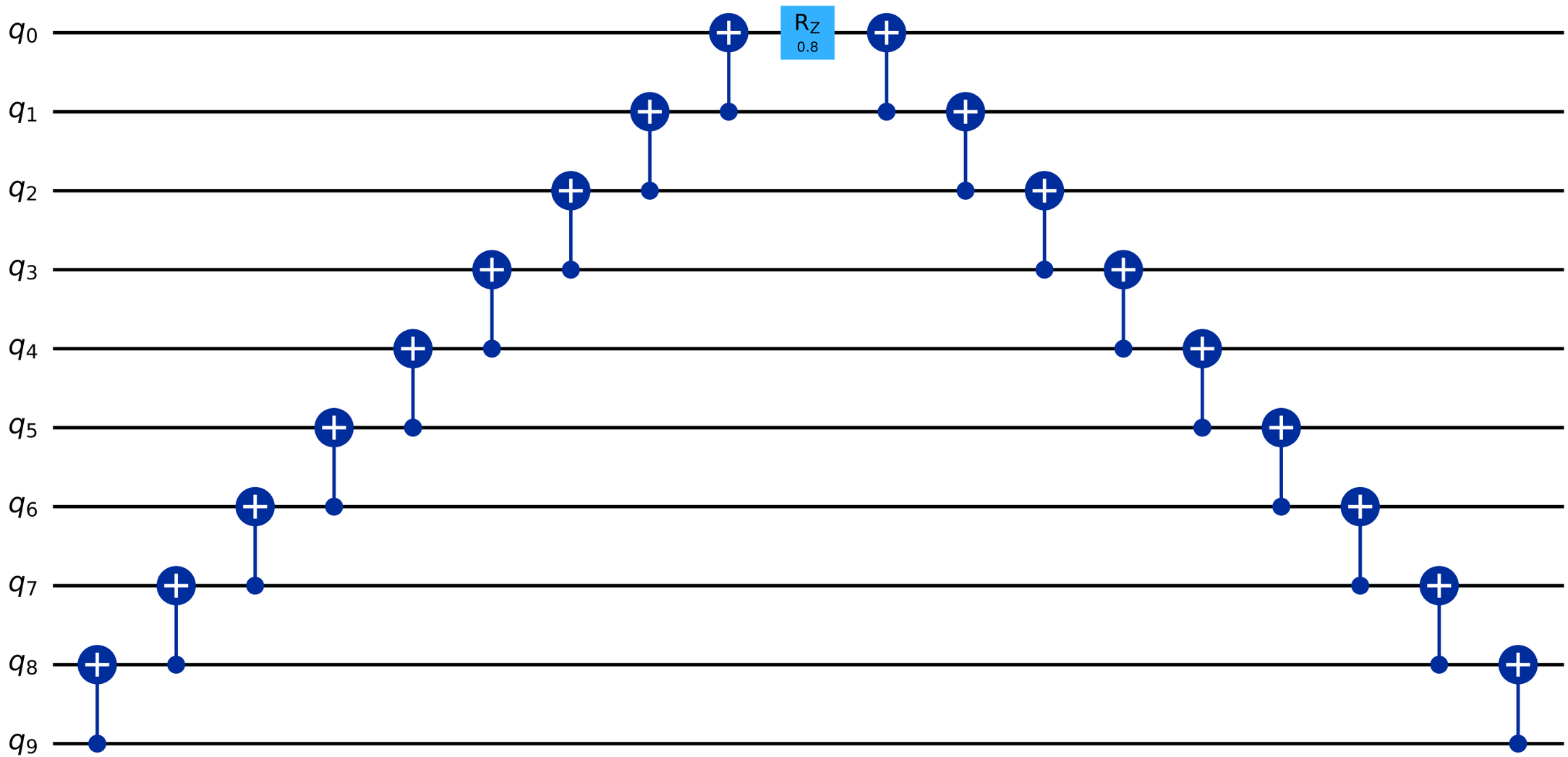
[Z8] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 44



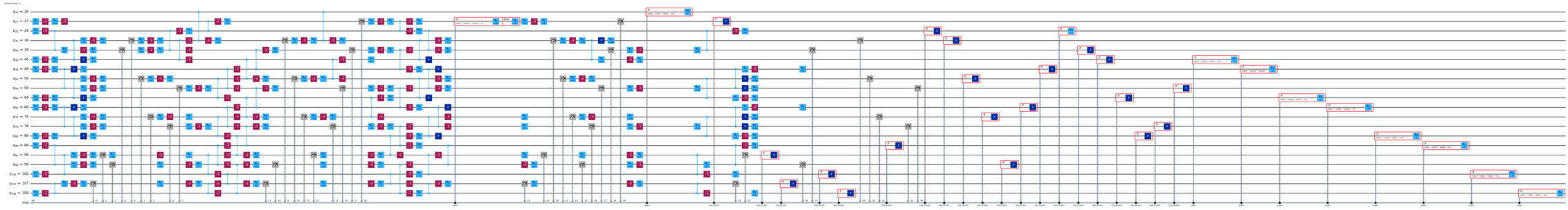
[Z19] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
expr=θ·Pi, θ=0.4 · P = 20 21 22 23 24 25 26 27 28 29 (10 logical qubits)
2q depth 6, 38 mid-circuit measurements, logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]



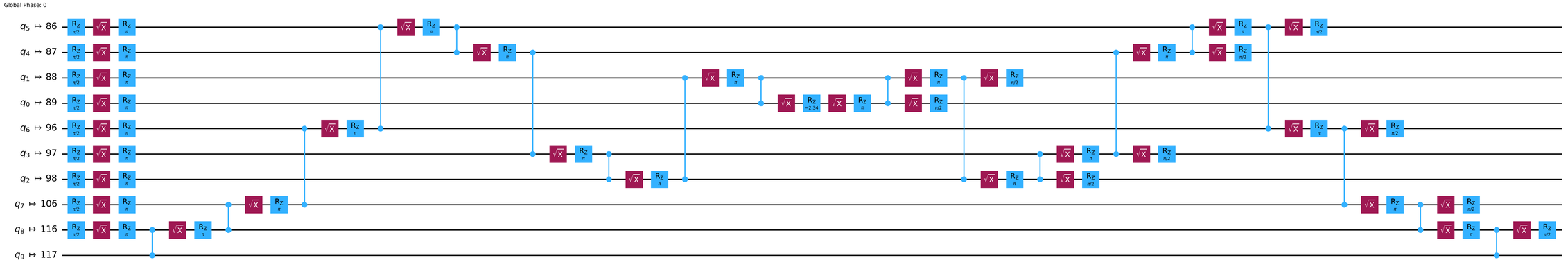
[Z10] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18



[210] DYNAMIC circuit TRANSPILED for ibm_nemostarter (ISA; cz:147 sx:114 cz:56 measure:38 if_else:30 x:15)
expi (-9.9), 6=0.4 · P = 21 22 23 24 25 26 27 28 29 · (10 logical qubits)
2q depth 6, total depth 44

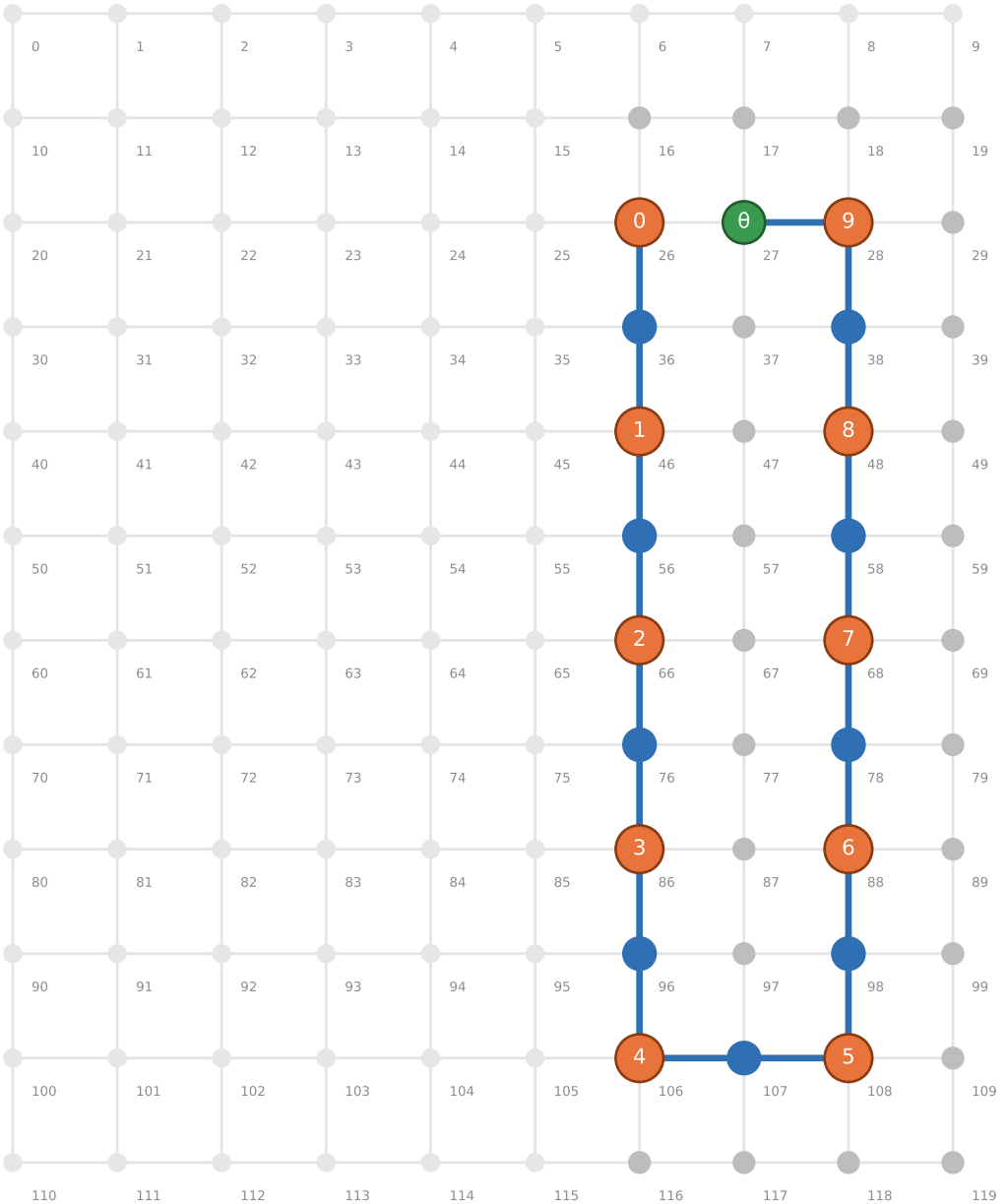


[Z10] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:45 sx:36 cz:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18, total depth 59

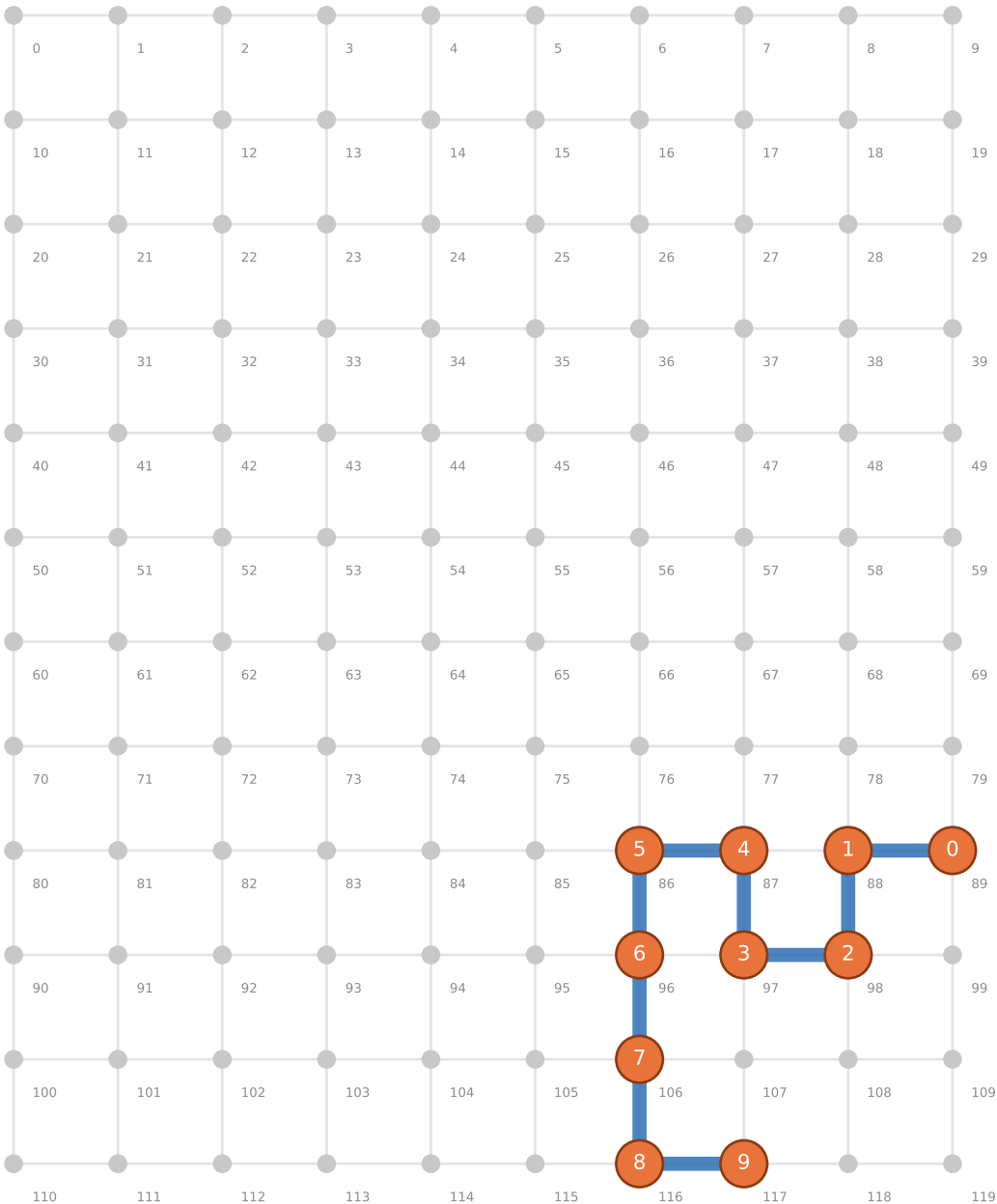


[Z10] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

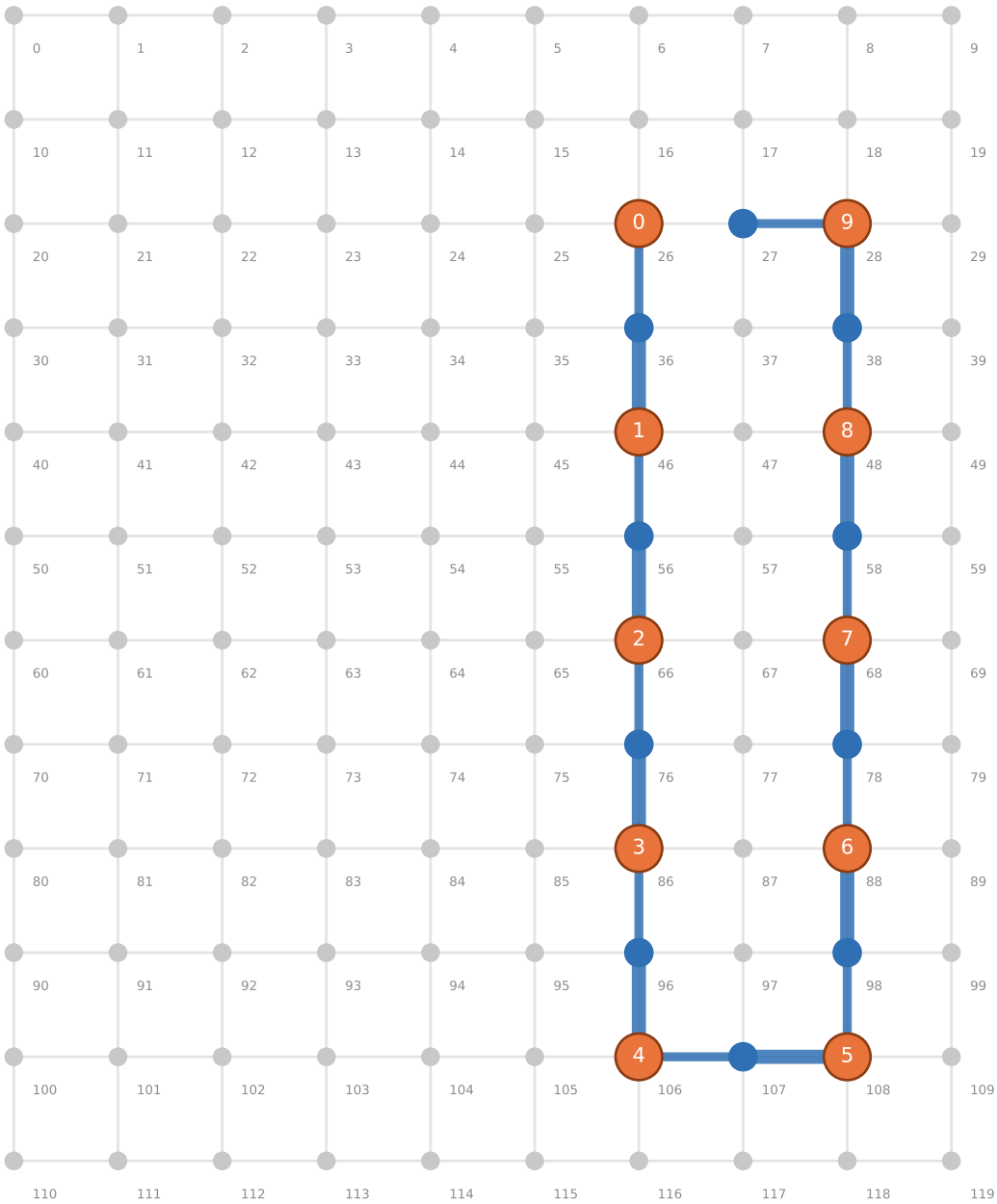
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
chain length 19, 9 ancillas, 0 |0> fillers



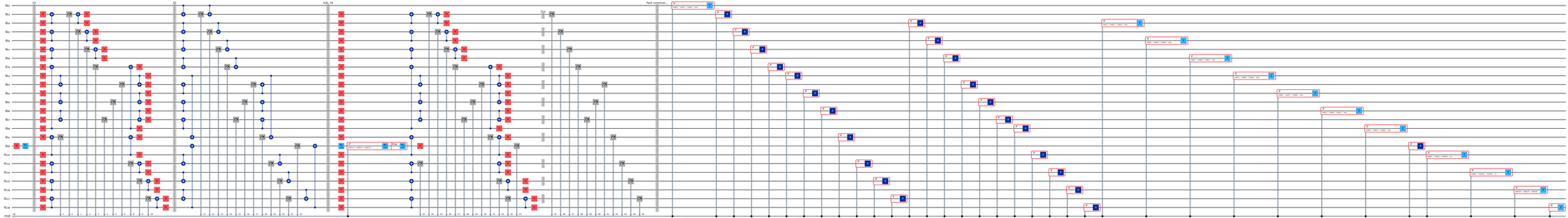
[Z10] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [89, 88, 98, 97, 87, 86, 96, 106, 116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 18, 2q gates 18 (+0 vs. untranspiled from routing)



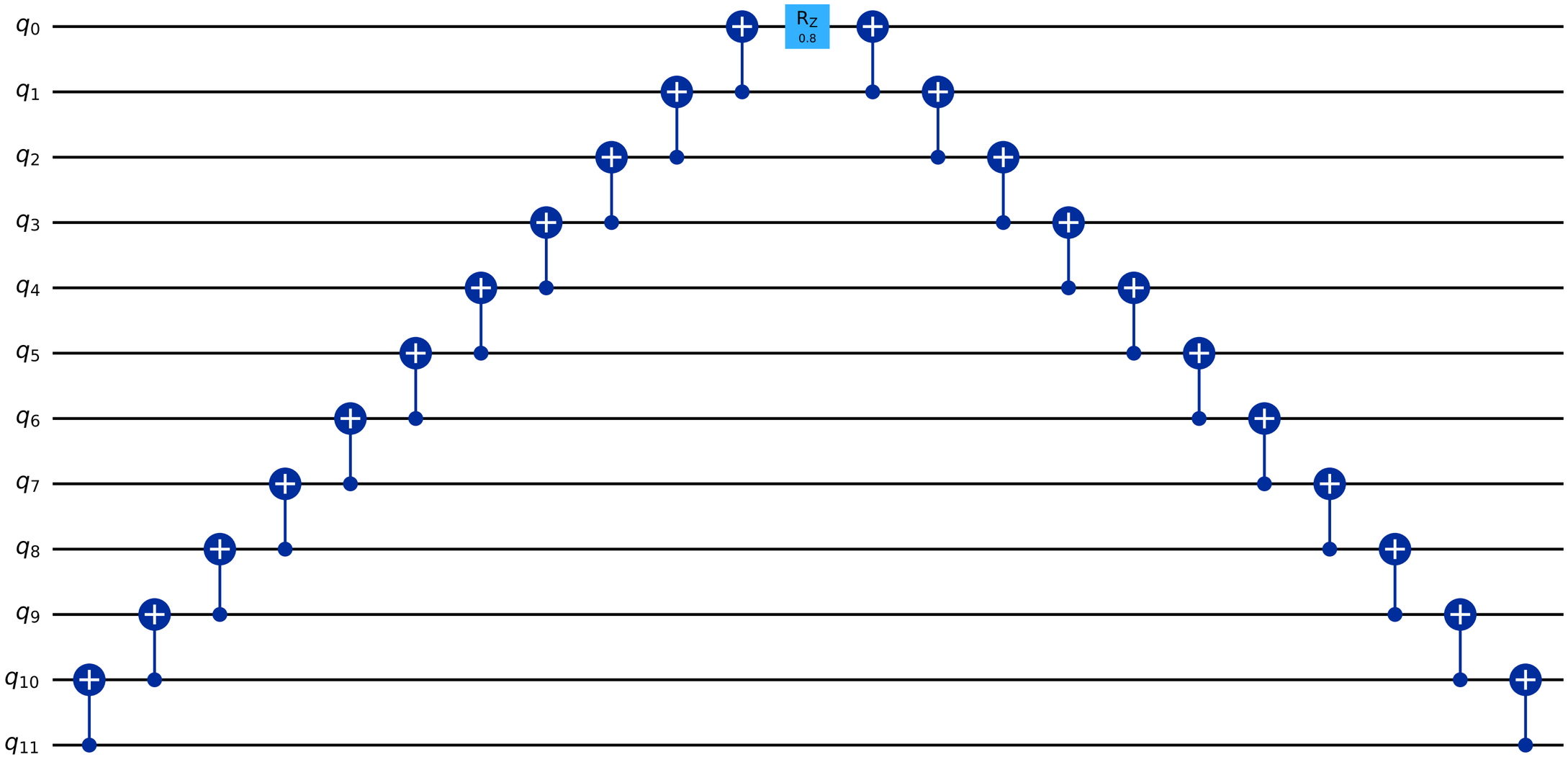
[Z10] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 56



[Z1] DYNAMIC constant-depth circuit (uncompiled, physical qubits of ibm_torocoder)
expr: k=5, N=6, R = 2^k − 2^(k−2) = 28
2q depth 6, 48 mid-circuit measurements, logical-ophysical [62, 64, 66, 68, 86, 88, 84, 82, 102, 104, 106, 108]

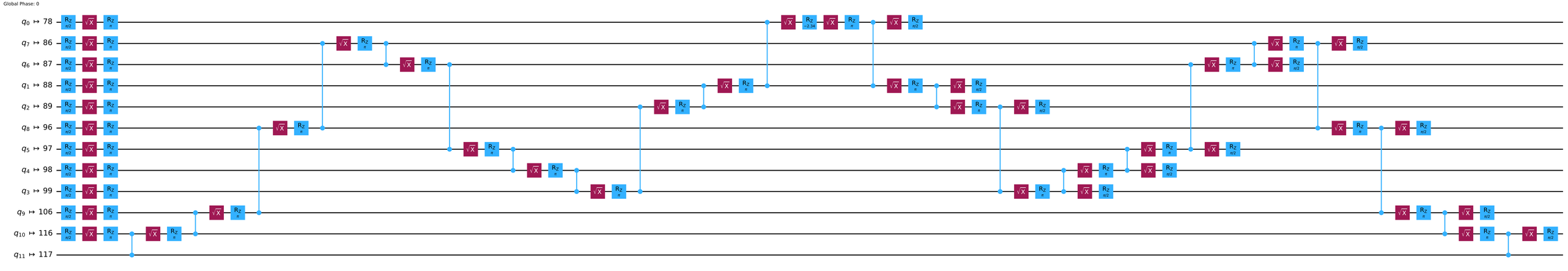


[Z12] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22



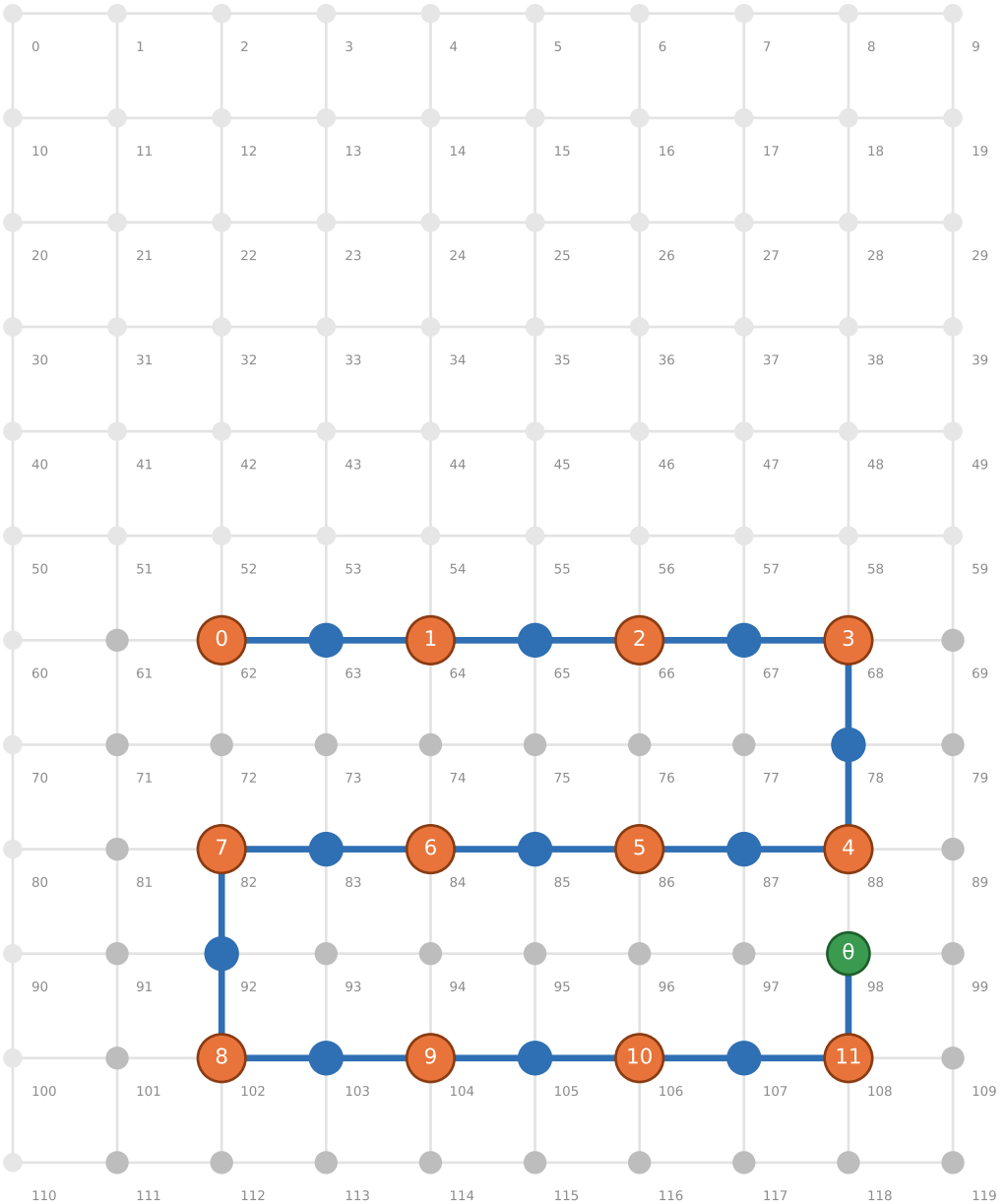
[2121] DYNAMIC circuit TRANSPILED for ibm_remoteaer (QA: n=180 sx:138 cz:68 measure:66 if_else:36 x:21)
depth: 8 % 9 ops: 8 : 8 : 25 25 21 22 23 24 25 26 27 28 29 210 211 - 112 logical qubits
26 depth 0, total depth: 67

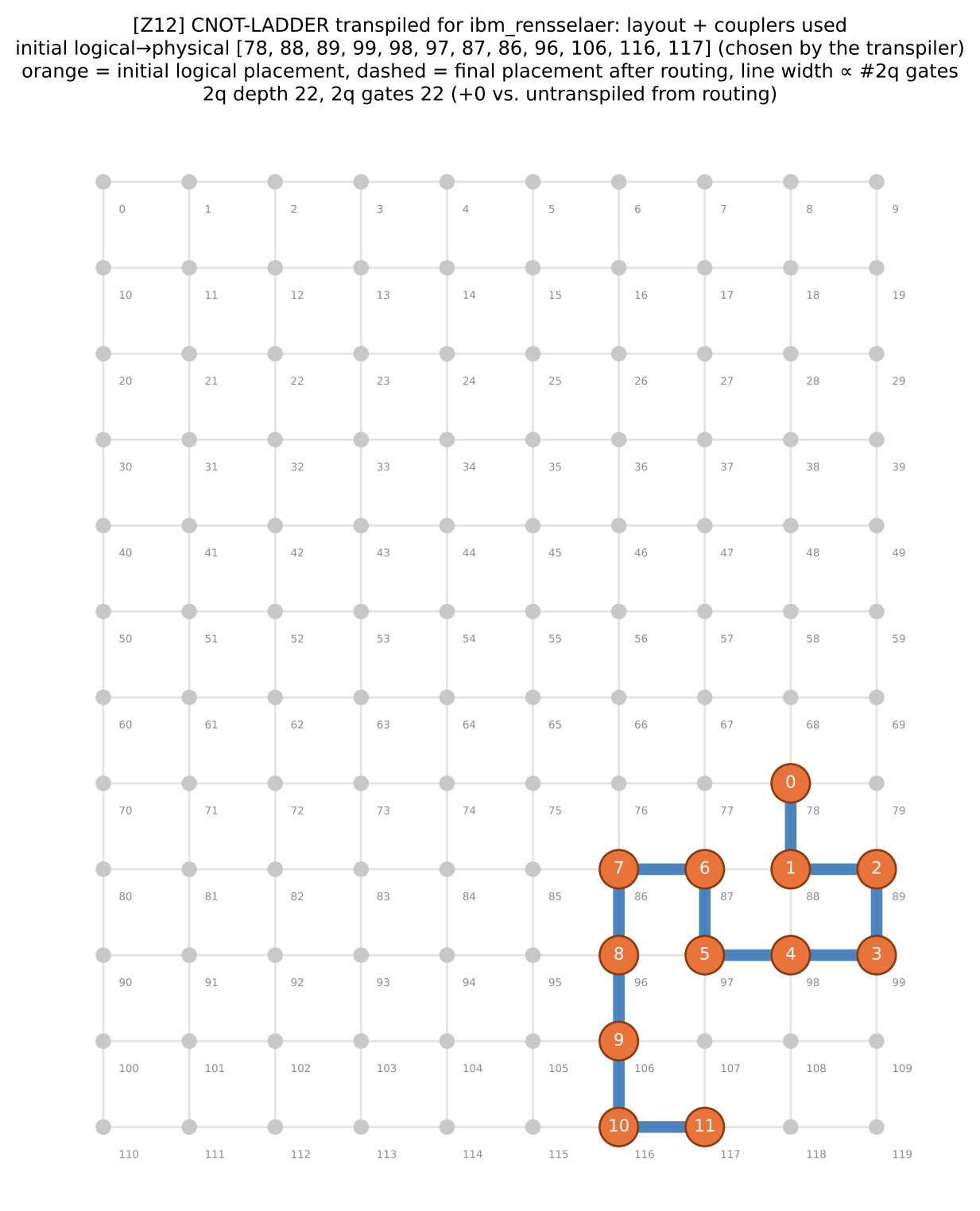
[Z12] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:55 sx:44 cz:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 (12 logical qubits)
2q depth 22, total depth 71



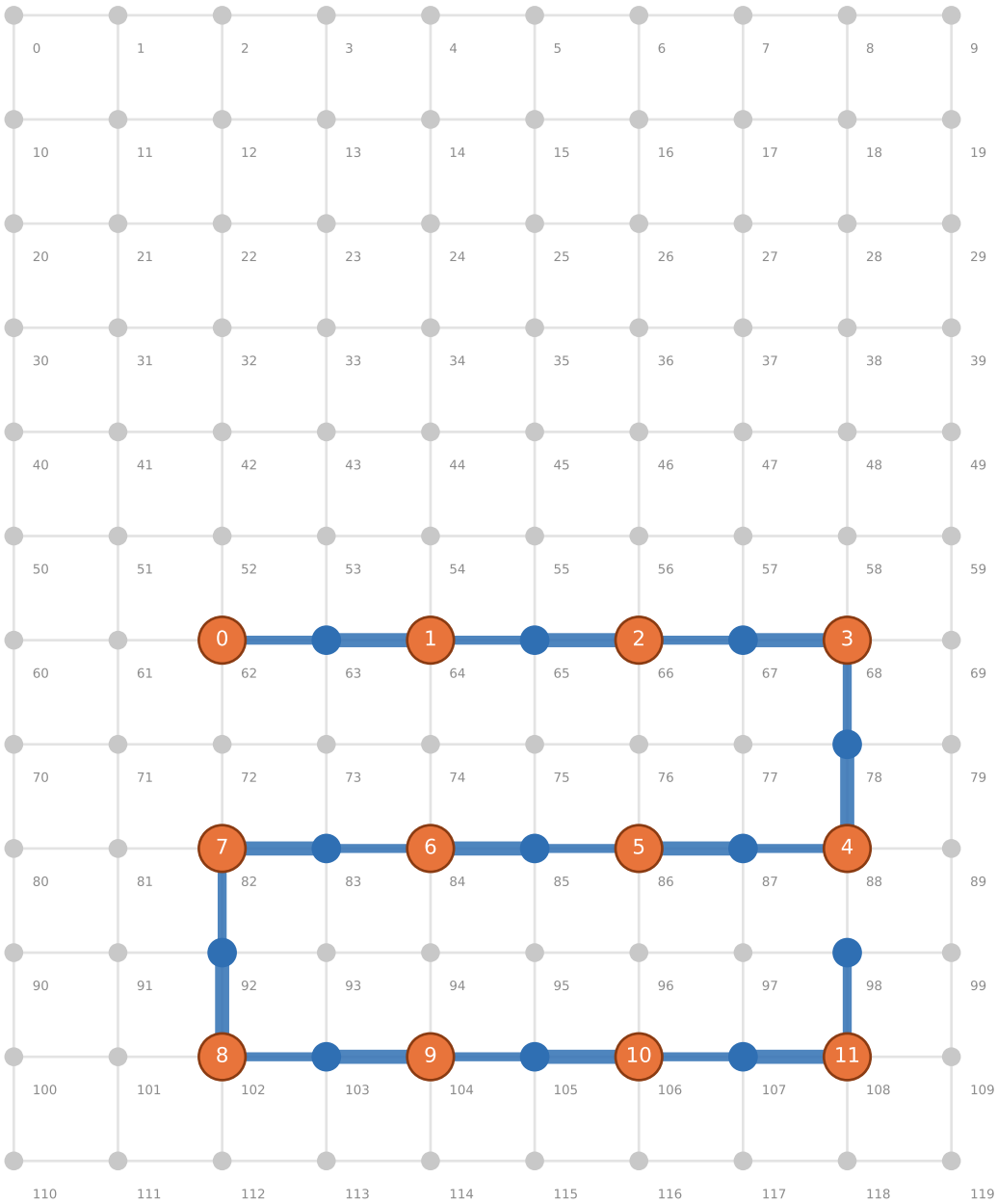
[Z12] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

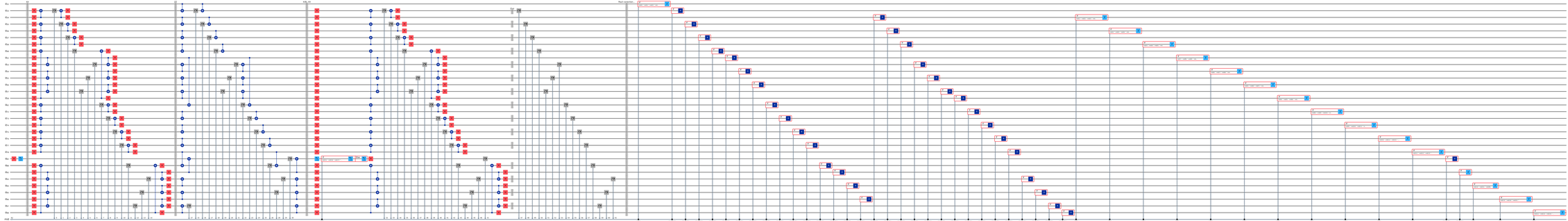
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11
chain length 23, 11 ancillas, 0 |0> fillers



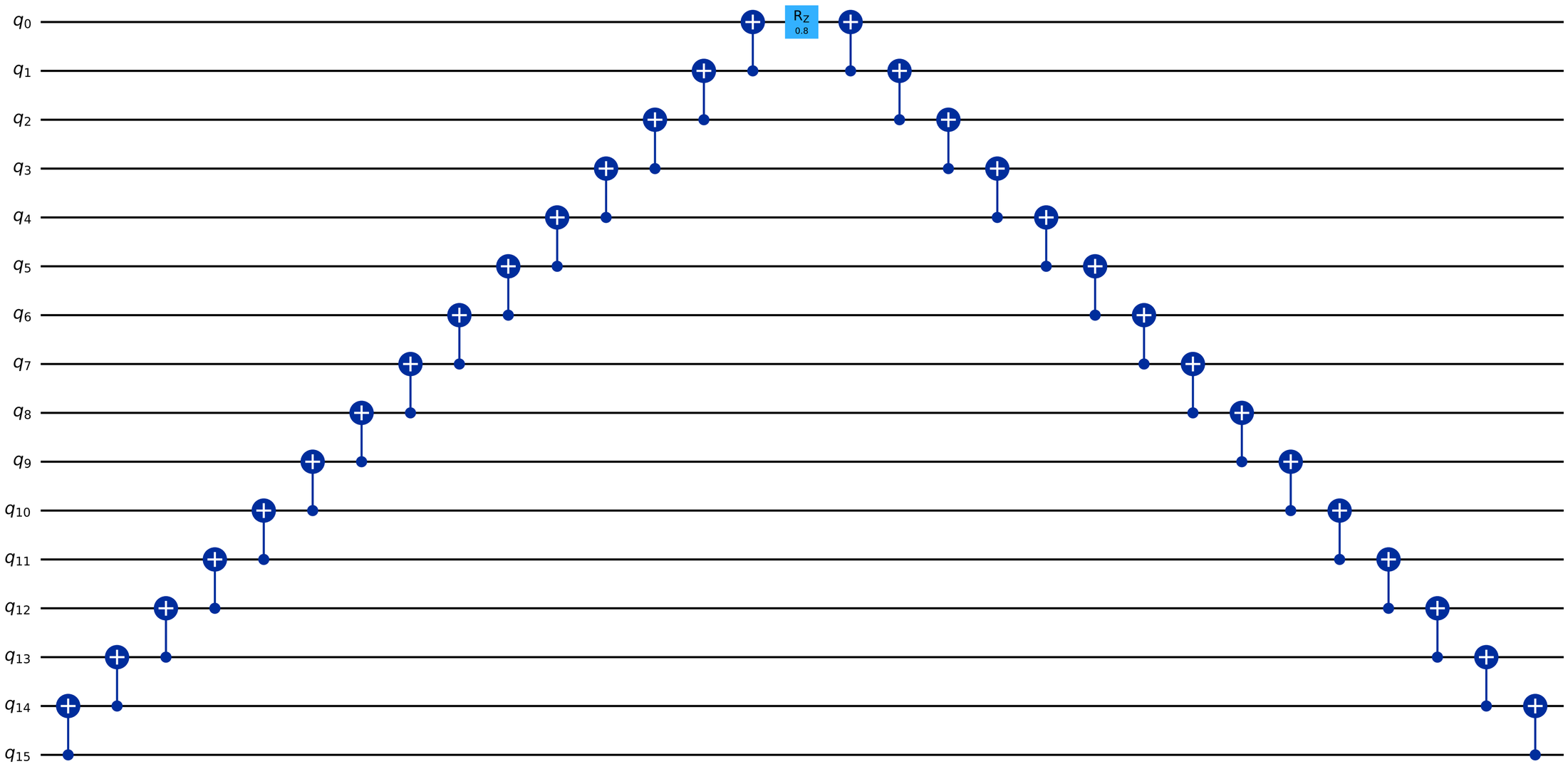

[Z12] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [78, 88, 89, 99, 98, 97, 87, 86, 96, 106, 116, 117] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 22, 2q gates 22 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
7 6 1 2
8 5 4 3
9
10 11

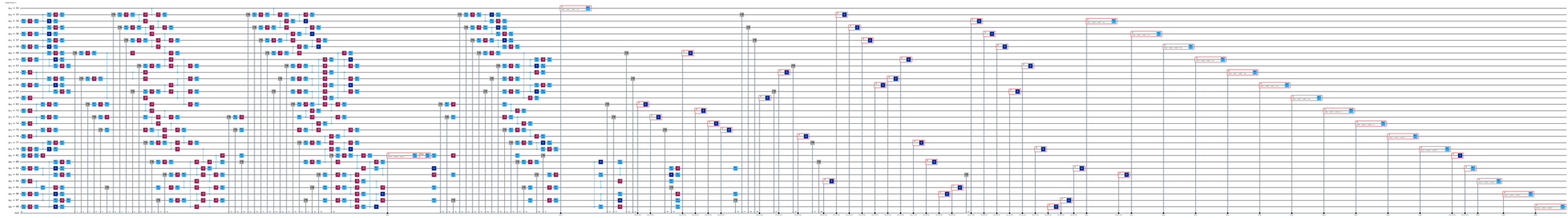
[Z12] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [62, 64, 66, 68, 88, 86, 84, 82, 102, 104, 106, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 68



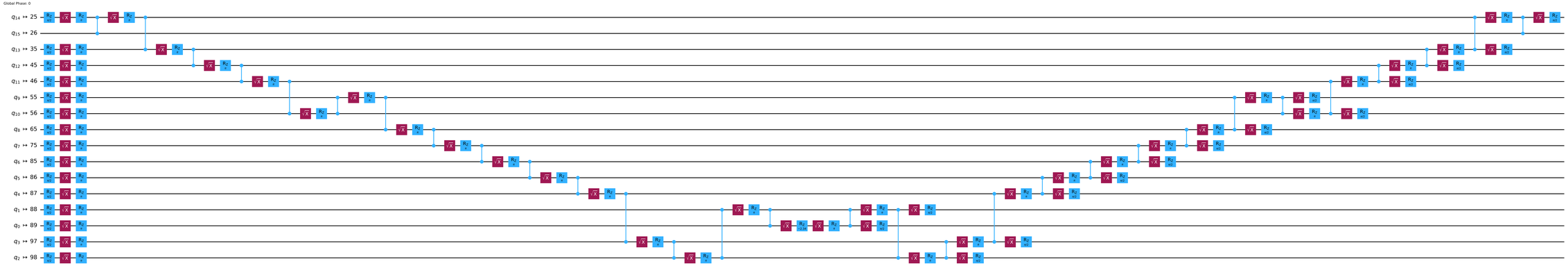

Multi-controlled gate decomposition
Pauli rotations

[Z16] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 30



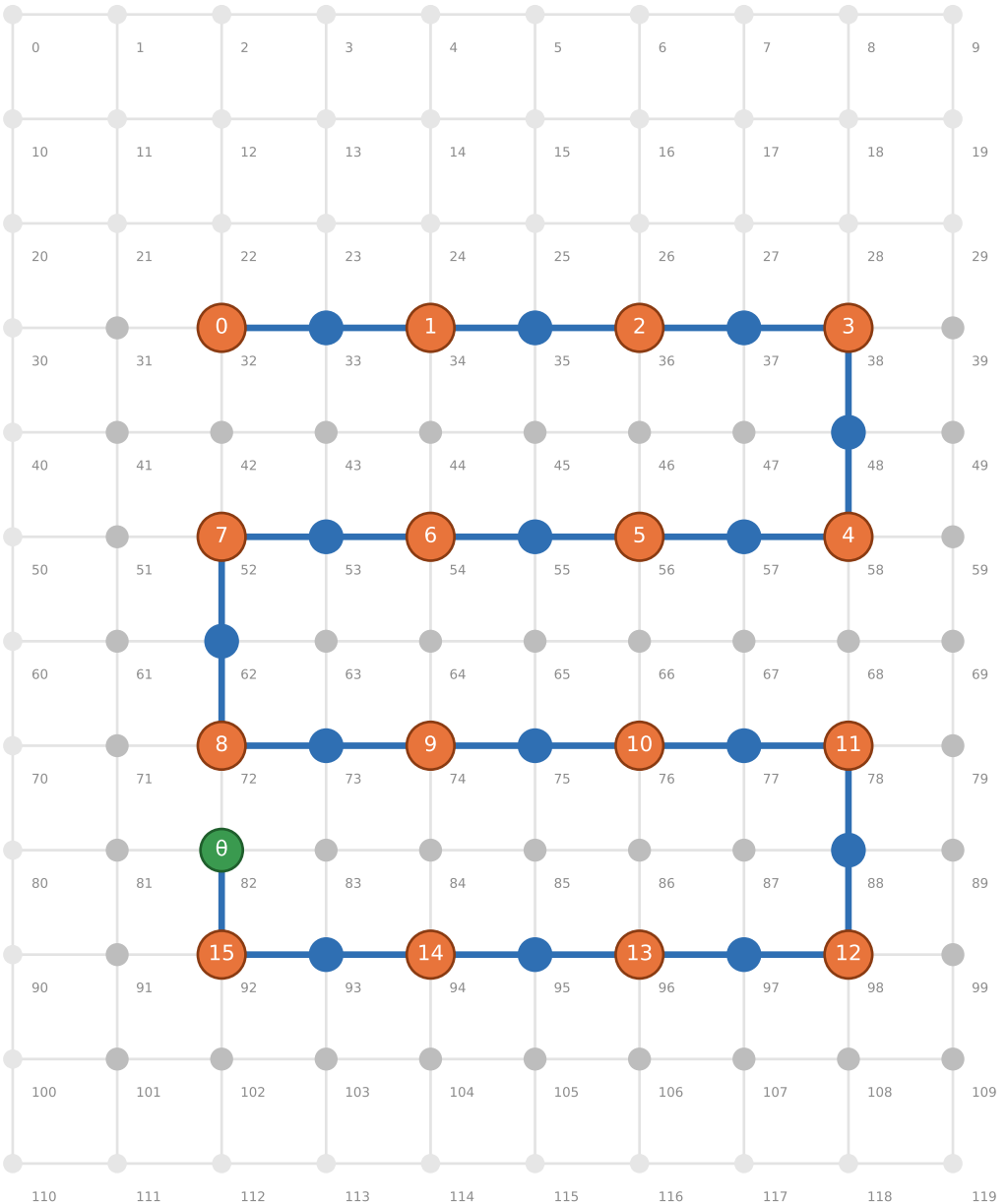

Initial Price: 32
q_0 = 32
q_1 = 34
q_2 = 35
q_3 = 36
q_4 = 37
q_5 = 38
q_6 = 48
q_7 = 52
q_8 = 53
q_9 = 54
q_10 = 55
q_11 = 56
q_12 = 57
q_13 = 58
q_14 = 72
q_15 = 73
q_16 = 74
q_17 = 75
q_18 = 76
q_19 = 77
q_20 = 78
q_21 = 79
q_22 = 80
q_23 = 81
q_24 = 82
q_25 = 83
q_26 = 84

[Z16] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:75 sx:60 cz:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30, total depth 95

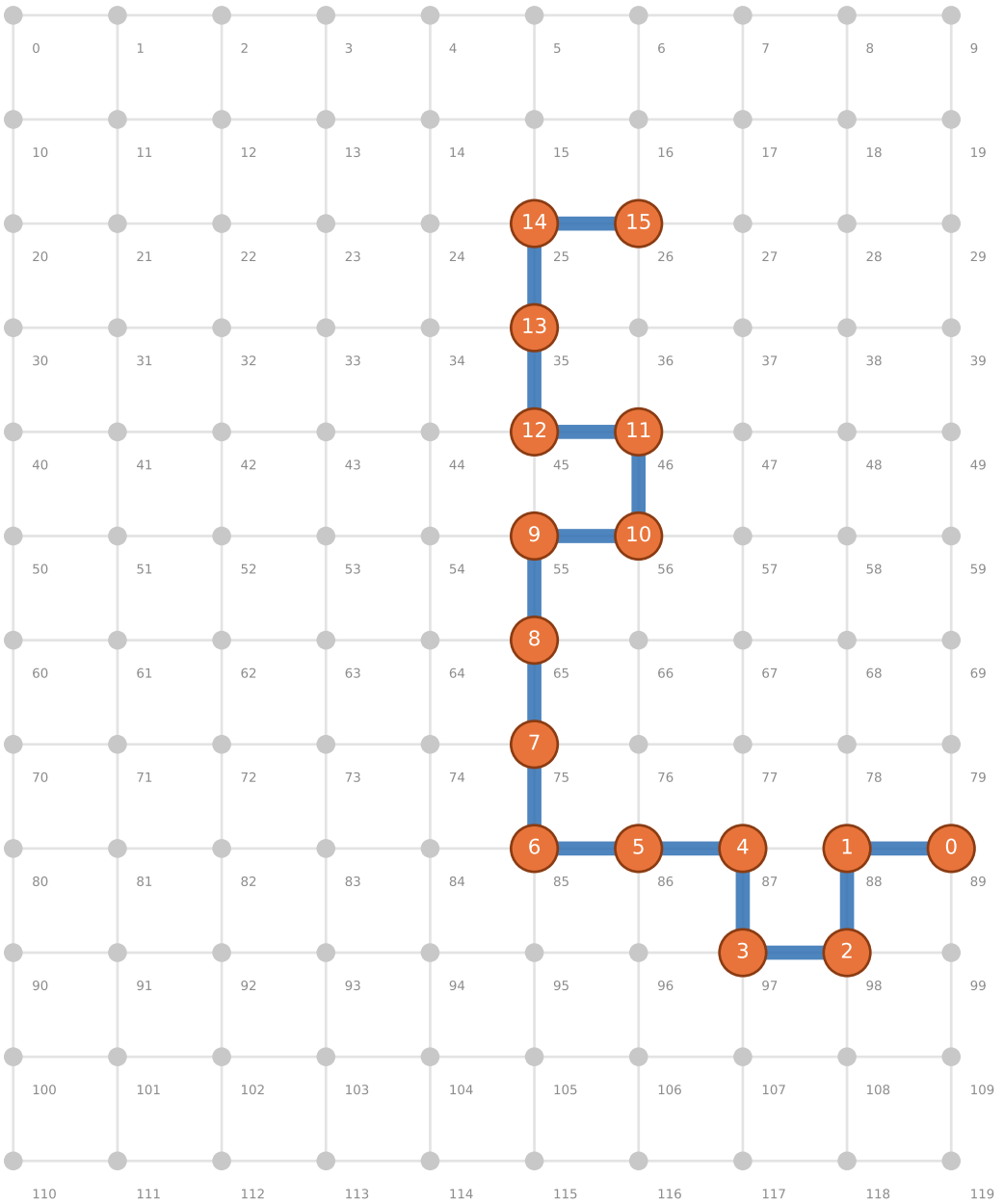


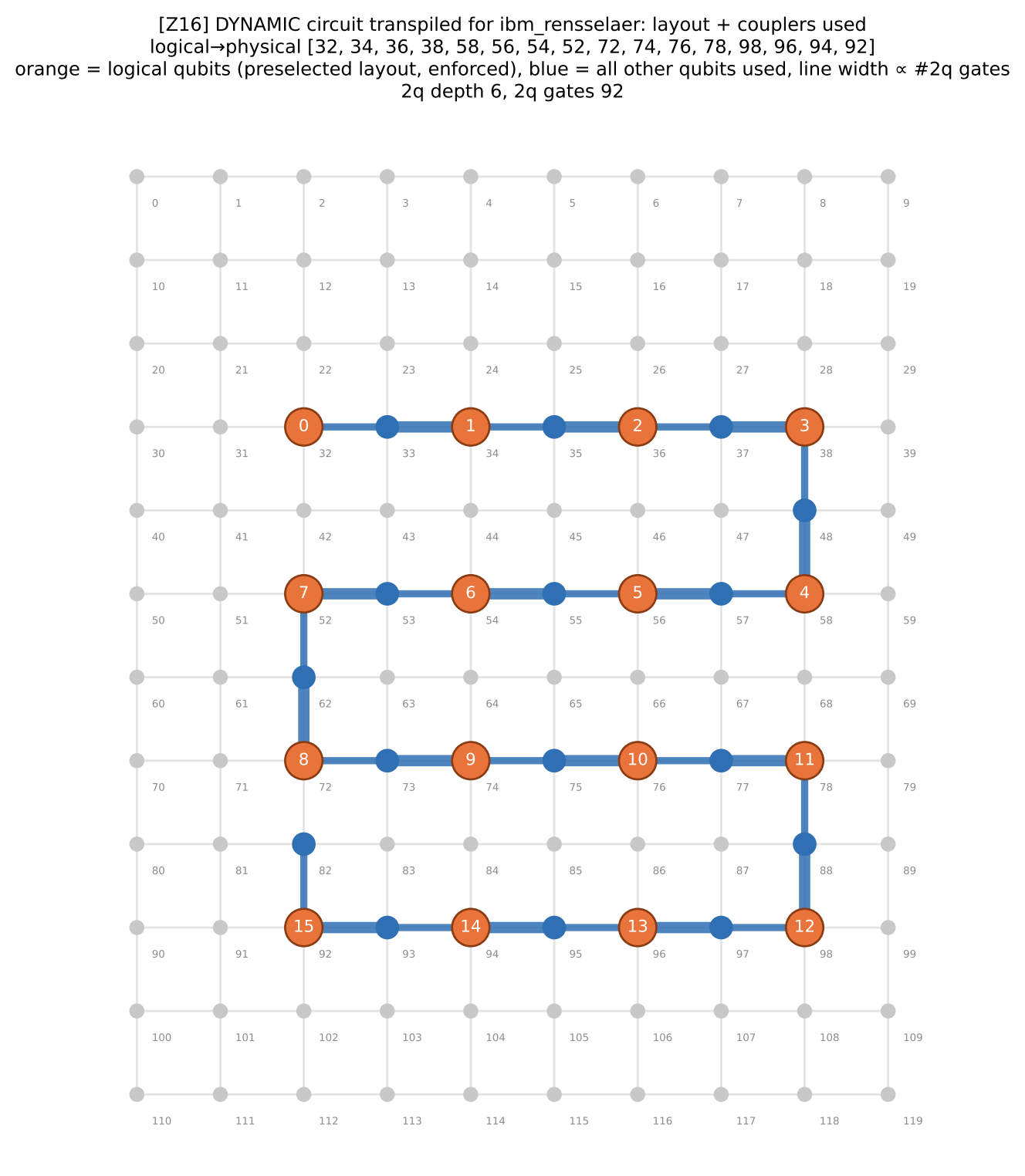
[Z16] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15
chain length 31, 15 ancillas, 0 |0> fillers

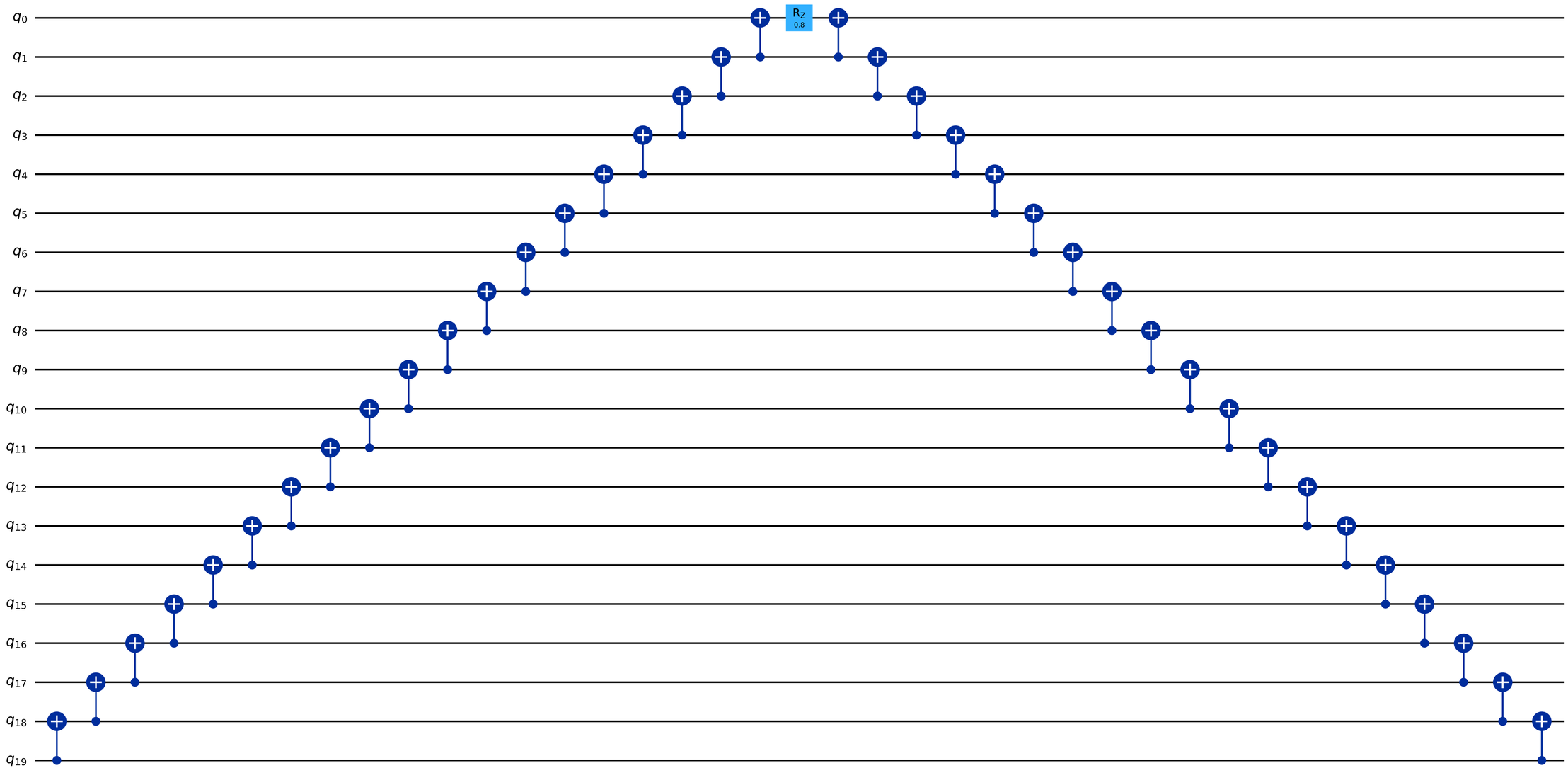


[Z16] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [89, 88, 98, 97, 87, 86, 85, 75, 65, 55, 56, 46, 45, 35, 25, 26] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 30, 2q gates 30 (+0 vs. untranspiled from routing)

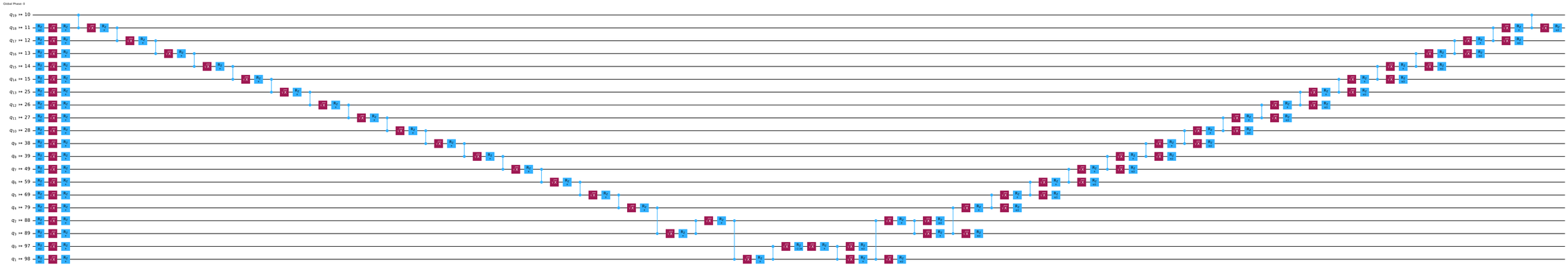



[Z16] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [32, 34, 36, 38, 58, 56, 54, 52, 72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 92
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0 1 2 3
7 6 5 4
8 9 10 11
15 14 13 12

[Z20] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 38

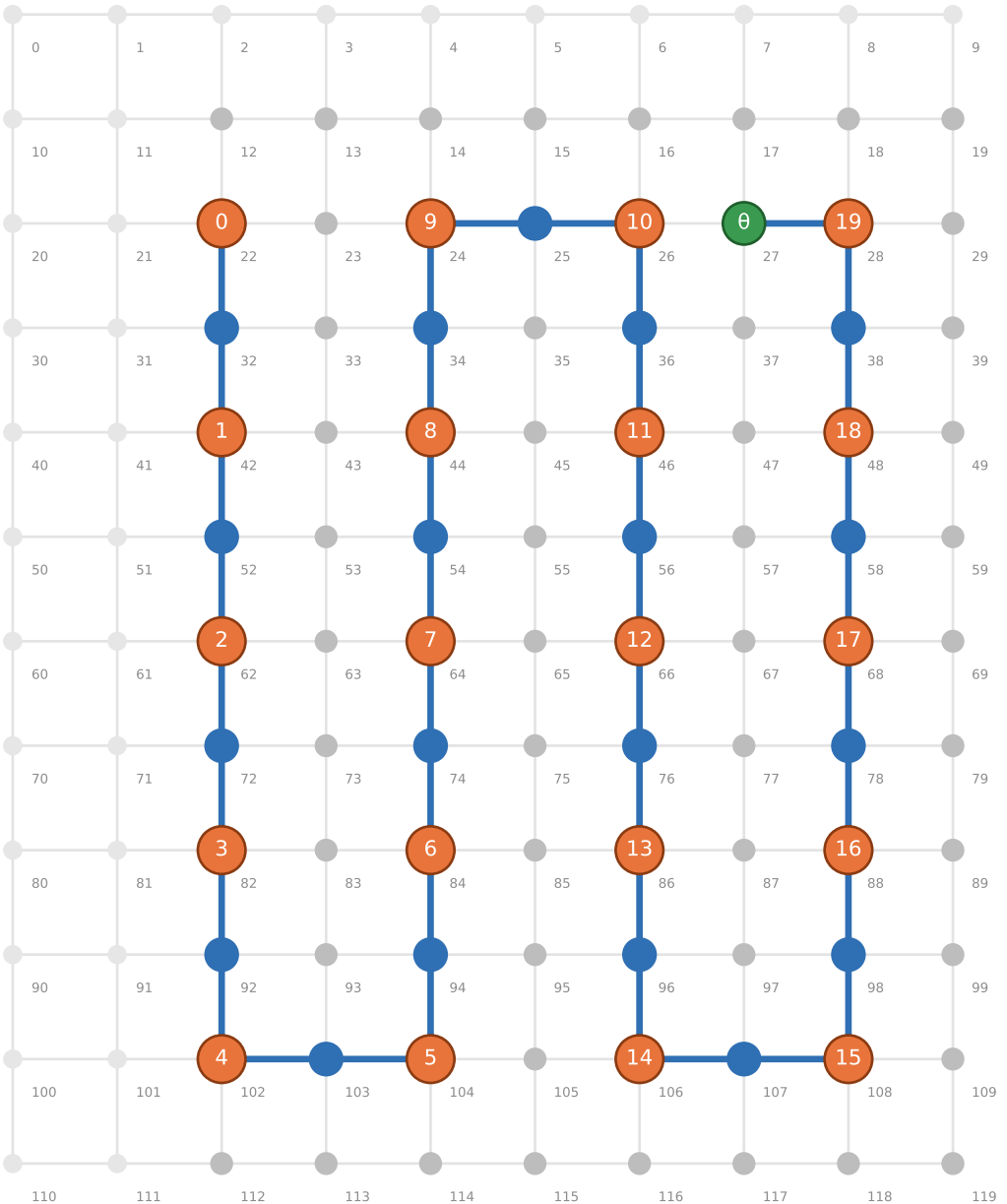


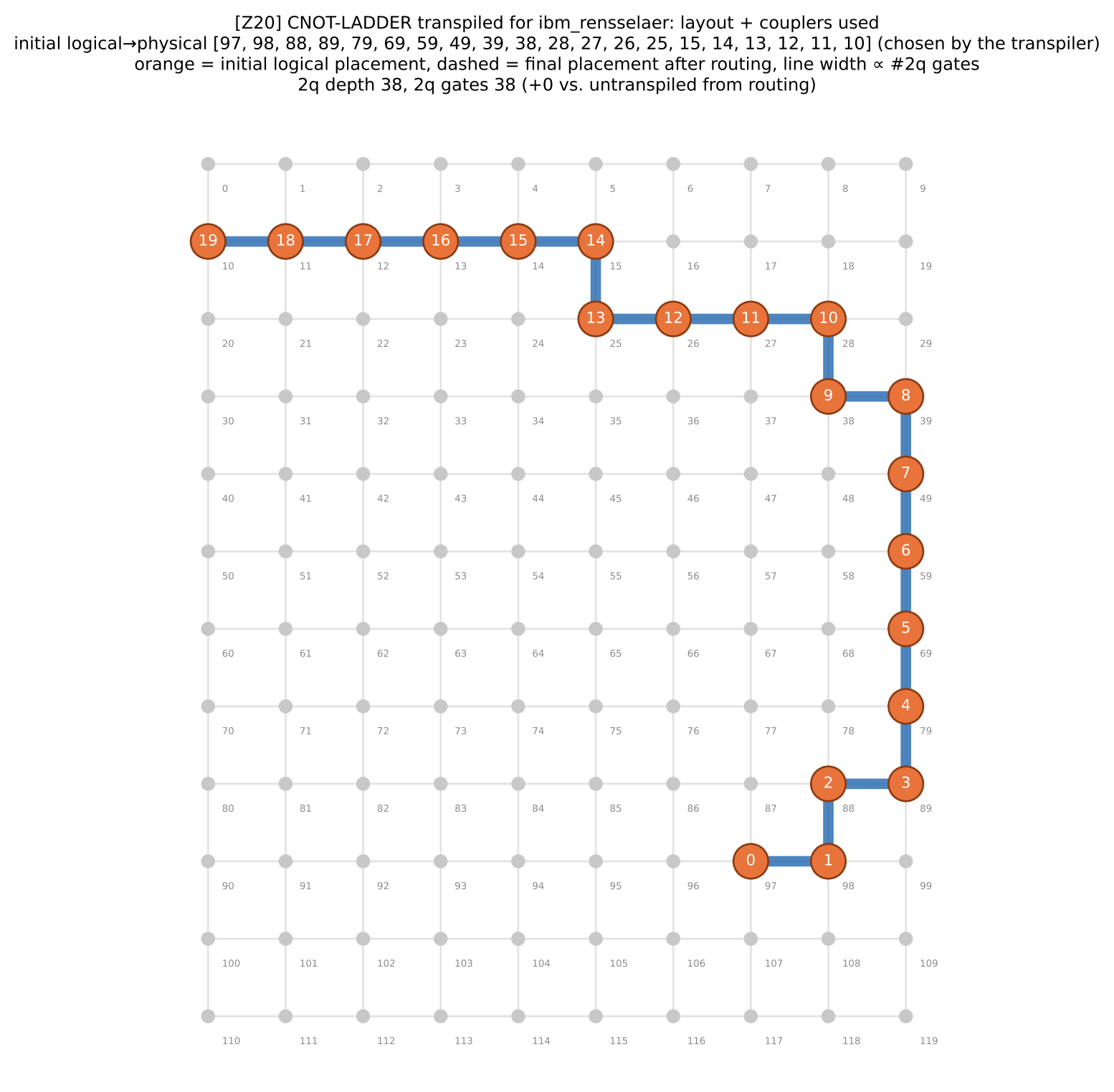
[Z20] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:95 sx:76 cz:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 (20 logical qubits)
2q depth 38, total depth 119



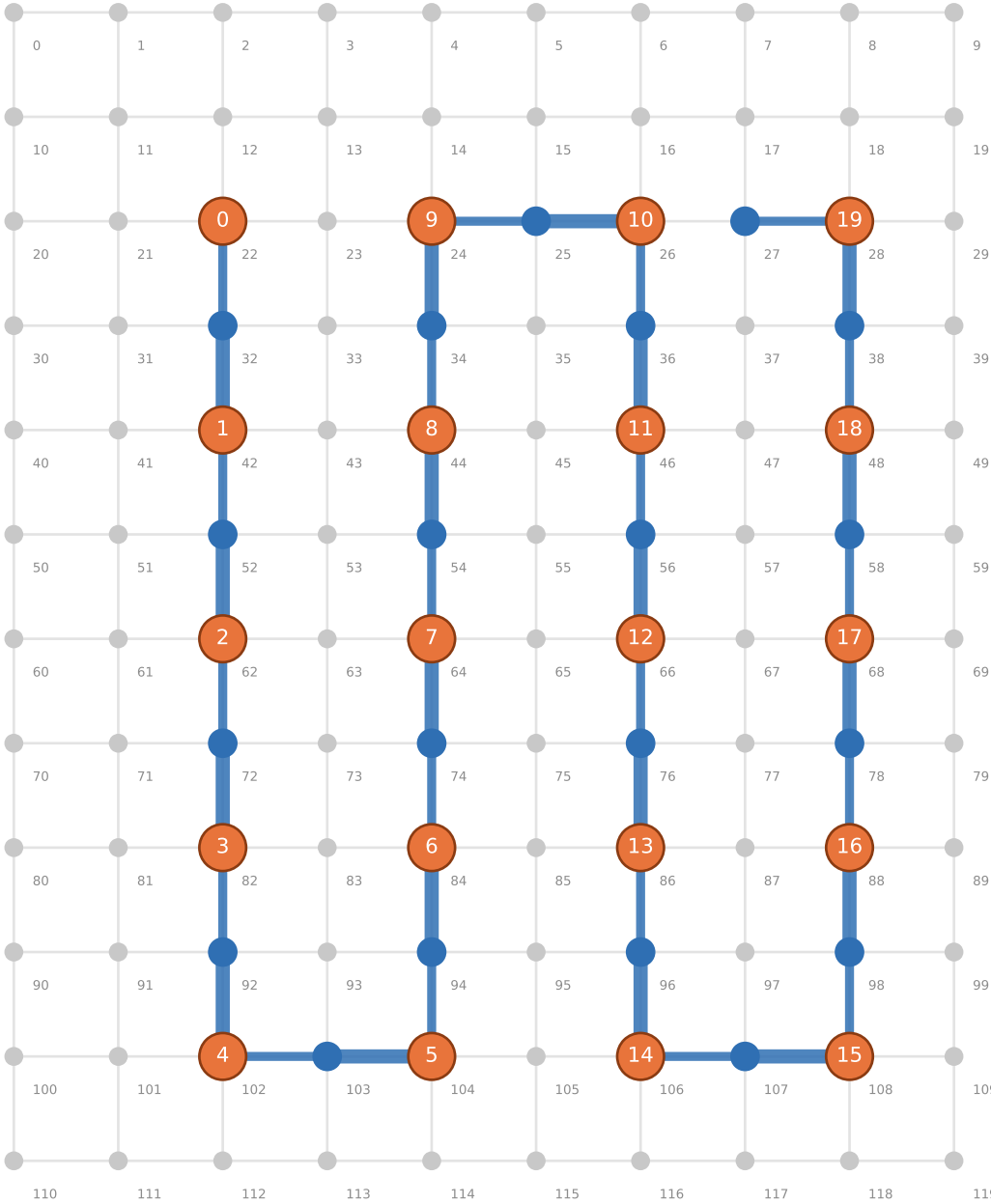
[Z20] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

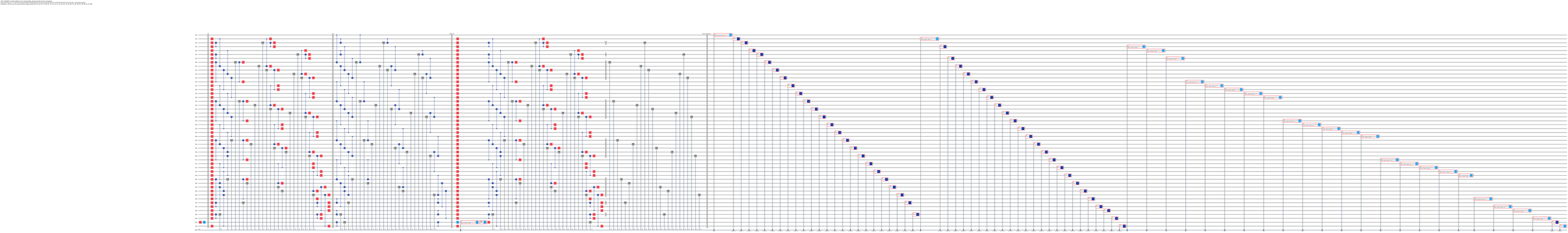
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19
chain length 39, 19 ancillas, 0 |0> fillers



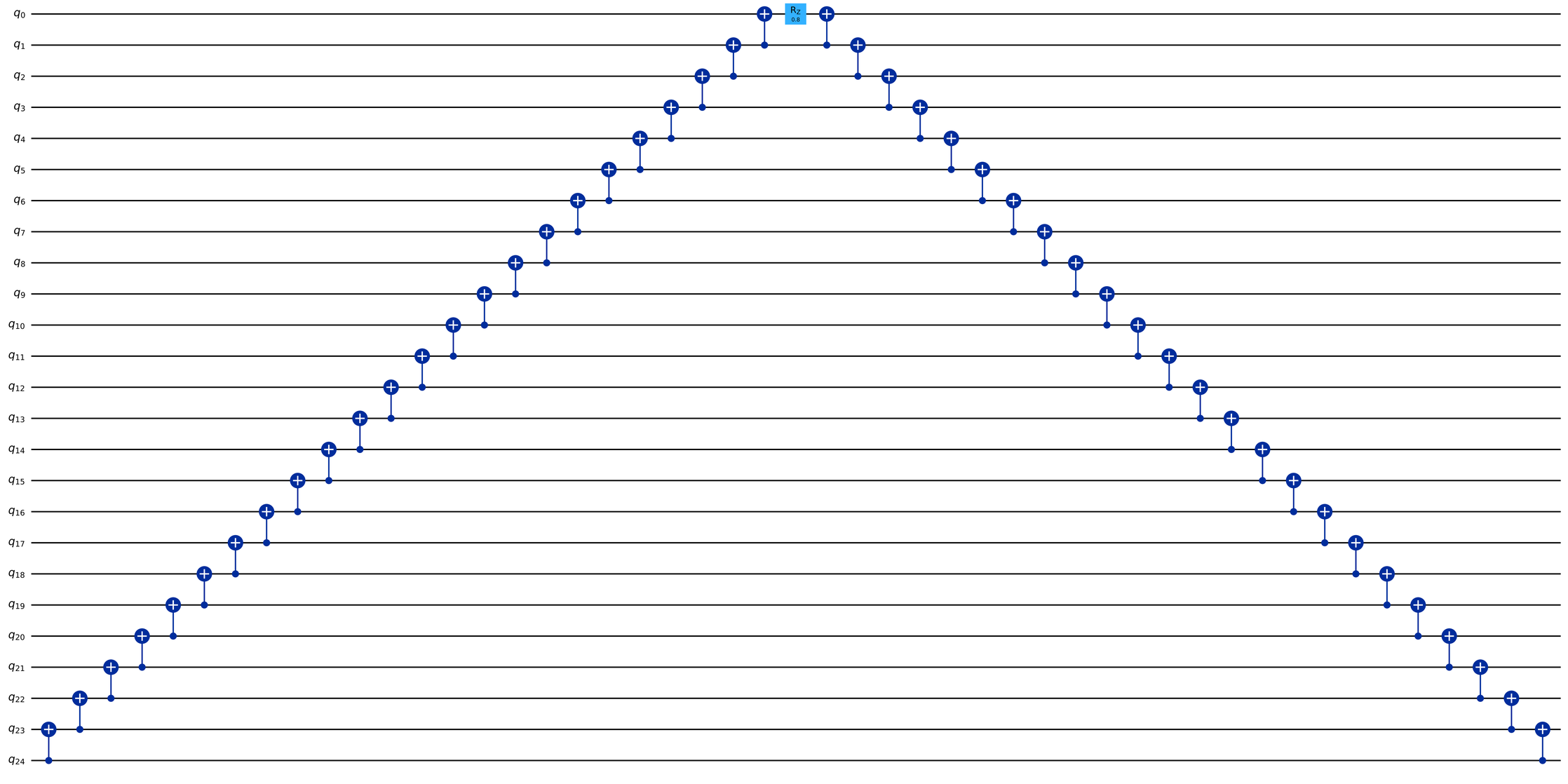

[Z20] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [97, 98, 88, 89, 79, 69, 59, 49, 39, 38, 28, 27, 26, 25, 15, 14, 13, 12, 11, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 38, 2q gates 38 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
19 18 17 16 15 14
10 11 12 13 14 15 16 17 18 19
13 12 11 10
20 21 22 23 24 25 26 27 28 29
9 8
30 31 32 33 34 35 36 37 38 39
7
40 41 42 43 44 45 46 47 48 49
6
50 51 52 53 54 55 56 57 58 59
5
60 61 62 63 64 65 66 67 68 69
4
70 71 72 73 74 75 76 77 78 79
2 3
80 81 82 83 84 85 86 87 88 89
0 1
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

[Z20] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [22, 42, 62, 82, 102, 104, 84, 64, 44, 24, 26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 116

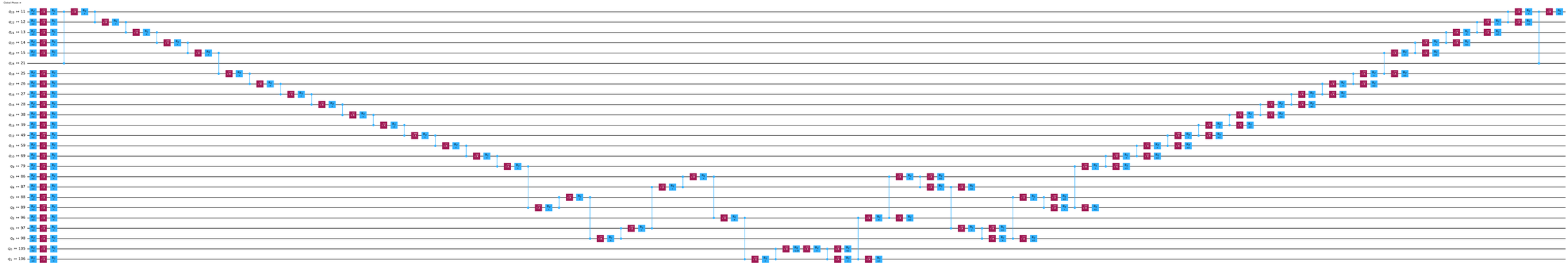




[Z25] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48

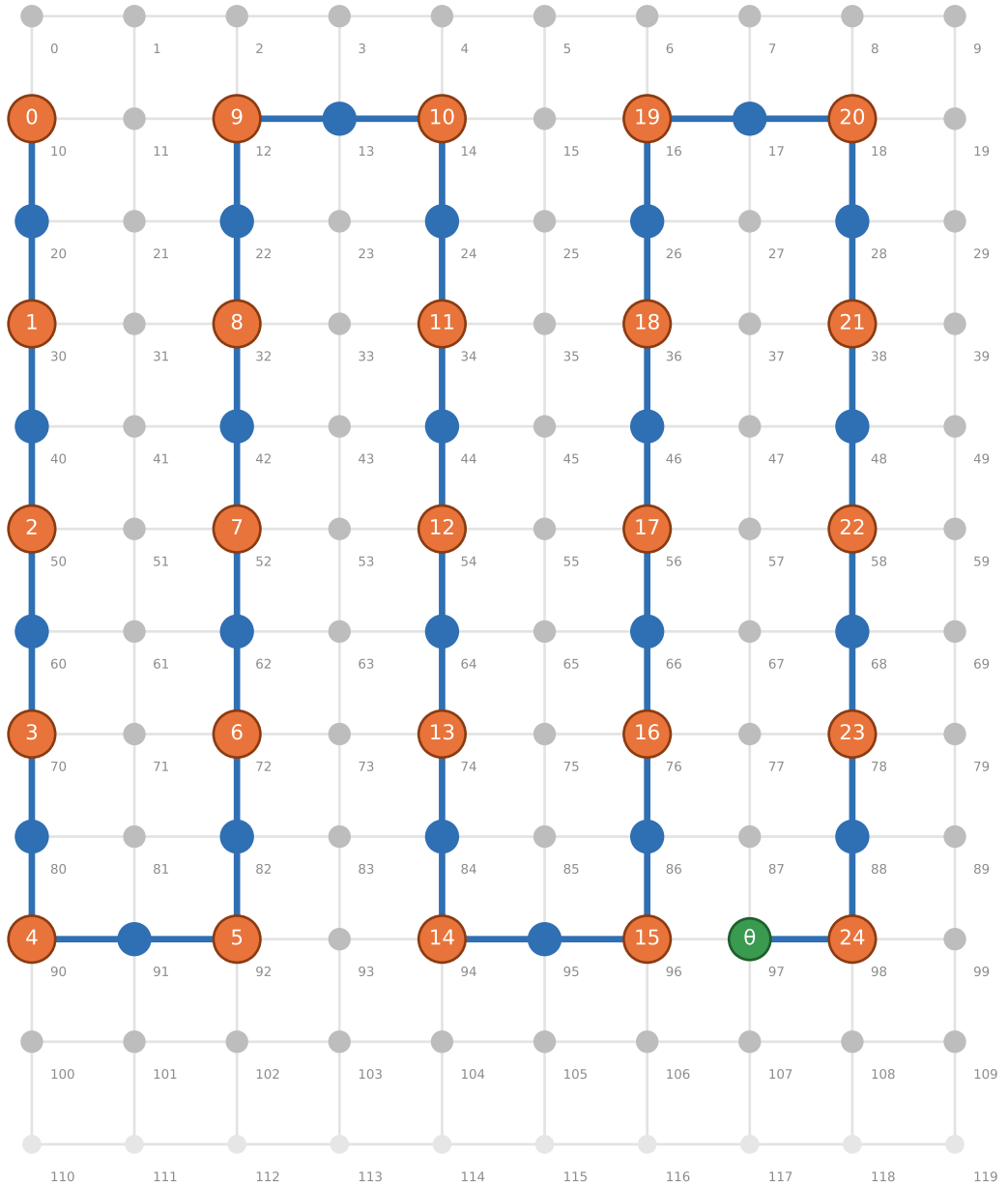


[225] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz/120 sx/96 cz/481)
expl-i 9 P1, th=0.4: P = 20 21 22 23 24 25 26 27 28 29 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 (25 logical qubits)
2q depth 48, total depth 149

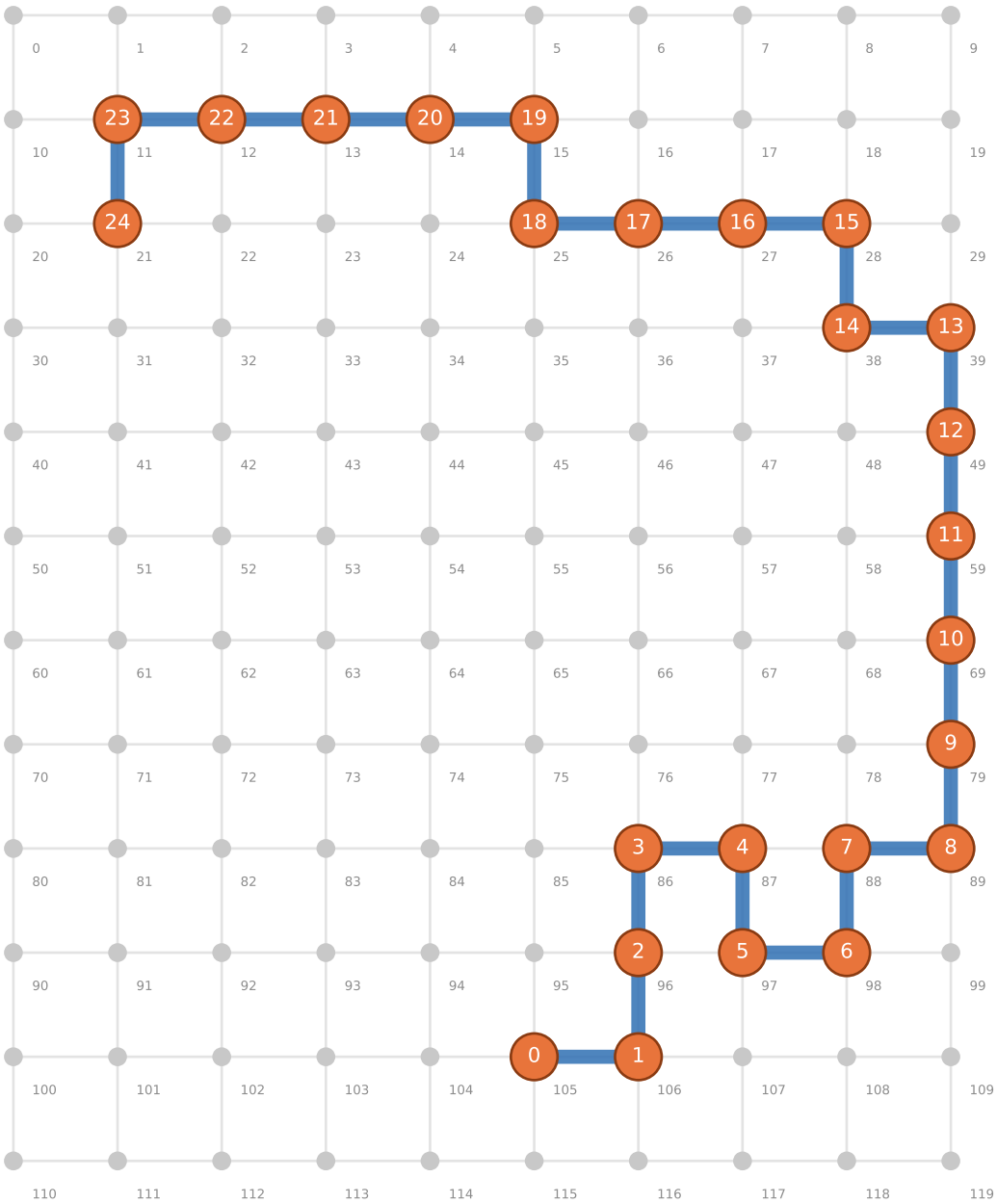


[Z25] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

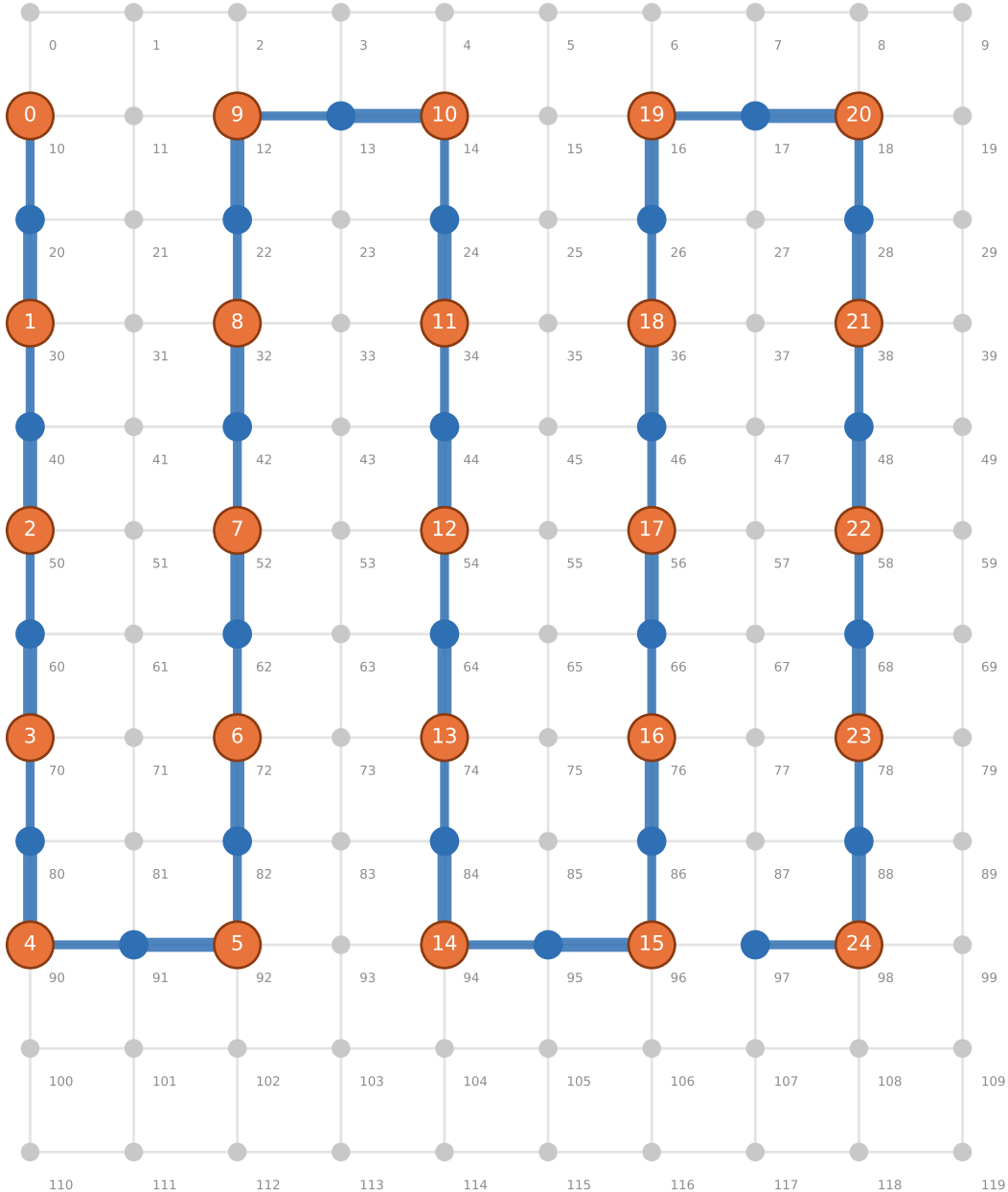
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24
chain length 49, 24 ancillas, 0 |0> fillers



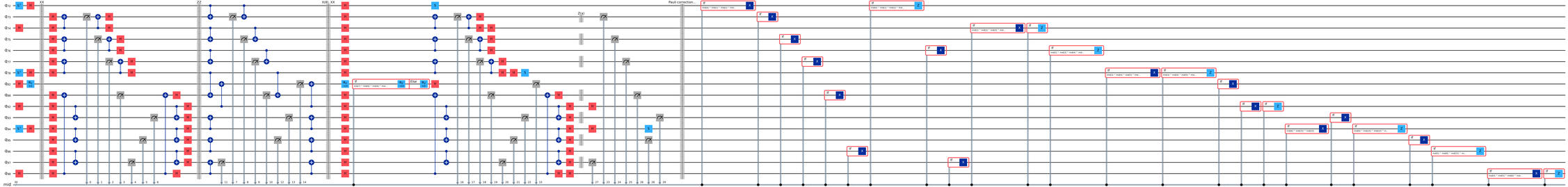
[Z25] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [105, 106, 96, 86, 87, 97, 98, 88, 89, 79, 69, 59, 49, 39, 38, 28, 27, 26, 25, 15, 14, 13, 12, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 48, 2q gates 48 (+0 vs. untranspiled from routing)



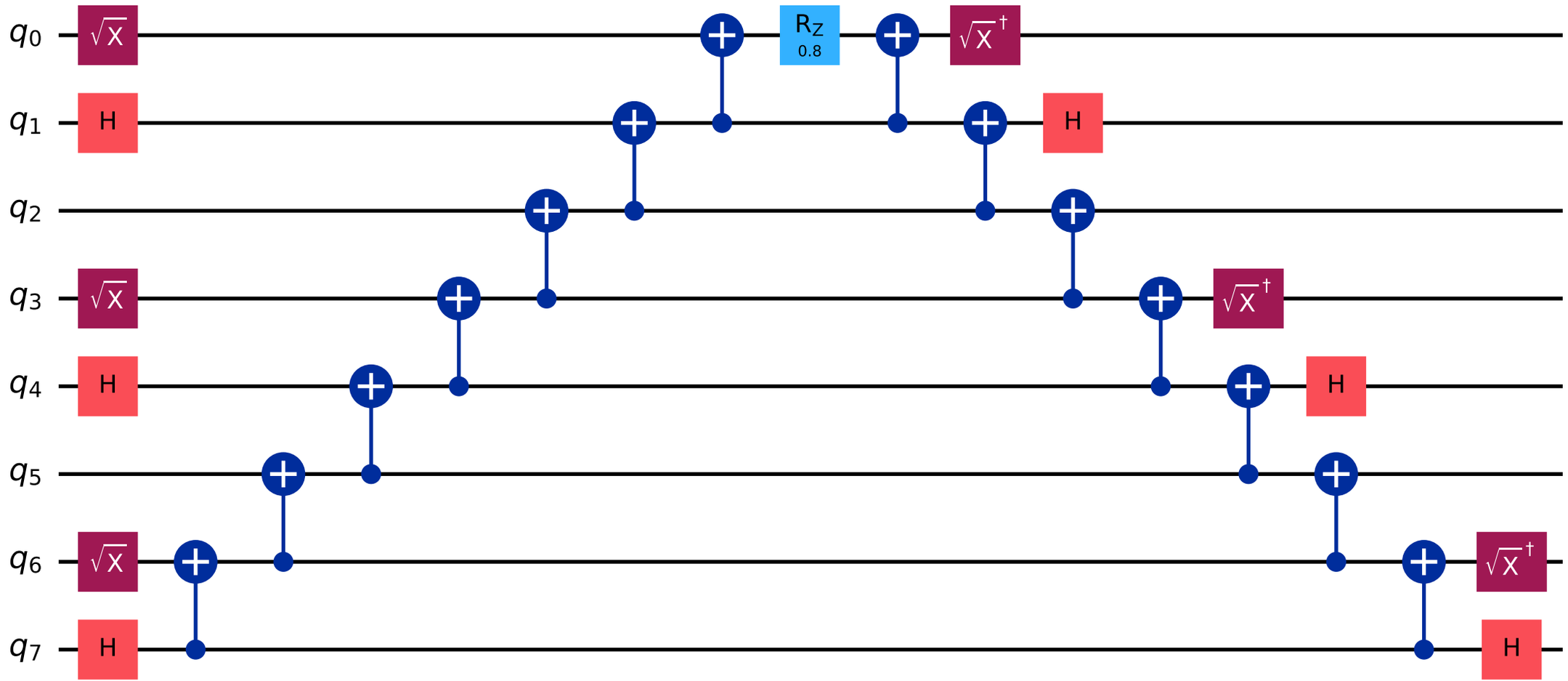
[Z25] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [10, 30, 50, 70, 90, 92, 72, 52, 32, 12, 14, 34, 54, 74, 94, 96, 76, 56, 36, 16, 18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 146



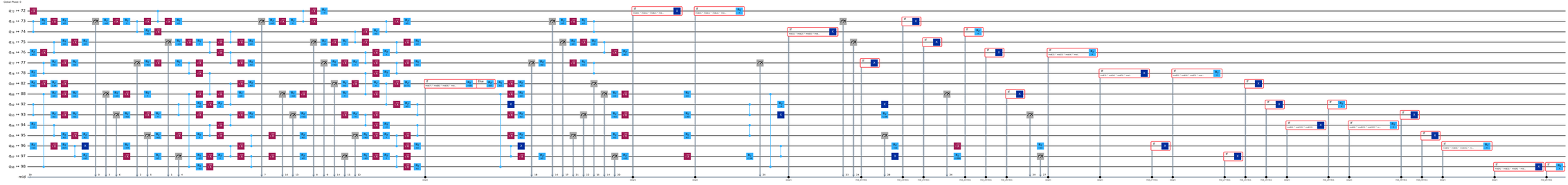
[XYZXYZXY] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P)ᵏ, θ=π/4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7 (8 logical-qubits)
2q depth 6, 30 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]



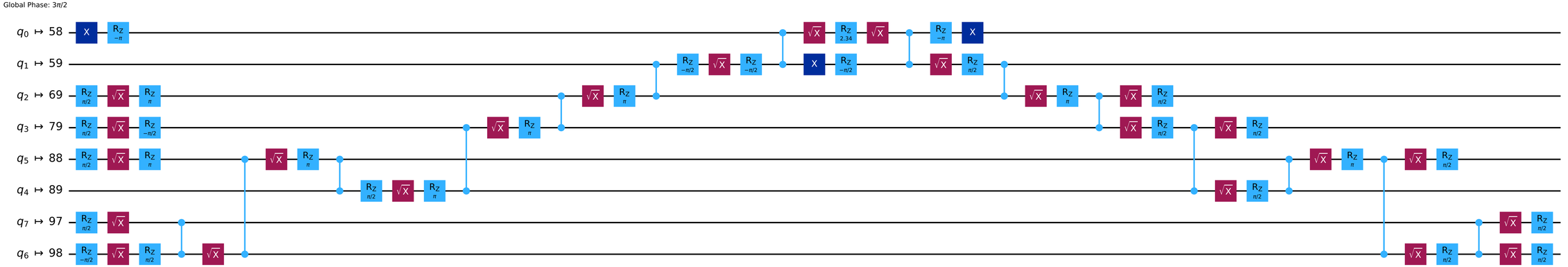
[XYZXYZXY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14



[XYZXYZXY] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz,sx,x,ecr,id,reset,measure,if_else,c_if,delay)
exp(-i θ P), θ=0.4 · P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7 · (8 logical qubits)
2q depth 6, total depth 43

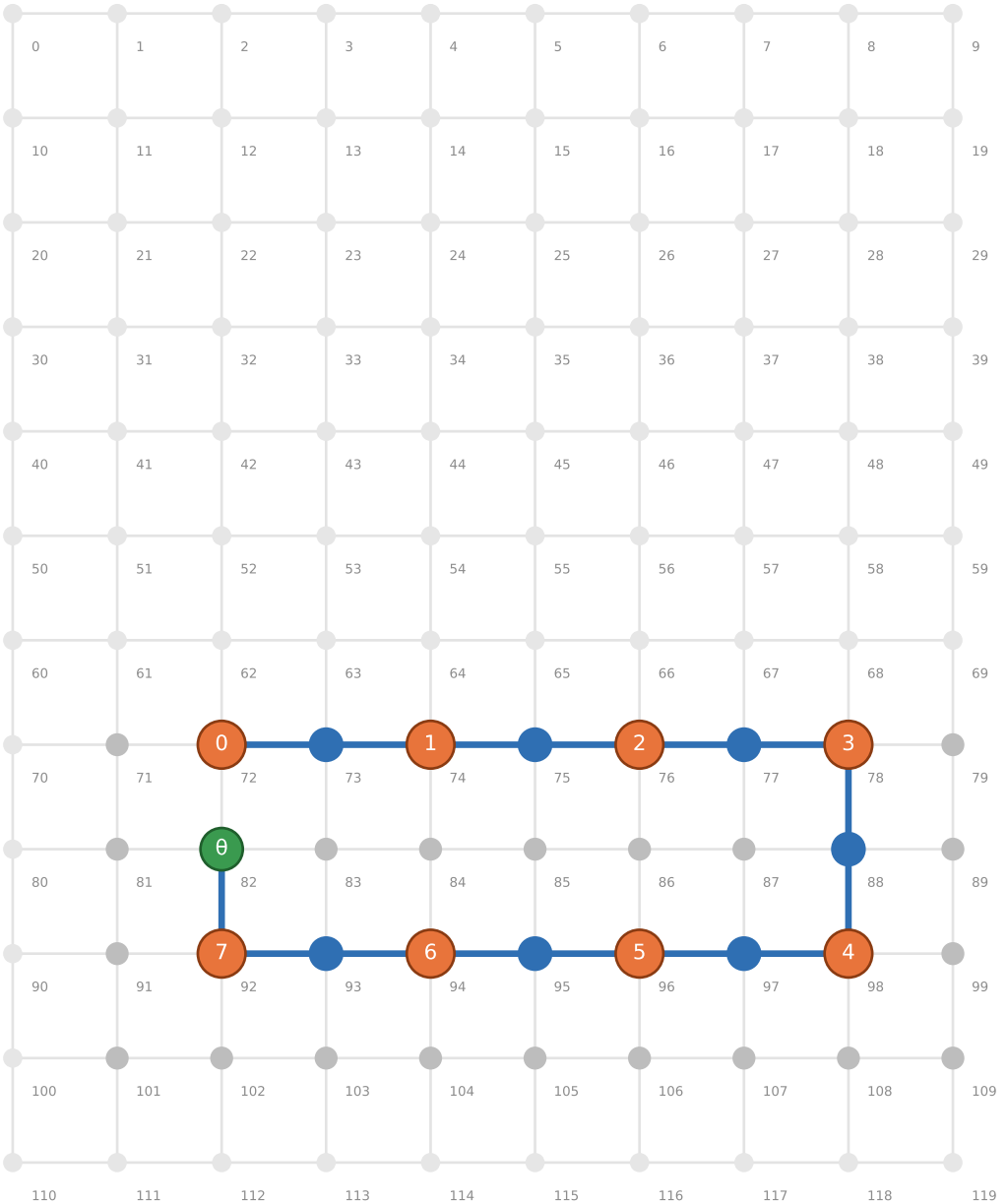


[XYZXYZXY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:31 sx:24 cz:14 x:3)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14, total depth 47

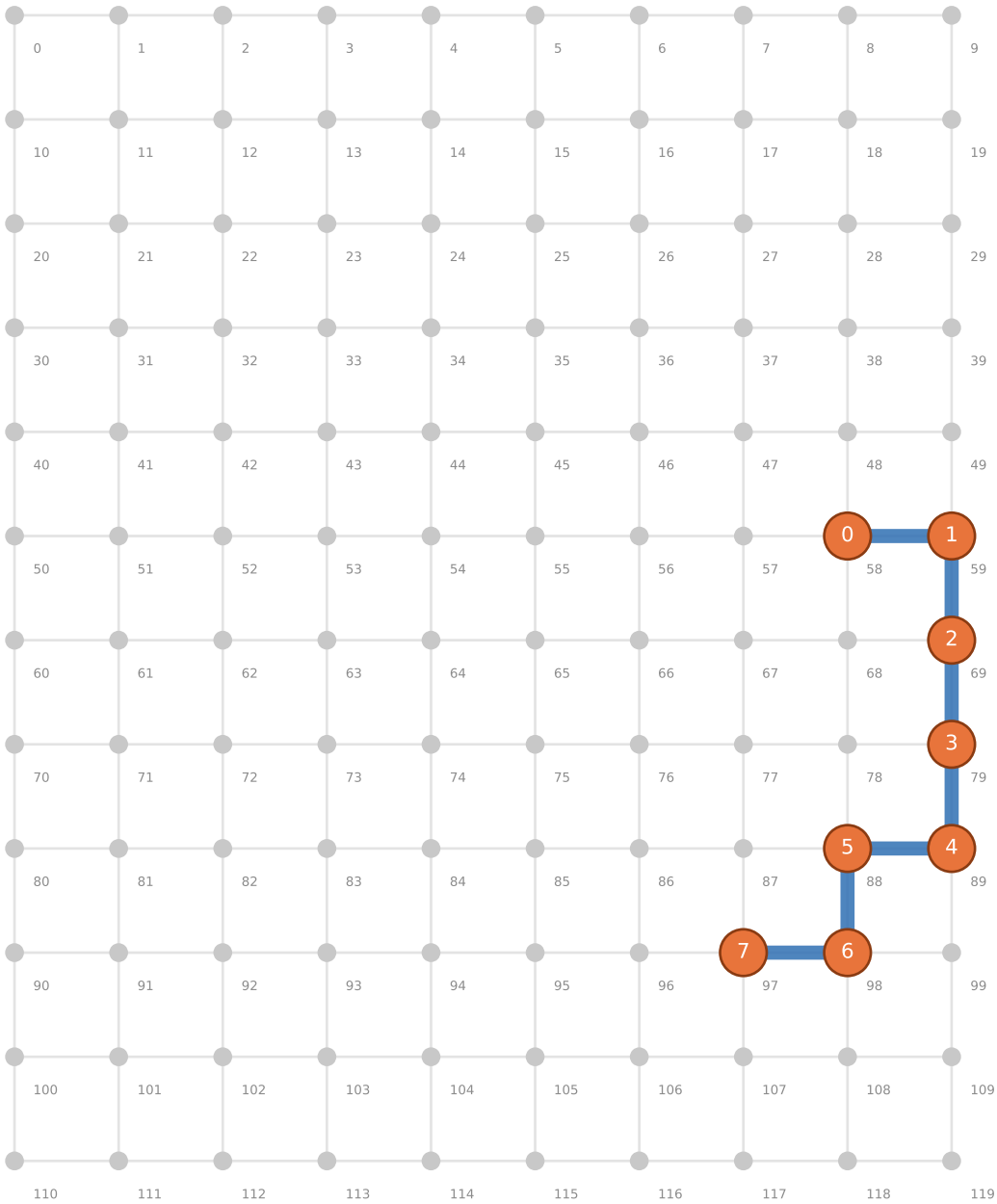


[XYZXYZXY] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

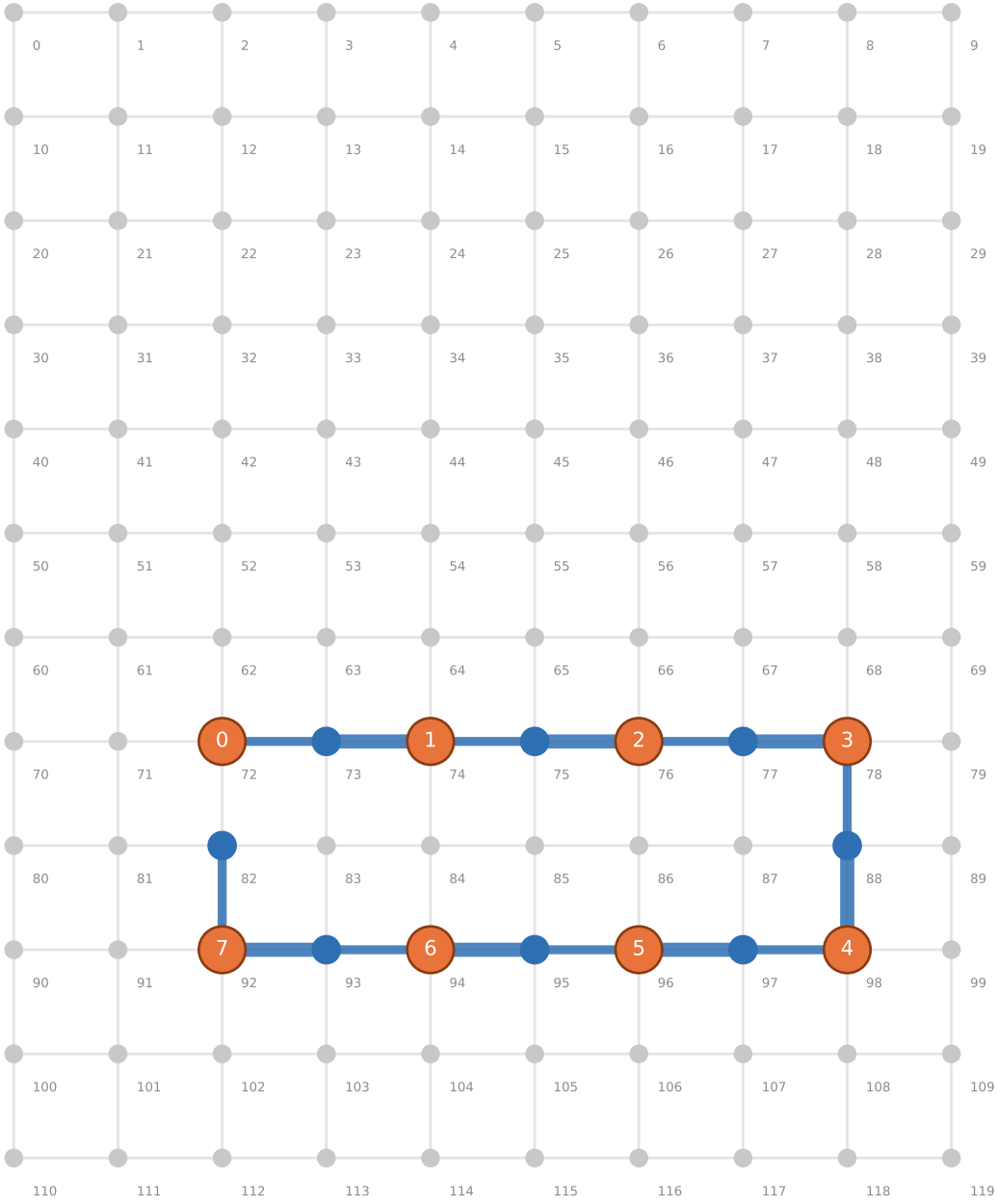
term 0: Y0 X1 Z2 Y3 X4 Z5 Y6 X7
chain length 15, 7 ancillas, 0 |0> fillers



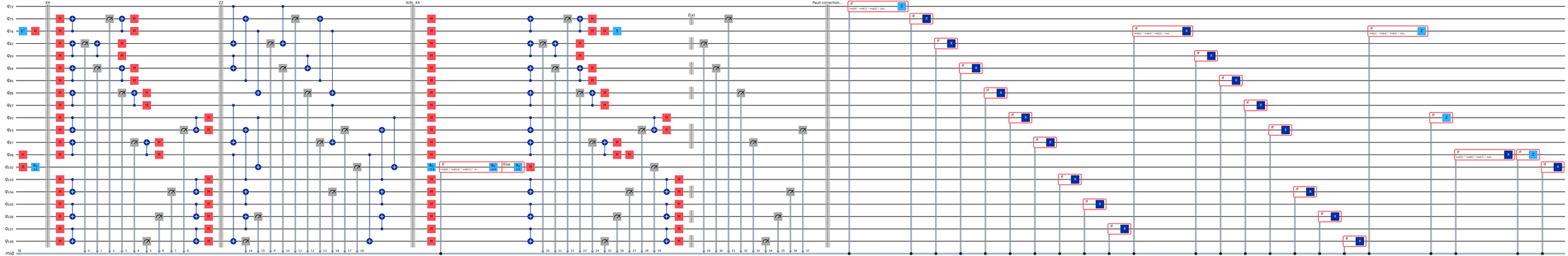
[XYZXYZXY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [58, 59, 69, 79, 89, 88, 98, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



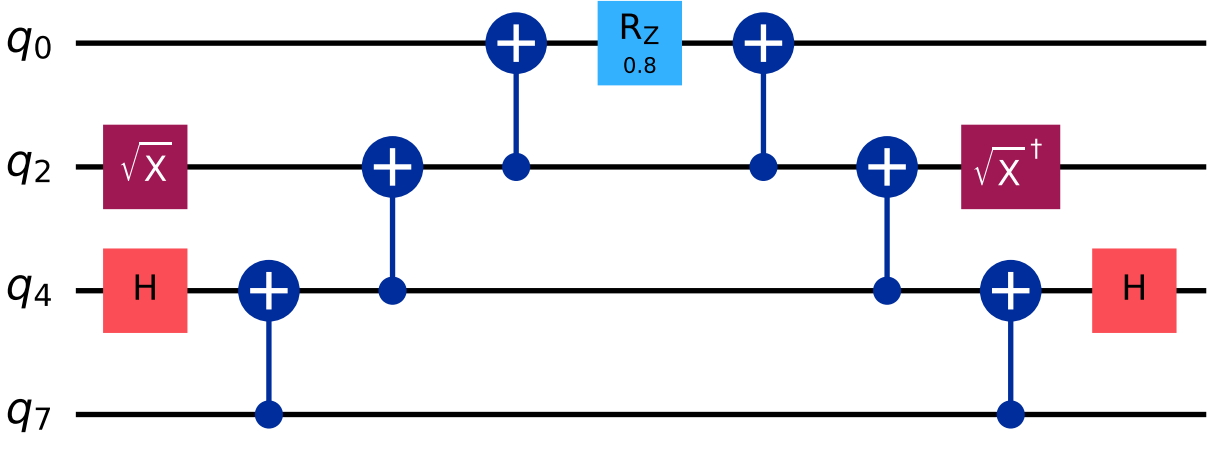
[XYZXYZXY] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 44



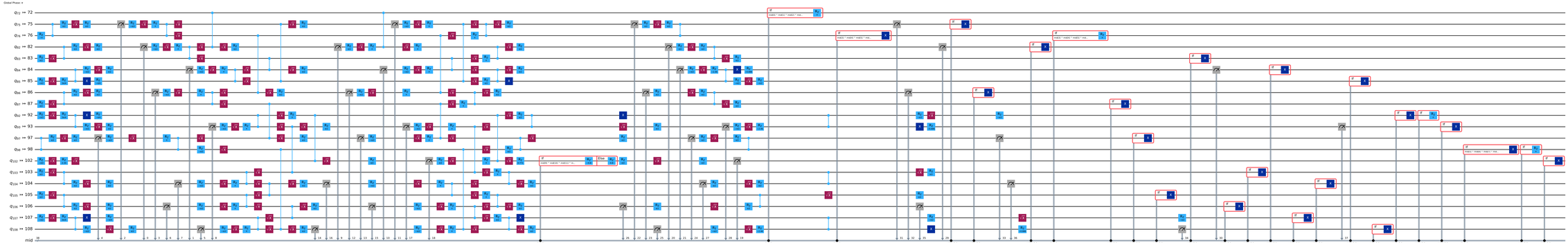
[ZiXiYiZ_gapped] DYNAMIC constant-depth circuit (untranspiled, physical qubits of ibm_rensselaer)
exp(-i θ P), θ=0.4: P = ZQ Y2 X4 Z7 (8 logical qubits)
2q depth 6, 38 mid-circuit measurements, logical→physical [72, 74, 76, 78, 98, 96, 94, 92]



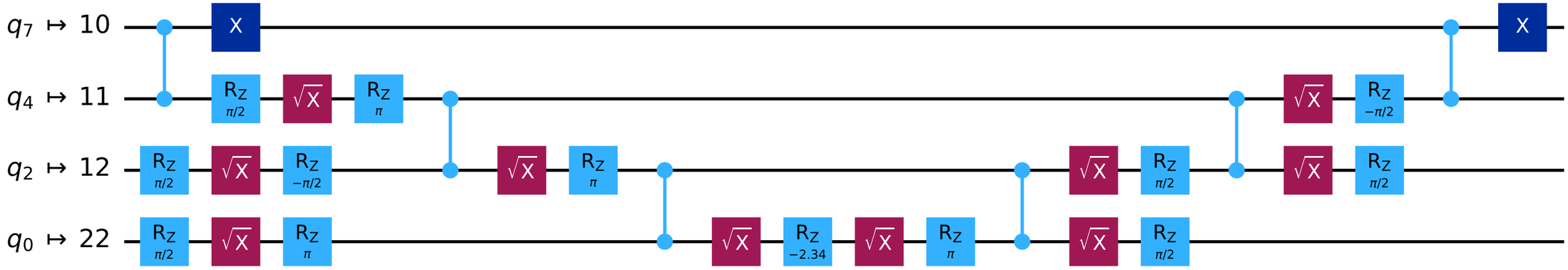
[ZIIXIYIZ_gapped] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6



[ZIXIYIZ_gapped] DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:137 sx:110 cz:56 measure:38 if_else:24 x:9)
exp(-i 9 P1_8×5.4 · P = ZIYZ XA ZT · 18 logical qubits)
2q depth 6, total depth 39

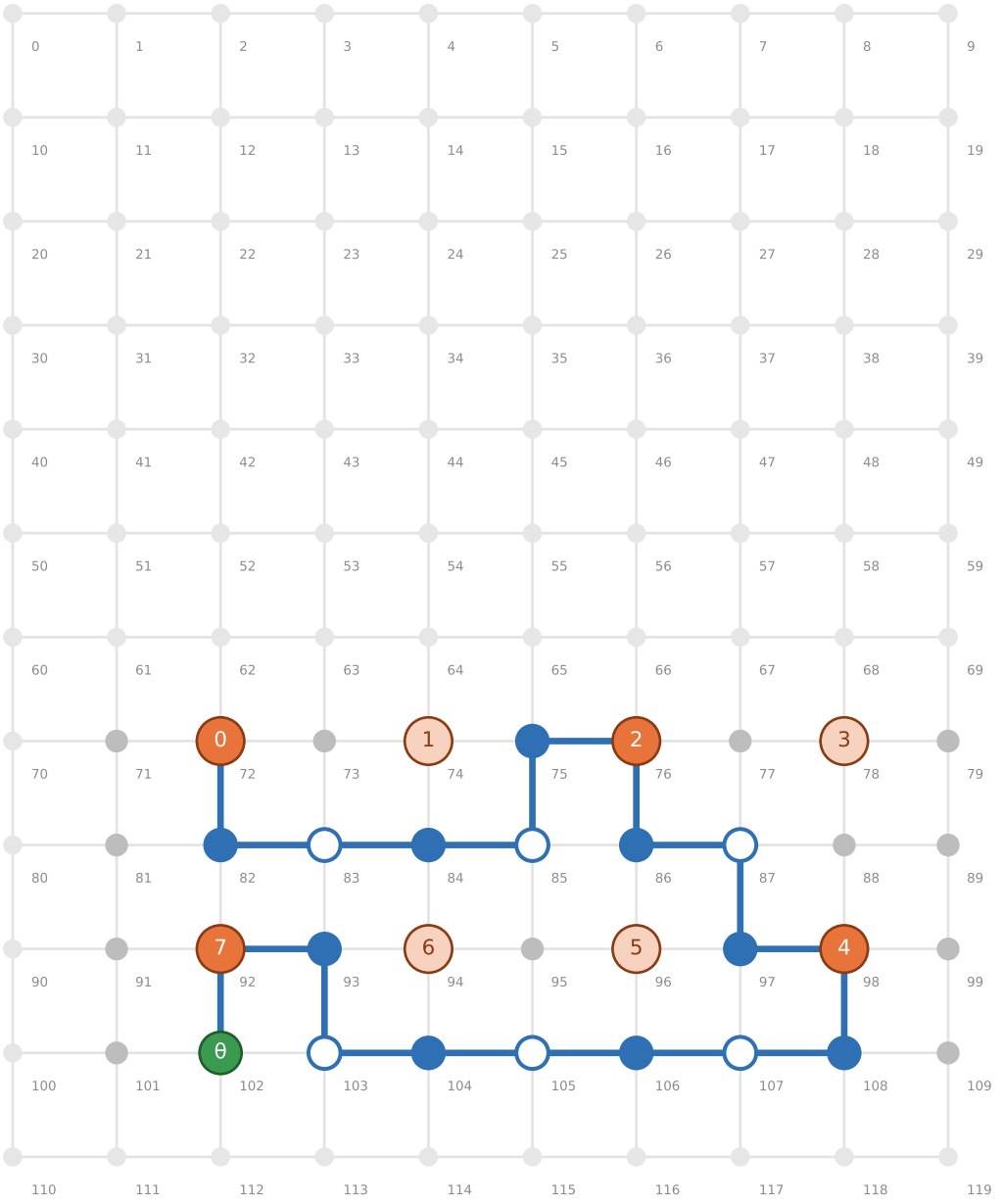


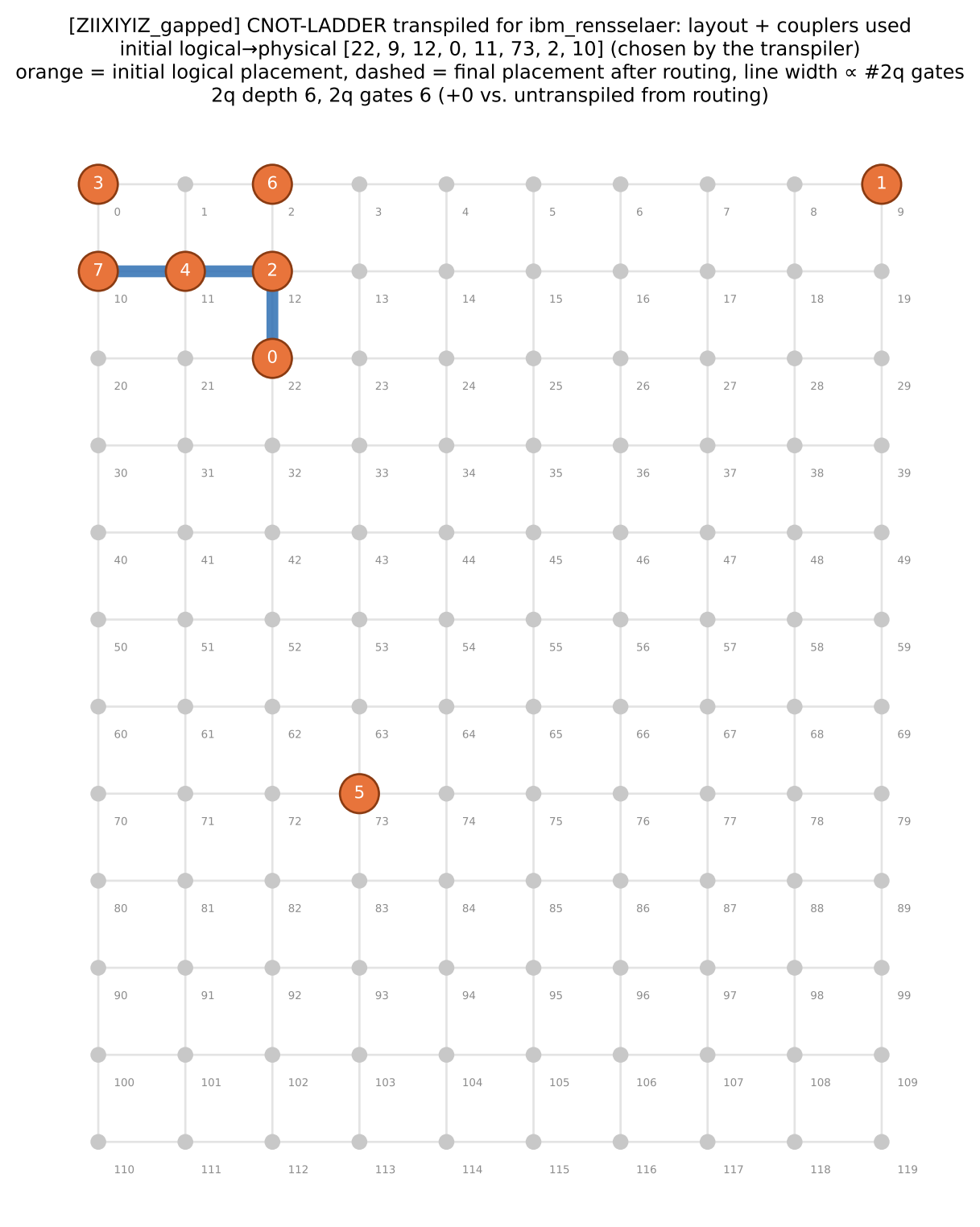
[ZIIXIYIZ_gapped] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:13 sx:10 cz:6 x:2)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
Global Phase: 0
2q depth 6, total depth 20



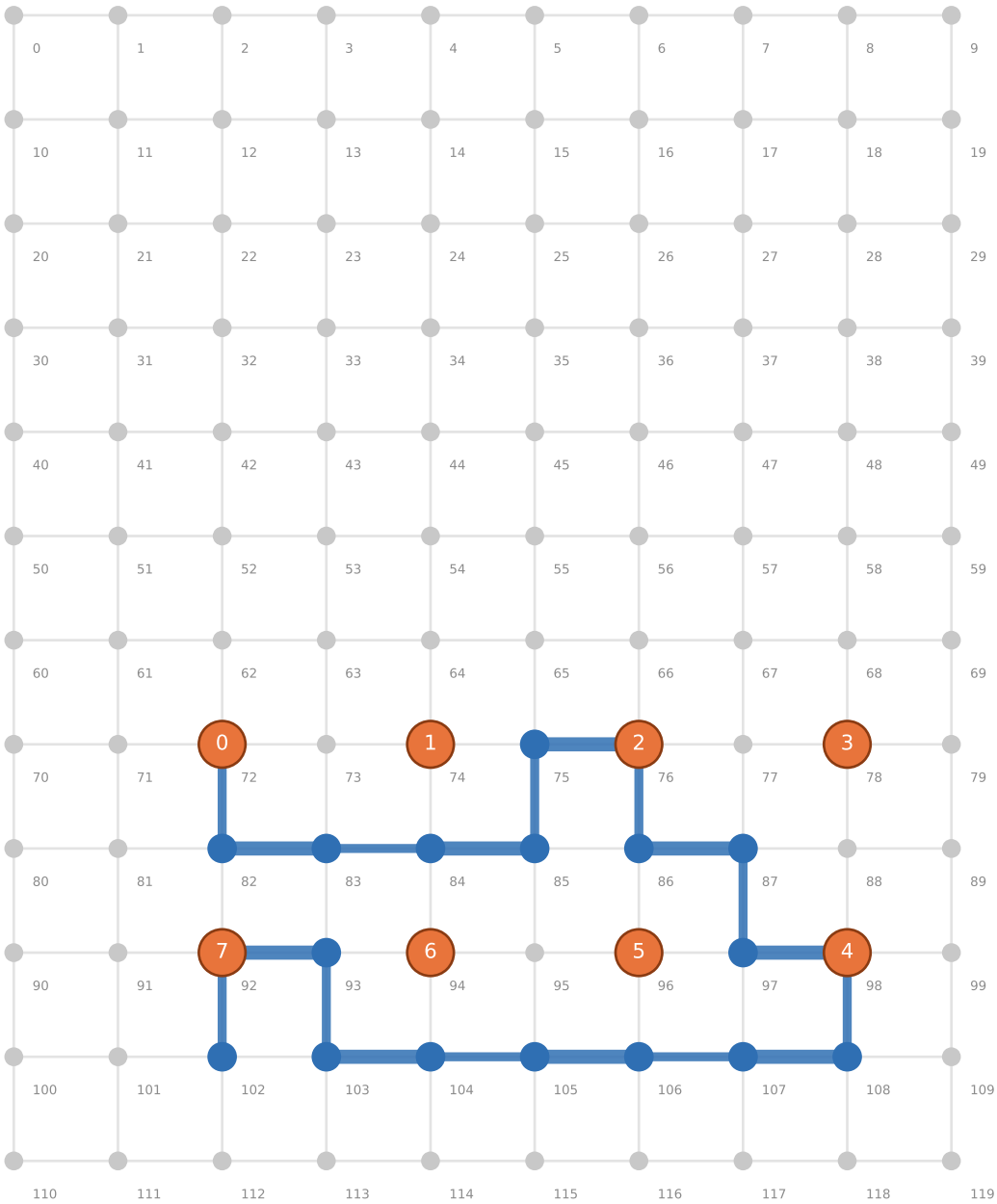
[ZIIXIYIZ_gapped] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Y2 X4 Z7
chain length 19, 9 ancillas, 6 |0> fillers

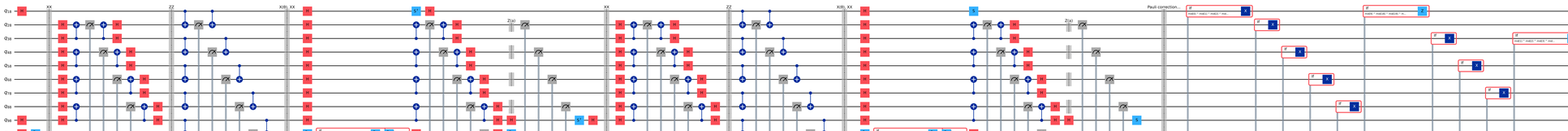



[ZIIXIYIZ_gapped] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 9, 12, 0, 11, 73, 2, 10] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
3
6
1
7
4
2
0
5
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

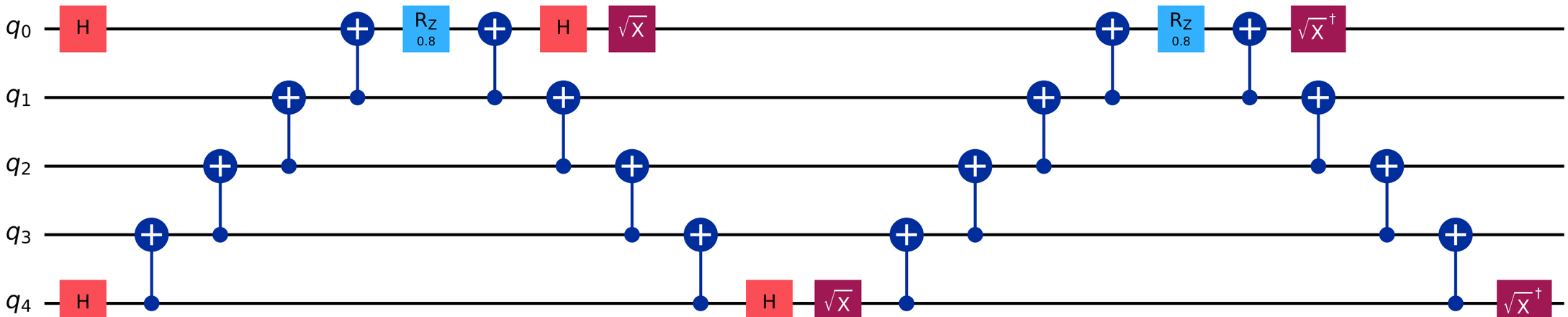
[ZIIXIYIZ_gapped] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [72, 74, 76, 78, 98, 96, 94, 92]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 56



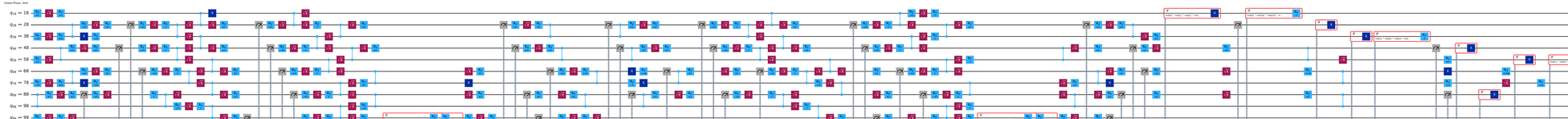
[XZZZX+YZZZY] DYNAMIC constant-depth circuit (unbranched, physical qubits of ibm_rensselaer)
exp(-i θ P)_θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 · (5 logical qubits)
2q depth 12, 36 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



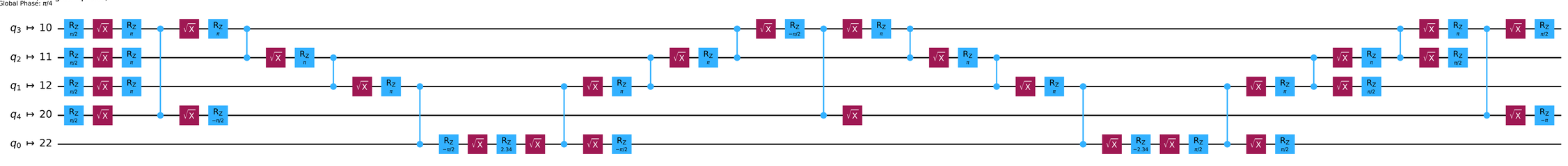
[XZZZX+YZZZY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 16



{XZZZX+YZZY}DYNAMIC circuit TRANSPILED for ibm_rensselaer (ISA: rz:124 sx:101 cz:52 measure:36 if_else:17 x:8)
expct=8.95, fn=0.4, P = 90.21 ZZ 23 X4 · 15 ZZ 22 23 Y4 · 15 (logical qubits)
2q depth 12, total depth 72

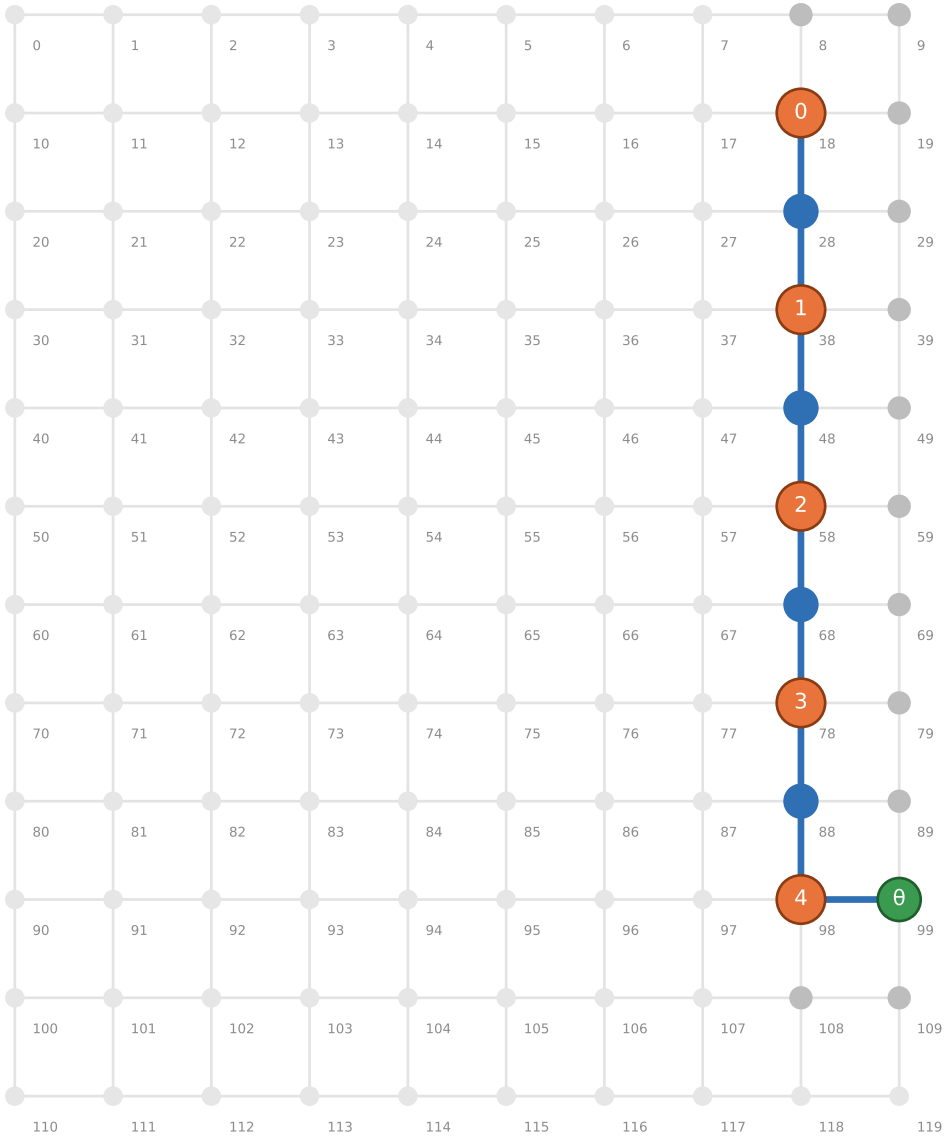

Global Phase: 3π/4
q$_{14}$ ** 18
q$_{24}$ ** 28
q$_{34}$ ** 38
q$_{44}$ ** 48
q$_{54}$ ** 58
q$_{64}$ ** 68
q$_{74}$ ** 78
q$_{84}$ ** 88
q$_{94}$ ** 98
q$_{99}$ ** 99
meas: 36
if (meas[0]==1)&&(meas[1]==1)&&(meas[2]==1)&&(meas[3]==1)&&(meas[4]==1)&&(meas[5]==1)&&(meas[6]==1)&&(meas[7]==1)&&(meas[8]==1)&&(meas[9]==1)&&(meas[10]==1)&&(meas[11]==1)&&(meas[12]==1)&&(meas[13]==1)&&(meas[14]==1)&&(meas[15]==1)&&(meas[16]==1)&&(meas[17]==1)&&(meas[18]==1)&&(meas[19]==1)&&(meas[20]==1)&&(meas[21]==1)&&(meas[22]==1)&&(meas[23]==1)&&(meas[24]==1)&&(meas[25]==1)&&(meas[26]==1)&&(meas[27]==1)&&(meas[28]==1)&&(meas[29]==1)&&(meas[30]==1)&&(meas[31]==1)&&(meas[32]==1)&&(meas[33]==1)&&(meas[34]==1)&&(meas[35]==1)

[XZZZX+YZZZY] CNOT-LADDER circuit TRANSPILED for ibm_rensselaer (ISA: rz:30 sx:28 cz:15)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 15, total depth 52

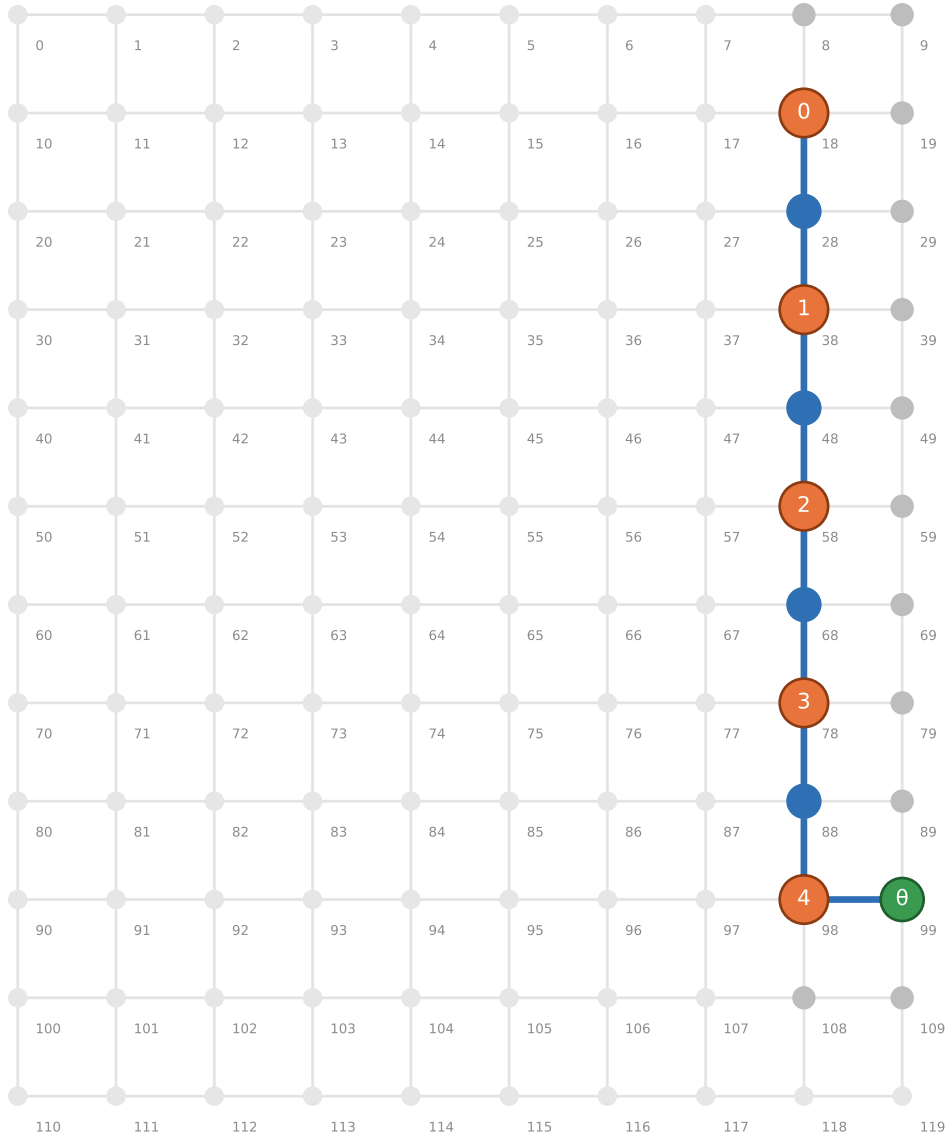


[XZZZX+YZZZY] DYNAMIC circuit layout on ibm_rensselaer (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

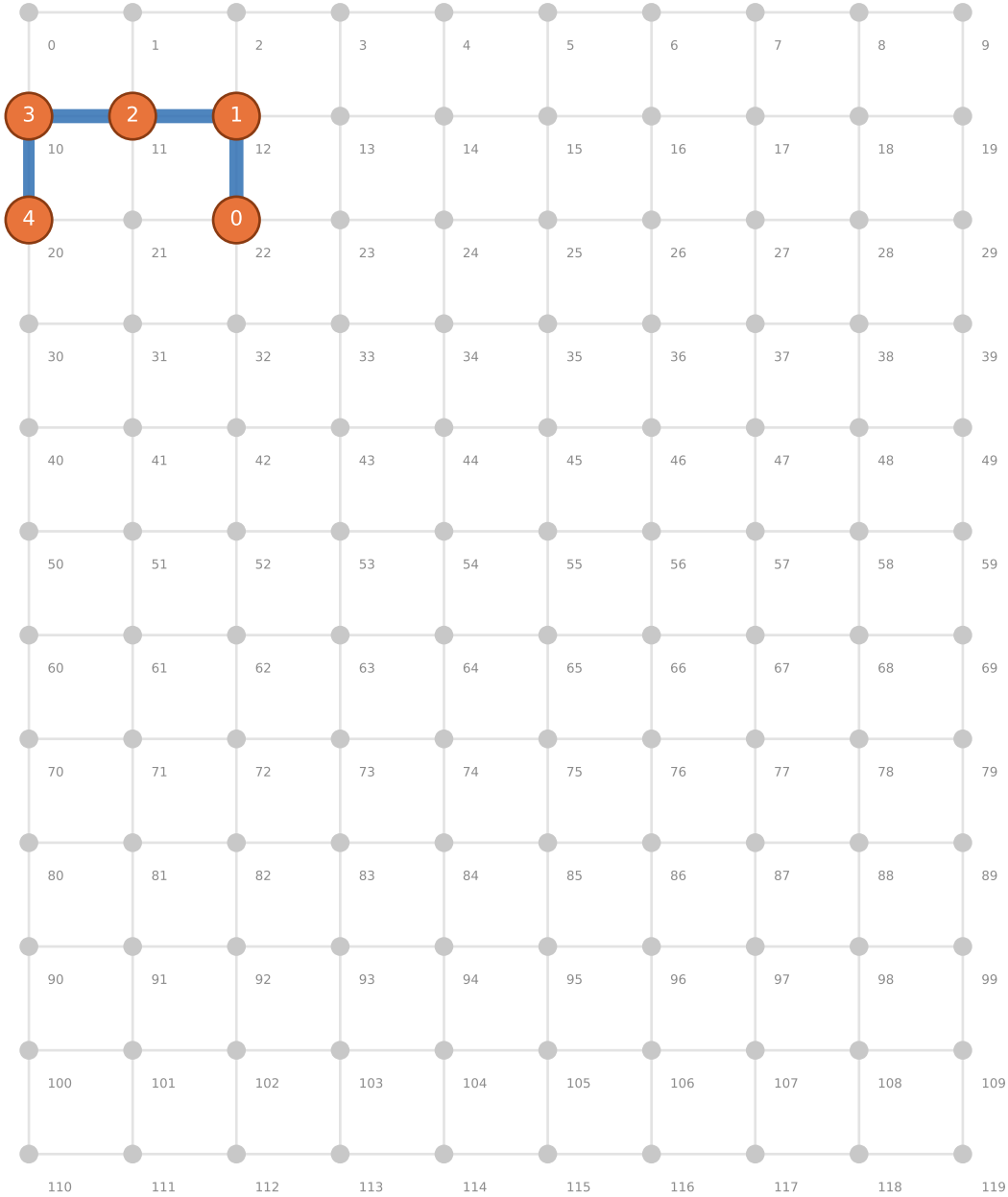
term 0: X0 Z1 Z2 Z3 X4
chain length 9, 4 ancillas, 0 |0> fillers



term 1: Y0 Z1 Z2 Z3 Y4
chain length 9, 4 ancillas, 0 |0> fillers



[XZZZX+YZZZY] CNOT-LADDER transpiled for ibm_rensselaer: layout + couplers used
initial logical→physical [22, 12, 11, 10, 20] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 15, 2q gates 15 (-1 vs. untranspiled from routing)



[XZZZX+YZZZY] DYNAMIC circuit transpiled for ibm_rensselaer: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 12, 2q gates 52

